

TABLE 18.5
Calculated Solubility Sphere for Rhodamin FB

Solvent	δ_D	δ_P	δ_H	Solubility	RED	V
Acetone	15.5	10.4	7.0	0	1.125	74.0
Benzene	18.4	0.0	2.0	0	1.991	89.4
1-Butanol	16.0	5.7	15.8	0*	0.999	91.5
Butyl acetate	15.8	3.7	6.3	0	1.517	132.5
gamma-Butyrolactone	19.0	16.6	7.4	0*	0.988	76.8
Cyclohexane	16.8	0.0	0.2	0	2.076	108.7
Diacetone alcohol	15.8	8.2	10.8	1*	1.001	124.2
Diethylene glycol	16.6	12.0	20.7	1	0.486	94.9
Diethylene glycol monomethyl ether	16.2	7.8	12.6	1	0.934	118.0
Diethylenetriamine	16.7	13.3	14.3	1	0.487	108.0
Dimethyl formamide	17.4	13.7	11.3	1	0.677	77.0
Dimethyl sulfoxide	18.4	16.4	10.2	1	0.741	71.3
Dipropylene glycol	16.5	10.6	17.7	1	0.570	130.9
Ethanol	15.8	8.8	19.4	1	0.732	58.5
Ethylene glycol	17.0	11.0	26.0	1	0.815	55.8
Ethylene glycol monoethyl ether	16.2	9.2	16.4	1	0.707	79.1
2-Ethyl hexanol	15.9	3.3	11.8	0	1.294	156.6
Glycerol	17.4	12.1	29.3	1	0.996	73.3
Methanol	15.1	12.3	22.3	1	0.589	40.7
Methylisoamyl ketone	16.0	5.7	4.1	0	1.530	142.8
Methyl-2-pyrrolidone	18.0	12.3	7.2	1*	1.042	96.5
1-Pentanol	15.9	4.5	13.9	0	1.138	108.6
1-Propanol	16.0	6.8	17.4	0*	0.889	75.2
Propylene glycol	16.8	9.4	23.3	1	0.772	73.6
Tetrahydrofuran	16.8	5.7	8.0	0	1.295	81.7
Toluene	18.0	1.4	2.0	0	1.902	106.8
Trichloroethylene	18.0	3.1	5.3	0	1.615	90.2
Water	15.5	16.0	42.3	0	1.965	18.0

Note: $\delta_D = 16.7$; $\delta_P = 17.5$; $\delta_H = 18.5$; $R_o = 12.2$; $FIT = 0.930$; $NO = 28$.

Source: Riedel, G., *Farbe and Lack*, 82(4), 281–287, 1976; Birdi, K.-S., Ed., *Handbook of Surface and Colloid Chemistry*, CRC Press, Boca Raton, FL, 1997, chap. 10.

energy parameter plot (see Figure 18.1). The irregular plot can presumably still be used as such, but the characterization of the dye in question is not useful in relation to prediction of interactions with other materials. The plot has given several individuals the impression that there are significant problems with the HSP approach when it is applied to this kind of material. This is not true. A computer analysis based on the pure solvent data given by Riedel¹⁶ confirms that a good “spherical characterization of Rhodamin FB is possible using the same data otherwise used in Figure 18.1.”⁵ The HSP data for this correlation are given in Table 18.5. The data fit as 0.93 for 28 data points. Figure 18.1 clearly shows that this HSP sphere covers more space than the data, with a significant portion in the high energy region where there are no liquids. Chapter 1, Equation 1.9 (with the constant 4) was used in this correlation, as it has been used in all the other HSP correlations in this book. The HSP correlations for water-soluble polymers and other high energy materials involved similar extrapolations into domains where there are no liquids. This procedure may be subject to revision at some future point in time, but for the present it seems to be the only procedure possible to maintain consistency in the HSP procedures developed. It should be remembered that many

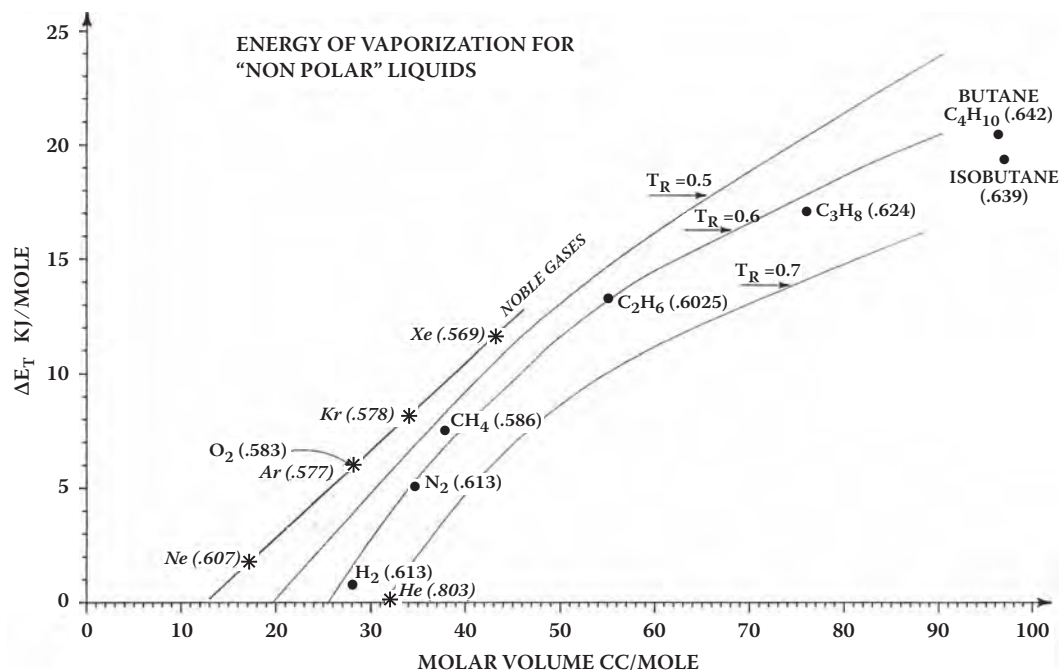


FIGURE 18.2 Cohesion energy for various low molecular weight materials as a function of molecular volume and reduced temperature (given by curves or in parentheses). (See text for discussion.)

(most) liquids with high HSP (water, methanol, glycols) also have low molecular volumes (V). This makes them "better" solvents than expected by comparison with all the other solvents (whose average V is closer to 100 cc/mol). This fact might give the impression that the constant 4 in Chapter 1, Equation 1.9 should be increased. This is discussed more in Chapter 2.

A unifying concept and procedure for the use of water in all testing is needed. The HSP considerations discussed in this book provide help toward reaching this goal.

GASES

HSP can also be used to improve understanding of the solubility behavior of gases. Solubility parameters are usually derived from data at the normal boiling points. HSP derived from these numbers seem to be in good qualitative agreement with expectations (even at 25°C), and in many cases quantitative agreement with physical behavior has also been found. Some examples are given by Barton.² Solubility parameter correlations for oxygen²⁸ (Chapter 13) and nitrogen²⁹ (Chapter 13) have been used as examples in this book. The δ_p and δ_H parameters for these two gases are zero. HSP for many gases where this is not true are reported in Chapter 13, Table 13.4. A specific example where this is not the case is carbon dioxide. Carbon dioxide is extensively discussed in Chapter 10 where the HSP are calculated for large variations in pressure and temperature. This same procedure is applicable to other gases. Chapter 3 also treats a method to calculate the three partial solubility parameters for gases.

In the process of calculating the HSP for gases, it was found necessary to extrapolate the data in Chapter 1, Figure 1.1 to lower molar volumes. Figure 18.2 is derived from this. This figure is worthy of some consideration from a theoretical point of view. The basis of the HSP is a corresponding states calculation for E_D as the energy of vaporization of a corresponding hydrocarbon solvent (same V and structure) at the same reduced temperature. This is, of course, 298.15 K divided

by the solvent's critical temperature. The reduced temperature at the boiling point is indicated in parentheses in Figure 18.2. Questions can be raised as to why the noble gases differ from the hydrocarbon solvents and whether the hydrocarbon solvents were the best choice as reference materials. Also, why is oxygen among the noble gases in cohesion behavior rather than rated with the other gases? At the time of choice for a reference for the dispersion bonding energies, there were ample data on latent heats for the hydrocarbons and the aliphatic hydrocarbons were considered as having δ_D and δ_H values equal to zero. This may not quite be true, but the corrections would be minor, and the necessary data for a revised reference are lacking. It appears that the currently chartered course of using hydrocarbon solvents as a basis will be maintained. Some additional considerations may be found in further study of the relation of HSP to the corresponding states theory of Prigogine and co-workers as discussed in Chapter 2. The behavior of the hydrocarbon solvents appears to be included within the Prigogine parameter for differences in size (ρ).

ORGANIC SALTS

The HSP of several organic salts have been compared with the HSP for the organic acids and organic bases from which they were made.³⁰ The result was that the organic salts always had considerably higher HSP than either of the components making them up. As examples, the salts made from formic acid and acetic acid combined individually with dimethyl ethanolamine had $\delta_D, \delta_P, \delta_H$ equal to 17.2; 21.5; 22.5 and 16.8; 19.8; 19.8, respectively, whereas the $\delta_D, \delta_P, \delta_H$ are 14.3; 11.9; 16.6 for formic acid, 14.5; 8.0; 13.5 for acetic acid, and 16.1; 9.2; 15.3 for dimethyl ethanolamine. All of these values are in $\text{MPa}^{1/2}$. This general relationship was also found for other salts formed by combinations of organic acids with a variety of amines. These values are reported in Appendix Table A.1 in dark type, as there are experimental data to justify the numbers. The HSP for the salts are generally close to those mentioned earlier. These values are high enough to make the salt entities insoluble in most polymers. Their affinities for water will be very high, however, both because of high HSP and also because of the charges associated with the salt groups. There was about 10% shrinkage in volume compared with the original volumes of the acids and bases. In some cases, the cohesion energy of the salts is high enough to make them solids rather than liquids. This study showed that organic salts can indeed be characterized by HSP. More work is necessary, however, with other types of salts. In particular, the acid groups found in nature, such as in hemicelluloses, deserve more attention (see Chapter 15 and Reference 24).

INORGANIC SALTS

The solubility of magnesium nitrate [$\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$] was evaluated in a standard set of solvents¹ and later correlated more precisely with HSP. The HSP derived from this are $\delta_D, \delta_P, \delta_H$, and R_o equal to 19.5; 22.1; 21.9, and 13.2, respectively, all in $\text{MPa}^{1/2}$. Nitrates are known to be among the most soluble of salts. Somewhat less soluble than the nitrates are chlorides. These are only partly soluble in a few organic liquids with very high HSP. Group contributions to the HSP from the nitrate group are expected to result in lower HSP, and, in particular, lower δ_D for the nitrate portion of a salt than would be expected from the group contributions from a chloride. This would lead to greater solubility of the nitrates in organic solvents, which is indeed the case. The δ_D parameter seems to be qualitatively capable of describing the behavior of metals to some extent. It may be possible to arrive at an approximate description of inorganic salt solubility in organic media (perhaps water, too) using HSP or some modification/extension thereof. The salting in and salting out of various polymers can perhaps provide clues to assign HSP in this connection. Finally, it should be noted that an excellent HSP correlation of the chemical resistance of an inorganic zinc silicate coating is reported in Chapter 12, Table 12.1.

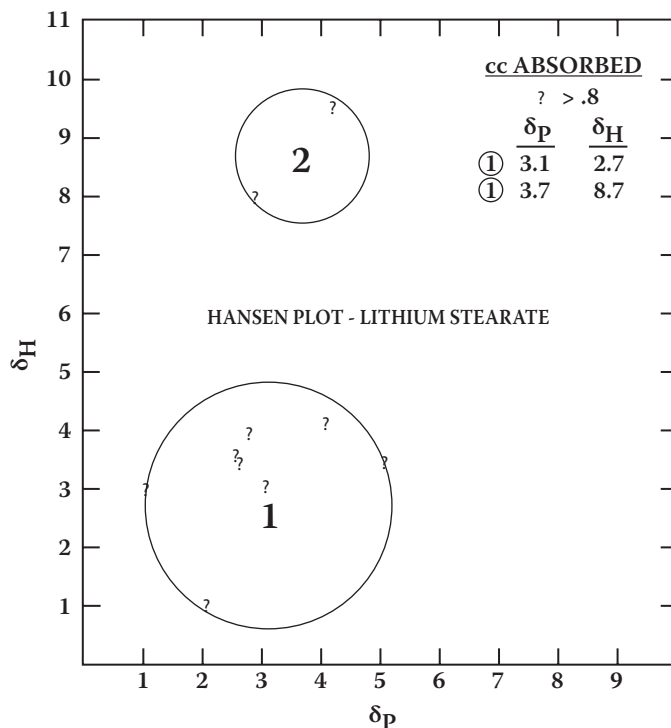


FIGURE 18.3 HSP plot for solvent uptake by lithium stearate.^{31,32} Units for δ_P and δ_H are $(\text{cal}/\text{cm}^3)^{1/2}$. (From Alan Beerbower, personal communication.)

ORGANOMETALLIC COMPOUNDS

No systematic studies of the HSP of organometallic compounds have been made. An exception is perhaps that shown in Figure 18.3 where Beerbower³¹ used data from Panzer³² to show that lithium stearate does indeed have two distinct regions of solvent uptake and that a HSP plot can show why. This example shows that one can calculate HSP values where the relevant data can be found in the literature and then test these with relevant experiments. Group contributions would be valuable. Metallic bonds differ in nature from those usually discussed in connection with organic compounds. A suspicion is that, at least in practice, the cohesion energy derived from the “metallic” bonding in organometallic compounds can be coupled with the dispersion parameter. There is also a question, for example, of whether metal atoms in the center of more complicated molecules are effectively shielded from any (surface) contact with a solvent. Surface contacts are clearly important, but it appears that the nature of the central atom also has an effect. Finally, it might be noted that Hildebrand and Scott presented a chapter on the solubility parameters of metals.³³ Unfortunately, we do not often deal with pure metals in this context, but rather metal oxides, for which no HSP work has been reported, as least not to the author's knowledge.

AROMAS AND FRAGRANCES

Aromas and fragrances are important in connection with packaging materials, foodstuffs, cosmetics, chewing gum, etc. A recent report³⁴ discussed HSP in connection with fragrances and aromas. It is clear that HSP exist for these materials, but very little work has been published in the area. One of the examples included in Reference 34 was the development of an artificial nose based on coating oscillating sensors, which oscillate more slowly when they gain weight. Matching HSP for the coating and material to be detected leads to increased weight gain and increased sensitivity. Other

examples where HSP could be systematically used include: counteracting undesirable odors using fragrances that have reasonably similar HSP; absorption of odors into plastics, coatings, sealants, etc.; development of packaging with designed HSP to either function as a barrier or as a sink; and an estimate of where a given aromatic material is likely to reside. The key to interpretation is, as usual, that similarity of HSP means higher affinity. It is thought that the masking effect is the result of adsorption of the fragrance on the regions of the sensory system where HSP are similar to those of the undesired odor. There may not be a complete match regarding steric adsorption, but the sensitive areas are covered anyway, and this barrier prevents the odor from arriving at the given sites.

ABSORPTION OF CHEMICALS IN PLASTICS

HSP correlations exist for chemical resistance, permeation phenomena, and uptake of solvents in many polymers. The recycling of polymeric containers has a potential problem in that the polymers used are able to readily absorb those chemicals whose HSP are not too different from their own. Once a chemical has absorbed into a polymer, and particularly if it is a rigid polymer with a relatively high glass transition temperature, it can be very difficult to get it out again. A relatively slow diffusion process is required to do this. See Chapter 16. It is suggested that an extensive HSP analysis be done for those polymers where potential misuse or contamination of containers prior to recycling is a possibility. This can point out which chemicals are most likely to present the greatest problems.

CHEMICAL RESISTANCE

Chemical resistance studies have generally been performed with too few liquids and without the necessary spread of HSP to allow the data to be correlated with confidence. In addition, attainment of equilibrium is not usually confirmed. These shortcomings mean that HSP correlations of chemical resistance must be done with great care. This has been discussed in Chapter 12 in more detail. An additional activity, which should be done for practical reasons, is to assign effective HSP to various test materials such as mustard, ketchup, and other given products that often appear in tests of chemical resistance. Such data will allow greater use of the correlations since guidelines for potential improvements can be obtained.

CONTROLLED RELEASE

HSP considerations can provide an extra formulating parameter for the controlled release of drugs. When the HSP relations between a drug and its surroundings are known, predictions of its behavior can be made. When there is a good match in the HSP values, the drug will be more soluble with the ability to move at some rate within a polymer matrix. On the other hand it can be surrounded by a matrix with similar HSP, and this may slow release more than desired. A poor match in HSP may leave holes and expose the drug for more rapid release. When the match is poor, the drug will not be able to permeate through its polymeric surroundings, but it can, of course, pass through open passageways. Drugs will also tend to adsorb at surfaces where there is a good HSP match.

It has been shown (personal communication, Andreas Gryczke, Degussa Pharma Polymers, Darmstadt) that calculations of HSP for a number of drugs and for a series of EUDRACIT® polymers provided a correlation confirming that the drugs are soluble in those polymers where the HSP match closely enough. When the drug and polymer HSP were within 10 MPa^{1/2} at the concentration studied (20% wgt.), what appeared to be true solubility was found. This was evidenced by x-ray diffraction and DSC measurement that confirmed the drugs were embedded completely amorphous (solid solution). A closer match would presumably be required to do this for higher concentrations of the drugs.

Table 18.6 contains HSP for some materials whose solubility in various media may have interest. It is possible to calculate HSP for essentially all drugs, although hydrochlorides and other salts

TABLE 18.6
Calculated HSP for Various Materials

Material	δ_D	δ_P	δ_H	V
Adrenaline	20.5	8.7	19.9	154.5
4-Aminopyridine	20.4	16.1	12.9	87.1
Ascorbic Acid	18.0	12.6	27.6	106.7
Caffeine	19.5	10.1	13.0	157.9
Cycloheximide	18.3	11.0	13.8	171.0
DDT	20.0	5.5	3.1	268.8
Dopamine	18.2	10.3	19.5	180.0
Ecstasy	18.0	5.1	6.1	202.9
Meclofenoxate (base only)	16.0	6.2	9.0	198.3
Norephedrin	18.0	10.7	24.1	141.9
2-Oxopyrrolidinacetamid	17.5	15.6	11.2	116.2
Quinine	19.0	4.6	11.0	310.7
Saccharin	21.0	13.9	8.8	206.8
Serotonin	18.0	8.2	14.4	144.4
Spermidin	16.7	11.2	12.0	155.6

Note: Units are MPa^{1/2}.

Source: Hansen, C.M., *Conference Proceedings, Pharmaceutical and Medical Packaging 2001*, Skov, H. R., Ed., Hexagon Holding, Copenhagen, 2001, pp. 20.1–20.10.

should be measured rather than calculated only, as there is very little experimental data on which to base an estimate with group contributions.

There are correlations of HSP with skin permeation presented in Chapter 15 that should allow an estimate of which drugs can enter via this route. Simple calculations confirm that dopamine, nicotine, skatole, and nitroglycerine, for example, have high affinity for skin.

NANOTECHNOLOGY

Controlling the orientation of molecules can be a key for switches and the like. An example has been given in Chapter 15 where anthracene units appended to a polymer molecule adopted one orientation in toluene at room temperature, but changed to a new orientation at temperatures above 38.5° in toluene. The configuration at the higher temperatures was also that adopted in tetrahydrofuran. This change of orientation is the result of a poorer solvent at the higher temperatures compared with a good one at room temperature. More details and additional examples are found in Chapter 15.

The following link reports what follows for studies of organically modified nanoclays (organoclay) in nanocomposites:

<http://www.hwi.buffalo.edu/ACA/ACA03/abstracts/text/W0383.html>

Quote:

W0383

Evaluating Organoclays for Nanocomposites by Small Angle Scattering. Derek L. Ho^{1,2} and Charles J. Glinka¹, ¹Center for Neutron Research, National Institute of Standards and Technology,

Gaithersburg, MD 20899, ²Dept. of Materials and Nuclear Engineering, Univ. of Maryland, College Park, MD 20742.

Understanding the interaction between organically modified clay (organoclay) platelets and organic solvent molecules as well as the corresponding structure of organoclays in a suspension is a critical step toward tailoring and characterizing nanocomposites formed by organoclays in a polymer matrix. Recently, nanocomposites composed of clays and polymers have been found to have improved mechanical properties as well as enhanced thermal stability. The improved properties are related to the degree of dispersal and exfoliation of the clay platelets in the polymer matrix. In order to understand and optimize potential processing conditions, organoclays were dispersed in a number of organic solvents covering a range of solubility parameters and characterized using small-angle neutron scattering and wide-angle x-ray scattering techniques. With Hansen's solubility parameters, $\delta_o^2 = \delta_d^2 + \delta_p^2 + \delta_h^2$, the correlation between the degree of exfoliation of organoclays and the solvent in which the clay platelets are dispersed/mixed has been analyzed. It has been found that the dispersion force of the solvent, reflected by δ_d , is the principal factor determining whether the clay platelets remain suspended in the solvent while the polar (δ_p) and hydrogen-bonding (δ_h) forces affect primarily the tactoid formation/structure of the suspended platelets. The organically modified clays studied in this work precipitated in any solvent with molecules with moderately strong hydrogen-bonding groups. The correlation found has been used to correctly identify a solvent, trichloroethylene, which completely exfoliates the organoclays studied in this work.

These and the many other examples in this handbook related to self-assembly (see above), surface adsorption, molecular orientation, and affinity among molecules and molecular segments should provide ample evidence that molecular guidance in nanotechnology endeavors can be found in the HSP concept.

THEORETICAL PROBLEMS AWAITING FUTURE RESOLUTION

POLYMER SOLUBILITY

The Flory chi parameter has been used to describe polymer-solvent interactions for many years.^{35,36} If this single parameter is to be complete in this function, it must include both the atomic/dispersion interactions as well as the specific interactions reflected by the δ_p and δ_H parameters. Attempts to calculate chi using HSP are reported in Chapter 2. More understanding is required before chi can be calculated with reasonable accuracy, but intensified efforts seem warranted. Zellers and coworkers have recently made an attempt to use this theory in conjunction with HSP studies.³⁷⁻⁴⁰ A major problem is the reliability of the chi parameter (and also HSP) values in the literature (see Chapter 2 for more details). Chapter 4 is a lengthy discussion of the use of HSP in thermodynamic models for polymer solutions.

The author's current view as expressed in Chapter 2 is that the Flory approach is a special case of the more general Prigogine corresponding states theory. This is in agreement with the view of Patterson discussed in Chapter 2. Furthermore, the very general applications of the HSP approach demonstrated in this book and elsewhere, and the apparent agreement in the treatment of specific interactions by both the HSP and Prigogine treatments, leads to confidence in the HSP approach. The geometric mean must be used in the Prigogine theory to arrive at this similarity of treatment. The statistical thermodynamic approach of Panayiotou and coworkers thoroughly discussed in Chapter 3 provides convincing evidence that the division of the cohesion energy density into three parts (at least) is the correct procedure to understand affinities among many types of materials. Finally, Chapter 4 deals with theories of vapor-liquid equilibrium in particular and how these relate to HSP.

SURFACE PHENOMENA

Surfaces can clearly be characterized by HSP as demonstrated in Chapter 6 and Chapter 7. The work of Beerbover contained in Reference 17 and Reference 23 has also shown applications of HSP to such varied phenomena as the work of adhesion of liquids on mercury, friction of polyethylene untreated and treated with sulfuric acid, the Reh binder effect — the crushing of aluminum oxide (Al_2O_3) under various liquids, and the Joffe effect — effect of liquid immersion on the fracture strength of soda-lime glass. Here again, the successful use of HSP to such applications might not have been anticipated had it been considered as a parameter for use in bulk systems only. A formalized unifying theory linking HSP to both bulk and surface phenomena is still lacking. Presently, the best that can be said is that the generality “like dissolves like” can be quantified in many cases. The extension of this, “like seeks like,” also seems to have been demonstrated. It is the surfaces of molecules which interact with each other (also in bulk and solution phenomena), so it is not surprising that cohesion parameters can be applied with success to both solubility and surface phenomena. Much more research needs to be done with these relations. A good starting point is the *Handbook of Surface and Colloid Chemistry*, edited by K.S. Birdi.⁴¹ If we consider chromatographic techniques as depending primarily on surface phenomena, mention should also be made of the extension of the three-parameter HSP approach to a five-parameter approach by Karger and co-workers.⁴² HSP characterizations of surfactants are also badly needed.

CONCLUSION

HSP have been shown useful in solvent selection; predicting polymer–polymer miscibility; characterizing the surfaces of polymers, fillers, and fibers; correlating permeation phenomena; characterizing organic salts and inorganic salts; gas solubility; etc. No other parameter can be assigned to such a range of materials spanning from gases and liquids, over surfaces, to inorganic salts. These results and the close relation with the Prigogine corresponding states theory of polymer solutions, and more recently to the work of Panayiotou summarized in Chapter 3, indicate that a still more general theory exists. This theory should quantify the adage “like seeks like,” i.e., include surface phenomena as well as bulk phenomena.

Specific areas needing more theoretical work related to HSP in the near future include better understanding of usage for predictions of behavior for water, gases, organic salts, inorganic salts, and organometallic compounds. Water remains special because of its low molecular volume and high δ_H . Most materials having HSP in the range of the customary test liquids can be studied using HSP with reasonable success. This is not fully the case for gases, many of which have much lower HSP than the well-studied liquids, and salts, many of which apparently have HSP very much higher than any of the liquids. Extensions of practical applications related to chemical resistance and the uptake of potentially dangerous materials in polymers are also required. Finally there is a great deal to be done in the areas of controlled drug release, improved understanding of some biological processes, and last but not least, the systematic use of HSP in nanotechnology to control the assembly and orientation of molecules or segments of molecules.

REFERENCES

1. Hansen, C.M., The universality of the solubility parameter, *Ind. Eng. Chem. Prod. Res. Dev.*, 8(1), 2–11, 1969.
2. Barton, A.F.M., *Handbook of Solubility Parameters and Other Cohesion Parameters*, CRC Press, Boca Raton, FL, 1983; 2nd ed., 1991.
3. Barton, A.F.M., *Handbook of Polymer-Liquid Interaction Parameters and Solubility Parameters*, CRC Press, Boca Raton, FL, 1990.

4. Hansen, C.M., Solubility parameters, in *Paint Testing Manual*, Manual 17, Koleske, J.V., Ed., American Society for Testing and Materials, Philadelphia, PA, 1995, pp. 383–404.
5. Hansen, C.M., Cohesion energy parameters applied to surface phenomena, *CRC Handbook on Surface Science*, Birdi, K.S., Ed., CRC Press, Boca Raton, FL, 1997, chap. 10.
6. Anonymous, Brochure: Co-Act — A Dynamic Program for Solvent Selection, Exxon Chemical International Inc., 1989.
7. Dante, M.F., Bittar, A.D., and Caillault, J.J., Program calculates solvent properties and solubility parameters, *Mod. Paint Coat.*, 79(9), 46–51, 1989.
8. Hansen, C.M. and Beerbower, A., Solubility parameters, in *Kirk-Othmer Encyclopedia of Chemical Technology*, Suppl. Vol., 2nd ed., Standen, A., Ed., Interscience, New York, 1971, pp. 889–910.
9. Hansen, C.M. and Andersen, B.H., The affinities of organic solvents in biological systems, *J. Am. Ind. Hyg. Assoc.*, 49(6), 301–308, 1988.
10. Wessling, R.A., The solubility of poly(vinylidene chloride), *J. Appl. Polym. Sci.*, 14, 1531–1545, 1970.
11. Grulke, E.A., Solubility parameter values, in *Polymer Handbook*, 3rd ed., Brandrup, J. and Immergut, E.H., Eds., John Wiley-Interscience, New York, 1989, pp. VII/519–559.
12. Burrell, H., Solubility of polymers, in *Kirk Othmer Encyclopedia of Polymer Science and Technology*, Vol. 12, 2nd ed., John Wiley & Sons, New York, 1970, pp. 618–626.
13. Hoy, K.L., New values of the solubility parameters from vapor pressure data, *J. Paint Technol.*, 42(541), 76–118, 1970.
14. Hoy, K.L., *Tables of Solubility Parameters*, Union Carbide Corp., Research and Development Dept., South Charleston, WV, 1985; 1st ed. 1969.
15. van Krevelen, D.W. and Hoftyzer, P.J., *Properties of Polymers: Their Estimation and Correlation with Chemical Structure*, 2nd ed., Elsevier, Amsterdam, 1976.
16. Riedel, G., The solubility of colorants in organic solvents (Löslichkeit von Farbstoffen in organischen Lösungsmitteln, in German), *Farbe und Lack*, 82(4), 281–287, 1976.
17. Beerbower, A., Environmental capability of liquids, in *Interdisciplinary Approach to Liquid Lubricant Technology*, NASA Publication SP-318, 1973, 365–431.
18. Fedors, R.F., A method for estimating both the solubility parameters and molar volumes of liquids, *Polym. Eng. Sci.*, 14(2), 147–154, 472, 1974.
19. Koenhen, D.N. and Smolders, C.A., The determination of solubility parameters of solvents and polymers by means of correlation with other physical quantities, *J. Appl. Polym. Sci.*, 19, 1163–1179, 1975.
20. Utracki, L., Personal communication, 1995.
21. Kähler, T. and Knudsen, S.L., Student Report, Technical University of Denmark, 1967.
22. Hansen, C.M., Characterization of surfaces by spreading liquids, *J. Paint Technol.*, 42(550), 660–664, 1970.
23. Beerbower, A., Boundary Lubrication — Scientific and Technical Applications Forecast, AD747336, Office of the Chief of Research and Development, Department of the Army, Washington, D.C., 1972.
24. Hansen, C.M. and Björkman, A., The ultrastructure of wood from a solubility parameter point of view, *Holzforschung*, 52(4), 335–344, 1998.
25. Hansen, C.M. and Knudtson, J., Spreading and wetting of an electrodeposition coating (Ausbreitung und Benetzung bei einem Elektrotauchlack, in German), *Farbe und Lack*, 2, 115–118, 1973.
26. *Progress in Organic Coatings*, Special issue devoted to Self-Stratifying Coatings, 28(3), July 1996.
27. Thulstrup, D., Characterization of surfaces by static contact angle measurements (lecture in Danish: Karakterisering af Overflader ved Statisk Kontaktvinkelmåling), Surface Properties and Modification of Plastics (Conference Notes), Ingeniørforening i Danmark, November 13, 1997, Copenhagen. See also DSM Materialenyt 2:97, Dansk Selskab for Materialebehandling og -forskning, Ingeniørforening i Danmark, Copenhagen, 1997, 63–69.
28. König, U. and Schuch, H., Molecular composition and permeability of plastics, (Konstitution und Permeabilität von Kunststoffen, in German), *Kunststoffe*, 67(1), 27–31, 1977.
29. Hansen, C.M., 25 years with solubility parameters (25 År med Opløselighedsparametrene, in Danish), *Dan. Kemi*, 73(8), 18–22, 1992.
30. Hansen, C.M., Some aspects of acid/base interactions (Einige Aspekte der Säure/Base-Wechselwirkung, in German), *Farbe und Lack*, 83(7), 595–598, 1977.

31. Beerbower, A., Personal communication, Predicting Wetting in the Oil-Brine-Rock System, Unedited manuscript for presentation at the National AIChE Meeting, Tulsa, OK, March 10–13, 1974.
32. Panzer, J., Components of solid surface free energy from wetting measurements, *J. Colloid Interface Sci.*, 44(1), 142–161, July 1973.
33. Hildebrand, J. and Scott, R.L., *The Solubility of Nonelectrolytes*, 3rd ed., Reinhold, New York, 1950.
34. Hansen, C.M., Solubility parameters for aromas and scents (Aromastof fers Opløselighedsparametre, in Danish), *Plus Process*, 11(9), 16–17, 1997.
35. Flory, P.J., *Principles of Polymer Chemistry*, Cornell University Press, New York, 1953.
36. Eichinger, B.E. and Flory, P.J., Thermodynamics of polymer solutions, *Trans. Faraday Soc.*, 64(1), 2035–2052, 1968; 64(2), 2053–2060, 1968; 64(3), 2061–2065, 1968; 64(4), 2066–2072, 1968.
37. Zellers, E.T., Three-dimensional solubility parameters and chemical protective clothing permeation. I. Modelling the solubility of organic solvents in Viton® gloves, *J. Appl. Polym. Sci.*, 50, 513–530, 1993.
38. Zellers, E.T., Three-dimensional solubility parameters and chemical protective clothing. II. Modelling diffusion coefficients, breakthrough times, and steady-state permeation rates of organic solvents in Viton® gloves, *J. Appl. Polym. Sci.*, 50, 531–540, 1993.
39. Zellers, E.T., Anna, D.H., Sulewski, R., and Wei, X., Critical analysis of the graphical determination of Hansen's solubility parameters for lightly crosslinked polymers, *J. Appl. Polym. Sci.*, 62, 2069–2080, 1996.
40. Zellers, E.T., Anna, D.H., Sulewski, R., and Wei, X., Improved methods for the determination of Hansen's solubility parameters and the estimation of solvent uptake for lightly crosslinked polymers, *J. Appl. Polym. Sci.*, 62, 2081–2096, 1996.
41. Birdi, K.S., *Handbook of Surface and Colloid Chemistry*, CRC Press, Boca Raton, FL, 1997.
42. Karger, B.L., Snyder, L.R., and Eon, C., Expanded solubility parameter treatment for classification and use of chromatographic solvents and adsorbents, *Anal. Chem.*, 50(14), 2126–2136, 1978.

Appendix A: Table A.1

Charles M. Hansen (with the help of Hanno Priebe)

COMMENTS TO TABLE A.1

Table A-1 is greatly expanded compared with the first edition of this handbook. There are more chemicals, and the information for each of them is much more comprehensive. There are two lists for chemical names. The first list follows the general nomenclature that is used in the first edition. This list follows usage that is commonly found in industrial contexts. The second list of names and also the structures have been compiled by Dr. Hanno Priebe, GE Healthcare. Most of these chemical names originate from AutoNom 2000 (Copyright © 1988-2006, Beilstein-Institut zur Förderung der Chemischen Wissenschaften licensed to Beilstein GmbH and MDL Information Systems GmbH. All rights reserved). These are obviously more correct in many cases.

In AutoNom the Beilstein ring system naming convention was chosen. In cases where AutoNom failed, the chemical names originate from ACD/Name freeware, version 8.0 (Advanced Chemistry Development, Inc., Toronto ON, Canada, www.acdlabs.com, 2005). In cases where the ACD/Name freeware failed, the name field was left empty.

Most of the HSP entries in Table A-1 are calculated values. Those entries having some experimental confirmation are the original 90 from Reference 1 to Reference 3, together with some from industrial experience at PPG Industries. These are indicated in dark type. The usual procedure for the calculations was to use the procedure given in Chapter 1. The figures were used to find δ_D from boiling point data. δ_P was estimated either with a dipole moment, where this could be found, or by group contributions. There are special problems where the dipole moment is zero or close to zero for symmetrical molecules. In some cases two sets of data are given. δ_H was almost always found by group contributions or by similarity to related compounds. It must be said that none of the parameters are precise. Even use of experimental data does not necessarily confirm the accuracy of the values. This is particularly true where a given HSP parameter is relatively high. Whether or not given solutes dissolve is not sufficient in most cases to narrow the potential error(s) in such cases. Where latent heats are known, Equations 1.6–1.8 do limit potential errors, and experience has shown that for all practical purposes, the values assigned are most useful. Notwithstanding, several parameters have been changed since the first edition appeared. These are indicated with a “*” in the first column of names. However, no changes in the older, published figures have been made for this reason, as this was judged unnecessary.

Many of the solvents for which experimental data were used to help assign the HSP were commercial in origin. This means that purity of the samples was that specified in the given case, but was usually very high, and that a mixture of isomers was present if isomers were possible. This is particularly true of the glycol ethers based on propylene oxide. For these chemicals and others where isomers are possible, the chemical names and structures indicated in Table A-1 are indicative of the type of compound present. It would be impractical to present all the possible isomers.

The HSP for the organic salts formed from equimolar mixtures of organic acids and amines are those reported in Reference 4. The assignments were made on the basis of which polymers or other solutes in a test series dissolved in the given salt. This was compared with how these same solutes behaved in a series of liquids with known HSP. When the same pattern of dissolving or not was found for both a salt and a known liquid, it was assumed that their HSP were the same (or at least very close).

The units used are MPa^{1/2} for the HSP and cc/mole for the molar volume.

REFERENCES

1. Hansen, C.M., The Three Dimensional Solubility Parameter and Solvent Diffusion Coefficient, Doctoral Dissertation, Danish Technical Press, Copenhagen, 1967.
2. Hansen, C.M., The Three Dimensional Solubility Parameter — Key to Paint Component Affinities I — Solvents, Plasticizers, Polymers, and Resins, *J. Paint Techn.*, 39(505), 104-117, 1967.
3. Hansen, C.M., and Skaarup, K., The Three Dimensional Solubility Parameter — Key to Paint Component Affinities III. — Independent Calculation of the Parameter Components, *J. Paint Techn.* 39(511), 511-514, 1967.
4. Hansen, C.M., Some Aspects of Acid/Base Interactions (in German) (Einige Aspekte der Säure/Base-Wechselwirkung), *Farbe und Lack*, 83(7), 595-598, 1977.

TABLE A.1

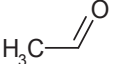
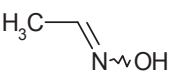
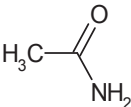
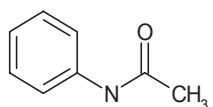
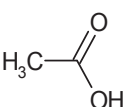
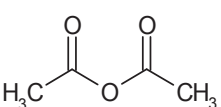
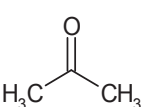
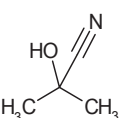
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1	Acetaldehyde*	Acetaldehyde	14.7	12.5	7.9	56.6
						
2	Acetaldoxime	Acetaldehyde oxime	16.3	4.0	20.2	61.2
						
3	Acetamide	Acetamide	17.3	18.7	22.4	60.8
						
4	Acetanilide	N-Phenyl-acetamide	20.6	13.3	12.4	110.9
						
5	Acetic Acid	Acetic acid	14.5	8.0	13.5	57.1
						
6	Acetic Anhydride	Acetic anhydride	16.0	11.7	10.2	94.5
						
7	Acetone	Propan-2-one	15.5	10.4	7.0	74.0
						
8	Acetonecyanhydrin	2-Hydroxy-2-methyl-propionitrile	16.6	12.2	15.5	94.0
						

TABLE A.1 (CONTINUED)

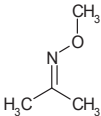
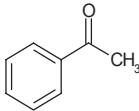
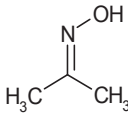
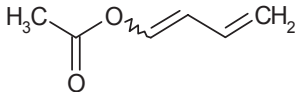
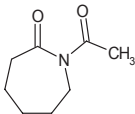
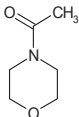
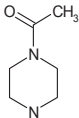
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
9	Acetonemethyloxime 	Propan-2-one O-methyl- oxime	14.7	4.6	4.6	96.7
10	Acetonitrile $\text{H}_3\text{C}-\equiv\text{N}$	Acetonitrile	15.3	18.0	6.1	52.6
11	Acetophenone 	1-Phenyl-ethanone	19.6	8.6	3.7	117.4
12	Acetoxime 	Propan-2-one oxime	16.3	3.7	10.9	80.2
765	1-Acetoxy-1,3-butadiene 	Acetic acid buta-1,3- dienyl ester	16.1	4.4	8.3	118.4
13	N-Acetyl Caprolactam 	1-Acetyl-azepan-2-one	18.9	8.7	4.8	155.0
14	N-Acetyl Morpholine 	1-Morpholin-4-yl- ethanone	18.3	5.3	7.8	115.6
15	N-Acetyl Piperidine 	1-Piperazin-1-yl- ethanone	18.5	10.0	6.5	125.8

TABLE A.1 (CONTINUED)

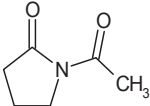
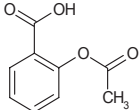
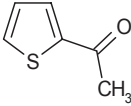
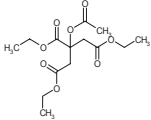
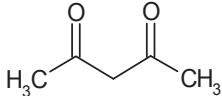
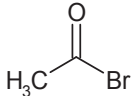
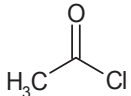
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
16	N-Acetyl Pyrrolidone 	1-Acetyl-pyrrolidin-2-one	17.8	13.1	8.3	127.0
1116	Acetyl Salicylic Acid 	2-Acetoxy-benzoic acid	19.0	6.6	9.3	133.5
785	2-Acetyl Thiophene 	1-Thiophen-2-yl-ethanone	19.1	12.2	9.3	108.0
1219	Acetyl Triethyl Citrate 	3-Acetoxy-3-ethoxycarbonyl-pentanedioic acid diethyl ester	16.6	3.5	8.6	279.9
17	Acetylacetone 	Pentane-2,4-dione	16.1	11.2	6.2	103.0
18	Acetyl bromide 	Acetyl bromide	16.7	10.6	5.2	74.0
19	Acetyl chloride 	Acetyl chloride	16.2	11.2	5.8	71.4
20	Acetylene (Ethyne) $\text{HC}\equiv\text{CH}$	Ethyne	14.4	4.2	11.9	42.1

TABLE A.1 (CONTINUED)

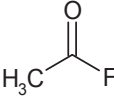
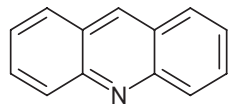
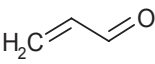
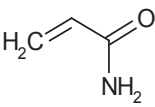
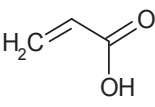
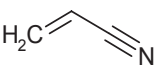
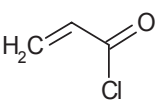
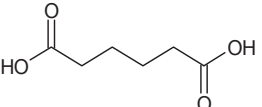
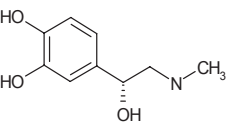
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
21	Acetylfluorid	Acetyl fluorid	14.7	14.0	5.7	62.0
						
860	Acridine	Acridine	21.7	5.9	2.0	163.0
						
22	Acrolein	Propenal	15.0	7.2	7.8	66.7
						
23	Acrylamide	Acrylamide	15.8	12.1	12.8	63.4
						
24	Acrylic Acid	Acrylic acid	17.7	6.4	14.9	68.5
						
25	Acrylonitrile	Acrylonitrile	16.0	12.8	6.8	67.1
						
26	Acrylylchloride	Acryloyl chloride	16.2	11.6	5.4	81.3
						
1152	Adipic Acid (1,6-Hexanedioic)	Hexanedioic acid	17.1	9.0	14.6	107.5
						
1195	Adrenaline	4-((R)-1-Hydroxy-2-methylamino-ethyl)-benzene-1,2-diol	20.5	8.7	19.9	154.5
						

TABLE A.1 (CONTINUED)

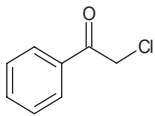
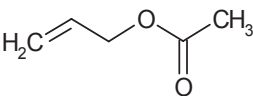
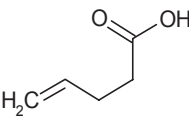
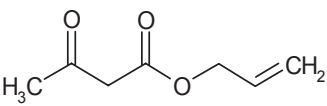
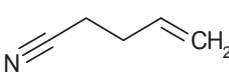
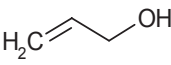
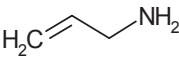
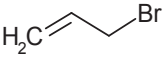
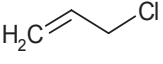
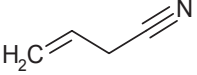
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1088	alfa-Chloro Acetophenone 	2-Chloro-1-phenyl- ethanone	20.0	9.6	5.7	116.8
27	Allyl Acetate 	Acetic acid allyl ester	15.7	4.5	8.0	108.5
28	Allyl Acetic Acid 	Pent-4-enoic acid	16.7	4.7	11.3	102.1
769	Allyl Acetoacetate 	3-Oxo-butyric acid allyl ester	15.9	6.9	8.6	137.8
29	Allyl Acetonitrile (4-Pentenitrile) 	Pent-4-enitrile	16.3	11.2	5.0	98.5
30	Allyl Alcohol 	Prop-2-en-1-ol	16.2	10.8	16.8	68.4
31	Allyl Amine 	Allylamine	15.5	5.7	10.6	74.9
32	Allyl Bromide (3-Bromoprene) 	3-Bromo-propene	16.5	7.3	4.9	86.5
33	Allyl Chloride 	3-Chloro-propene	17.0	6.2	2.3	82.3
34	Allyl Cyanide 	But-3-enitrile	16.0	14.3	5.6	80.5

TABLE A.1 (CONTINUED)

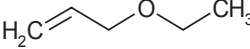
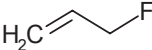
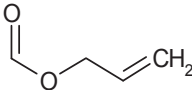
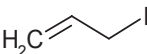
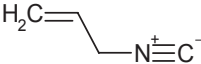
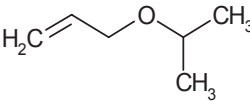
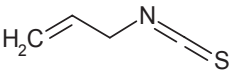
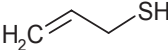
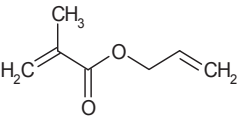
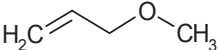
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
35	Allyl Ethylether 	3-Ethoxy-propene	15.0	4.8	5.1	112.6
36	Allyl Fluoride 	3-Fluoro-propene	14.9	6.4	1.0	71.5
37	Allyl Formate 	Formic acid allyl ester	15.7	5.4	8.8	91.0
38	Allyl Iodide 	3-Iodo-propene	17.4	6.4	3.3	90.8
39	Allyl Isocyanide 	3-Isocyano-propene	16.1	13.0	5.4	84.5
967	Allyl Isopropyl Ether 	3-Isopropoxy-propene	15.0	4.1	7.1	129.0
40	Allyl Isothiocyanate 	3-Isothiocyanato-propene	17.0	11.3	8.5	98.0
42	Allyl Mercaptan 	Prop-2-ene-1-thiol	16.4	6.2	7.9	80.2
1018	Allyl Methacrylate 	2-Methyl-acrylic acid allyl ester	15.2	4.1	7.5	134.8
43	Allyl Methylether 	3-Methoxy-propene	15.0	4.3	5.9	93.7

TABLE A.1 (CONTINUED)

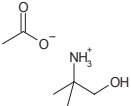
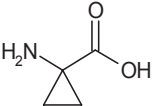
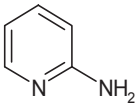
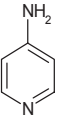
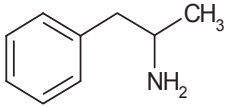
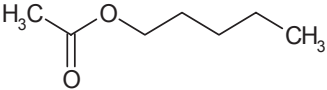
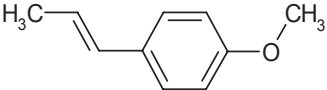
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1181	2-Amino-2-Methyl-1-Propanol/Acetic Acid	Acetate 2-hydroxy-1,1-dimethyl-ethyl-ammonium	17.2	22.5	23.3	
						
1206	1-Aminocyclopropane Carboxylic Acid	1-Amino-cyclopropane-carboxylic acid	17.0	6.3	13.6	93.7
						
1090	2-Amino Pyridine	Pyridin-2-ylamine	20.4	8.1	12.2	94.1
						
1205	4-Amino Pyridine	Pyridin-4-ylamine	20.4	16.1	12.9	87.1
						
44	Ammonia NH ₃	Ammonia	13.7	15.7	17.8	20.8
1179	Amphetamine	1-Methyl-2-phenyl-ethylamine	17.5	4.3	6.3	148.1
						
45	Amyl Acetate	Acetic acid pentyl ester	15.8	3.3	6.1	148.0
						
699	Anethole (Trans)	1-Methoxy-4-((E)-propenyl)-benzene	19.0	4.3	8.7	150.0
						

TABLE A.1 (CONTINUED)

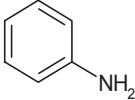
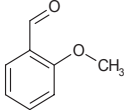
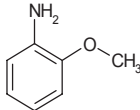
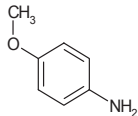
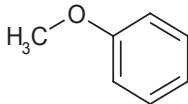
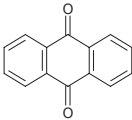
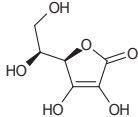
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
46	Aniline 	Phenylamine	19.4	5.1	10.2	91.5
1113	Anisaldehyde (2-Methoxy Benzaldehyde) 	2-Methoxy-benzaldehyde	19.4	11.9	8.3	120.8
949	Anisidine (2-Methoxyaniline) 	2-Methoxy-phenylamine	19.5	5.4	11.4	112.2
47	p-Anisidine (Methoxy Aniline) 	4-Methoxy-phenylamine	19.9	6.5	11.3	113.3
48	Anisole 	Methoxy-benzene	17.8	4.1	6.7	119.1
1099	9,10-Anthraquinone 	Anthraquinone	20.3	7.6	4.8	145.6
1163	Ascorbic Acid 	(R)-5-((S)-1,2-Dihydroxy-ethyl)-3,4-dihydroxy-5H-furan-2-one	18.0	12.6	27.6	106.7

TABLE A.1 (CONTINUED)

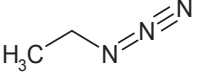
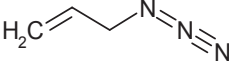
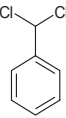
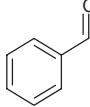
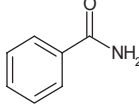
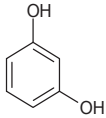
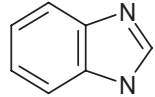
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
49	Azidoethane 	Azido-ethane	15.9	8.9	12.9	79.0
50	3-Azidopropene 	3-Azido-propene	16.8	7.7	13.4	83.0
807	Benzal Chloride 	Dichloromethyl-benzene	19.9	6.6	2.4	134.2
51	Benzaldehyde 	Benzaldehyde	19.4	7.4	5.3	101.5
1055	Benzamide 	Benzamide	21.2	14.7	11.2	90.3
52	Benzene 	Benzene	18.4	0	2.0	89.4
53	1,3-Benzenediol 	Benzene-1,3-diol	18.0	8.4	21.0	87.5
894	Benzimidazole 	1H-Benzoimidazole	20.6	14.9	11.0	102.7

TABLE A.1 (CONTINUED)

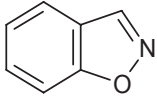
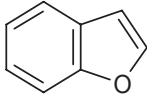
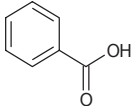
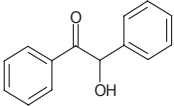
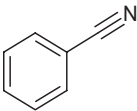
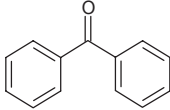
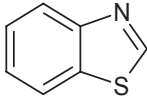
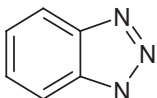
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
893	Benzisoxazole	Benzo[d]isoxazole	20.6	11.5	8.8	100.7
						
892	2,3-Benzofuran (Cumaron)	Benzofuran	18.7	5.1	5.7	110.2
						
54	Benzoic Acid	Benzoic acid	18.2	6.9	9.8	113.1
						
891	Benzoin	2-Hydroxy-1,2-diphenyl-ethanone	19.9	9.7	10.7	187.7
						
55	Benzonitrile	Benzonitrile	17.4	9.0	3.3	102.6
						
56	Benzophenone	Diphenyl-methanone	19.6	8.6	5.7	164.2
						
962	Benzothiazole	Benzothiazole	20.6	5.2	8.4	108.5
						
1053	1,2,3-Benzotriazole	1H-Benzotriazole	18.7	15.6	12.4	96.2
						

TABLE A.1 (CONTINUED)

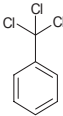
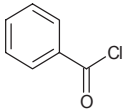
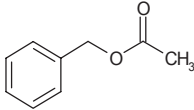
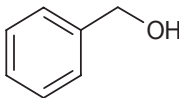
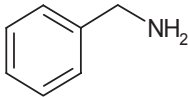
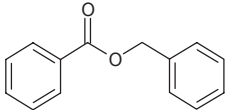
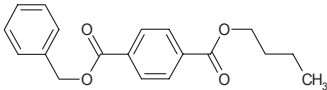
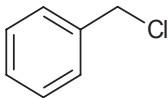
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
866	Benzotrichloride 	Trichloromethyl- benzene	20.2	6.6	3.2	142.1
57	Benzoyl Chloride 	Benzoyl chloride	20.7	8.2	4.5	116.0
903	Benzyl Acetate 	Acetic acid benzyl ester	18.3	5.7	6.0	142.8
58	Benzyl Alcohol 	Phenyl-methanol	18.4	6.3	13.7	103.6
971	Benzyl Amine 	Benzylamine	19.2	4.6	11.7	109.2
1149	Benzyl Benzoate 	Benzoic acid benzyl ester	20.0	5.1	5.2	191.2
59	Benzyl Butyl Phthalate 	Terephthalic acid 1- benzyl ester 4-butyl ester	19.0	11.2	3.1	306.0
60	Benzyl Chloride 	Chloromethyl-benzene	18.8	7.1	2.6	115.0

TABLE A.1 (CONTINUED)

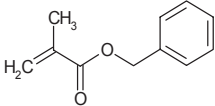
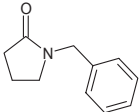
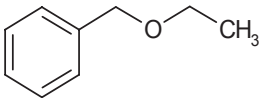
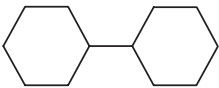
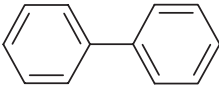
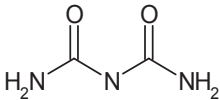
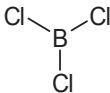
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
61	Benzyl Methacrylate 	2-Methyl-acrylic acid benzyl ester	16.8	4.1	4.1	167.8
62	N-Benzyl Pyrrolidone 	1-Benzyl-pyrrolidin-2-one	18.2	6.1	5.6	160.0
887	Benzylethyl Ether 	Ethoxymethyl-benzene	18.4	3.8	3.8	144.2
867	Bicyclohexyl 	Bicyclohexyl	18.6	0	0	188.5
868	Biphenyl 	Biphenyl	19.7	1.0	2.0	155.1
1066	Biuret 	Dicarbonimidic diamide	20.0	14.6	18.8	70.3
64	Borine Carbonyl $\text{H}_3\text{B}^- \equiv \text{O}^+$		17.0	10.2	6.9	41.8
1172	Boron Trichloride 		16.6	2.5	0	85.3
1170	Bromine (P From Group Cont.)	Bromine	18.2	14.9	0	51.5
1213	Bromine (P From Dipole Moment)	Bromine	18.2	2.1	0	51.5

TABLE A.1 (CONTINUED)

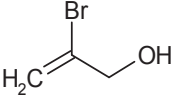
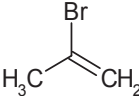
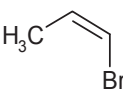
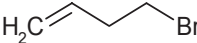
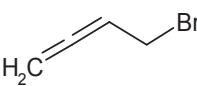
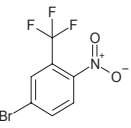
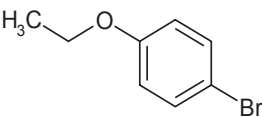
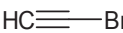
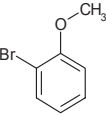
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
65	2-Bromo Allyl Alcohol	2-Bromo-prop-2-en-1-ol	17.1	9.9	16.2	84.5
						
66	2-Bromo Propene	2-Bromo-propene	16.1	6.0	4.9	88.8
						
67	1-Bromo Propene (CIS)	(Z)-1-Bromo-propene	16.3	6.4	5.0	84.7
						
68	4-Bromo-1-Butene	4-Bromo-but-1-ene	16.5	6.0	4.5	102.0
						
69	4-Bromo-1,2-Butadiene	4-Bromo-buta-1,2-diene	17.0	6.5	4.7	93.3
						
854	5-Bromo-2-Nitrobenzotrifluorid	4-Bromo-1-nitro-2-trifluoromet yl-benzene	20.0	6.0	4.9	150.1
						
815	1-Bromo-4-Ethoxy Benzene	1-Bromo-4-ethoxy-benzene	19.5	7.7	5.3	140.0
						
70	Bromoacetylene	Bromo-ethyne	15.7	9.9	5.6	67.7
						
800	o-Bromoanisole	1-Bromo-2-methoxy-benzene	19.8	8.4	6.7	124.5
						

TABLE A.1 (CONTINUED)

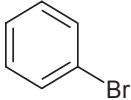
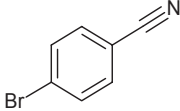
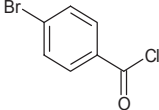
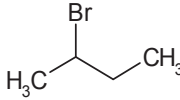
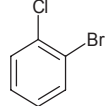
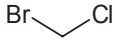
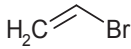
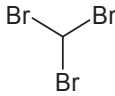
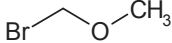
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
71	Bromobenzene	Bromo-benzene	20.5	5.5	4.1	105.3
						
806	p-Bromobenzonitrile	4-Bromo-benzonitrile	20.4	9.3	5.8	113.8
						
805	p-Bromobenzoyl Chloride	4-Bromo-benzoyl chloride	20.2	6.5	5.5	137.2
						
729	2-Bromobutane	2-Bromo-butane	16.3	7.7	4.4	109.5
						
802	o-Bromochlorobenzene	1-Bromo-2-chloro- benzene	20.3	7.7	4.6	116.9
						
72	Bromochloromethane	Bromo-chloro-methane	17.3	5.7	3.5	65.0
						
73	Bromoethylene	Bromo-ethene	16.1	6.3	2.3	71.6
						
74	Bromoform	Tribromo-methane	21.4	4.1	6.1	87.5
						
748	Bromomethyl Methyl Ether	Bromo-methoxy- methane	16.9	8.5	7.0	81.6
						

TABLE A.1 (CONTINUED)

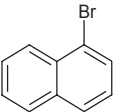
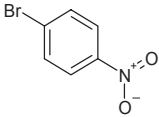
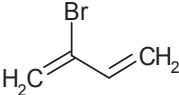
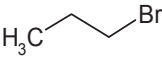
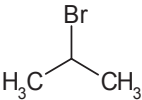
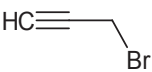
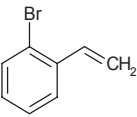
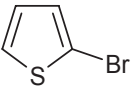
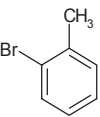
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
75	1-Bromonaphthalene 	1-Bromo-naphthalene	20.3	3.1	4.1	140.0
803	p-Bromonitrobenzene 	1-Bromo-4-nitro- benzene	20.9	9.9	5.9	103.7
76	Bromoprene 	2-Bromo-buta-1,3-diene	16.9	6.4	4.7	95.2
1124	1-Bromopropane 	1-Bromo-propane	16.4	7.9	4.8	90.9
1125	2-Bromopropane 	2-Bromo-propane	16.1	8.3	4.7	93.6
77	3-Bromopropyne 	3-Bromo-propyne	18.1	6.5	5.3	75.3
710	o-Bromostyrene 	1-Bromo-2-vinyl- benzene	19.5	5.2	5.3	130.0
78	2-Bromothiophene 	2-Bromo-thiophene	20.1	5.2	4.6	96.8
713	o-Bromotoluene 	1-Bromo-2-methyl- benzene	19.3	5.0	4.2	119.0

TABLE A.1 (CONTINUED)

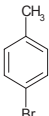
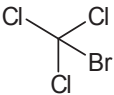
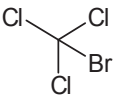
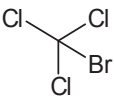
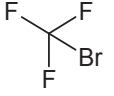
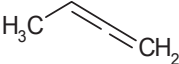
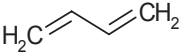
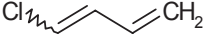
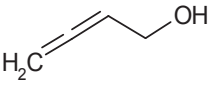
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
714	p-Bromotoluene 	1-Bromo-4-methyl- benzene	19.3	6.8	4.1	122.9
743	Bromotrichloro Methane (P from Dipole Moment) 	Bromo-trichloro- methane	18.3	0.8	0	99.2
744	Bromotrichloro Methane (P and H from Group cont.) 	Bromo-trichloro- methane	18.3	8.1	6.0	99.2
745	Bromotrichloro Methane (P from Group cont.) 	Bromo-trichloro- methane	18.3	8.1	0	99.2
79	Bromotrifluoromethane (Freon 1381) 	Bromo-trifluoro methane	9.6	2.4	0	97.0
80	1,2-Butadiene 	Buta-1,2-diene	14.7	1.7	6.2	82.3
81	1,3-Butadiene 	Buta-1,3-diene	14.8	2.8	5.6	83.2
82	1,3-Butadiene-1-Chloro 	1-Chloro-buta-1,3-diene	15.6	8.5	2.0	92.2
83	2,3-Butadiene-1-ol 	Buta-2,3-dien-1-ol	16.2	6.6	16.8	76.5

TABLE A.1 (CONTINUED)

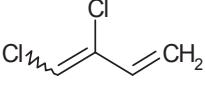
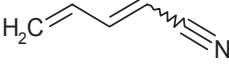
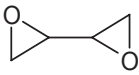
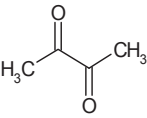
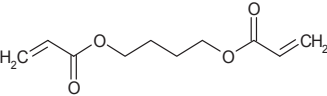
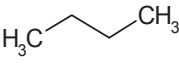
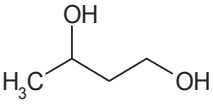
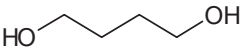
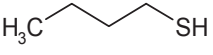
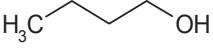
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
84	1,3-Butadiene-1,2-Di-Chloro	1,2-Dichloro-buta-1,3-diene	17.0	10.7	2.8	102.5
						
85	Butadiene-4-Cyano	Penta-2,4-dienenitrile	16.2	11.7	5.2	93.7
						
86	Butadienedioxide	[2,2']Bioxiranyl	18.3	14.4	6.2	77.8
						
87	Butadione	Butane-2,3-dione	15.7	5.1	6.8	87.8
						
88	1,4-Butandiol Diacrylate	Acrylic acid 4-acryloyloxy-butyl ester	16.8	9.1	4.2	194.4
						
89	Butane	Butane	14.1	0	0	101.4
						
90	1,3-Butanediol	Butane-1,3-diol	16.6	10.0	21.5	89.9
						
730	1,4-Butanediol	Butane-1,4-diol	16.6	15.3	21.7	88.9
						
91	1-Butanethiol	Butane-1-thiol	16.3	5.3	4.5	107.8
						
92	1-Butanol	Butan-1-ol	16.0	5.7	15.8	91.5
						

TABLE A.1 (CONTINUED)

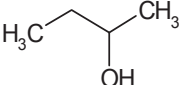
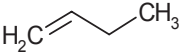
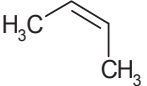
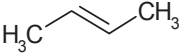
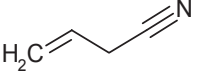
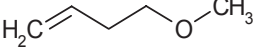
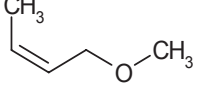
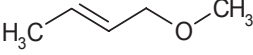
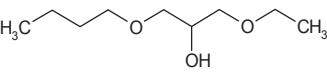
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
92	2-Butanol 	Butan-2-ol	15.8	5.7	14.5	92.0
94	1-Butene 	But-1-ene	13.2	1.3	3.9	94.3
95	2-Butene (cis) 	(Z)-But-2-ene	14.7	1.3	4.1	90.3
96	2-Butene (trans) 	(E)-But-2-ene	14.6	0	2.9	92.9
97	3-Butenenitrile 	But-3-enenitrile	16.2	14.3	5.6	80.5
98	1-Butenyl Methyl Ether 	4-Methoxy-but-1-ene	15.1	5.3	5.2	111.9
99	2-Butenyl Methyl Ether (cis) 	(Z)-1-Methoxy-but-2-ene	15.2	3.4	5.0	118.8
100	2-Butenyl Methyl Ether (trans) 	(E)-1-Methoxy-but-2-ene	15.3	4.4	4.3	110.4
101	Butoxy Ethoxy Propanol Commercial 	1-Butoxy-3-ethoxy-propan-2-ol	15.5	6.5	10.2	185.9

TABLE A.1 (CONTINUED)

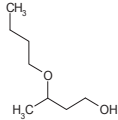
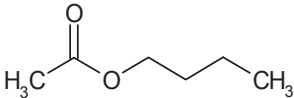
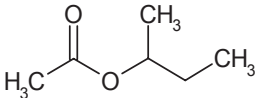
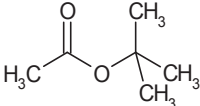
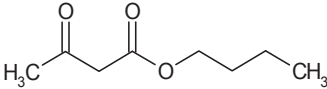
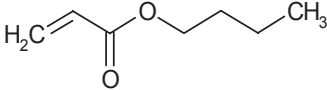
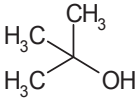
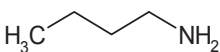
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
728	3-Butoxybutanol	3-Butoxy-butan-1-ol	15.9	5.5	10.6	166.3
						
102	n-Butyl Acetate	Acetic acid butyl ester	15.8	3.7	6.3	132.5
						
103	Sec-Butyl Acetate	Acetic acid sec-butyl ester	15.0	3.7	7.6	133.6
						
963	tert-Butyl Acetate	Acetic acid tert-butyl ester	15.4	6.2	6.2	134.1
						
768	n-Butyl Aceto Acetate	3-Oxo-butyric acid butyl ester	16.6	5.8	7.3	164.3
						
104	n-Butyl Acrylate	Acrylic acid butyl ester	15.6	6.2	4.9	143.8
						
611	t-Butyl Alcohol	2-Methyl-propan-2-ol	15.2	5.1	14.7	95.8
						
105	n-Butyl Amine	Butylamine	16.2	4.5	8.0	99.0
						

TABLE A.1 (CONTINUED)

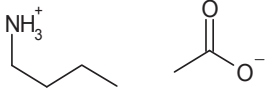
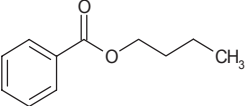
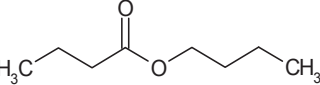
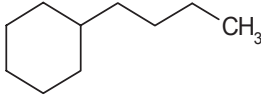
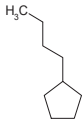
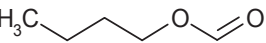
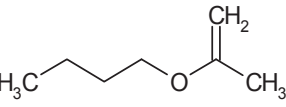
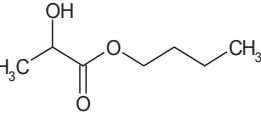
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1182	N-Butyl Amine/Acetic Acid	Acetate butyl-ammonium;	16.0	20.3	18.4	
						
718	Butyl Benzoate*	Benzoic acid butyl ester	18.3	5.6	5.5	178.0
						
774	n-Butyl Butyrate	Butyric acid butyl ester	15.6	2.9	5.6	166.7
						
722	n-Butyl Cyclohexane	Butyl-cyclohexane	16.2	0	0.6	176.7
						
723	n-Butyl Cyclopentane	Butyl-cyclopentane	16.4	0	1.0	162.0
						
927	Butyl Formate	Formic acid butyl ester	15.7	6.5	9.2	114.8
						
106	Butyl Isopropenyl Ether	1-Isopropenyloxy-butane	14.8	5.3	5.0	145.7
						
107	Butyl Lactate	2-Hydroxy-propionic acid butyl ester	15.8	6.5	10.2	149.0
						

TABLE A.1 (CONTINUED)

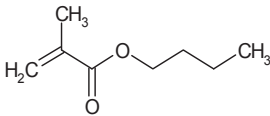
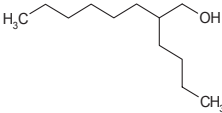
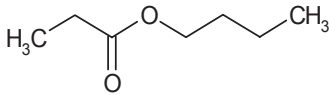
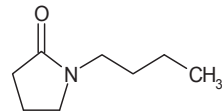
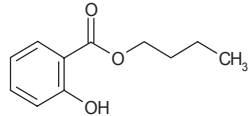
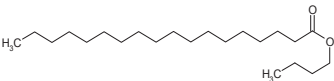
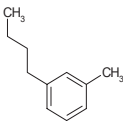
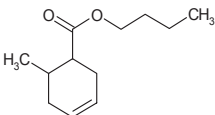
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
108	n-Butyl Methacrylate	2-Methyl-acrylic acid butyl ester	15.6	6.4	6.6	159.4
						
717	2-Butyl Octanol	2-Butyl-octan-1-ol	16.1	3.6	9.3	224.2
						
1006	n-Butyl Propionate	Propionic acid butyl ester	15.7	5.5	5.9	149.7
						
109	N-Butyl Pyrrolidone	1-Butyl-pyrrolidin-2-one	17.5	9.9	5.8	148.0
						
764	n-Butyl Salicylate	2-Hydroxy-benzoic acid butyl ester	17.9	4.8	11.7	181.7
						
223	Butyl Stearate	Octadecanoic acid butyl ester	14.5	3.7	3.5	382.0
						
724	3-n-Butyl Toluene	1-Butyl-3-methyl- benzene	17.4	0.1	1.0	173.7
						
747	Butyl-6-Methyl-3-Cyclohexene Carboxylate	6-Methyl-cyclohex-3- enecarboxylic acid butyl ester	16.1	2.5	5.7	209.1
						

TABLE A.1 (CONTINUED)

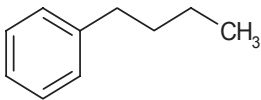
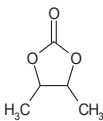
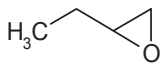
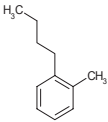
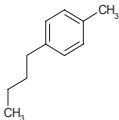
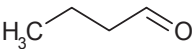
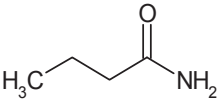
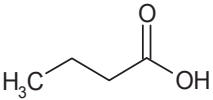
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
725	n-Butylbenzene 	Butyl-benzene	17.4	0.1	1.1	157.0
110	2,3-Butylene Carbonate 	4,5-Dimethyl- [1,3]dioxolan-2-one	18.0	16.8	3.1	105.5
111	Butyleneoxide 	2-Ethyl-oxirane	16.3	6.2	5.9	87.5
719	o-n-Butyltoluene 	1-Butyl-2-methyl- benzene	17.6	0.1	1.0	171.3
720	p-n-Butyltoluene 	1-Butyl-4-methyl- benzene	17.4	0.1	1.0	174.2
112	2-Butynedinitrile 	But-2-ynedinitrile	15.2	16.2	8.0	78.4
1060	Butyraldehyde* 	Butyraldehyde	15.6	10.1	6.2	90.5
1072	n-Butyramide 	Butyramide	16.9	13.7	12.3	98.4
114	Butyric Acid 	Butyric acid	14.9	4.1	10.6	110.0

TABLE A.1 (CONTINUED)

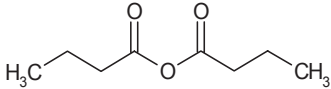
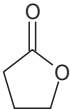
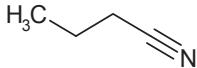
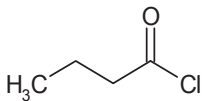
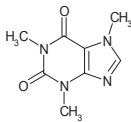
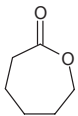
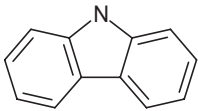
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
767	Butyric Anhydride	Butanoic anhydride	16.0	6.3	7.7	164.4
						
115	Gamma-Butyrolactone	Dihydro-furan-2-one	19.0	16.6	7.4	76.8
						
116	Butyronitrile	Butyronitrile	15.3	12.4	5.1	87.3
						
117	Butyrylchloride	Butyryl chloride	16.8	9.4	4.8	103.6
						
1200	Caffeine	1,3,7-Trimethyl-3,7-dihydro-purine-2,6-dione	19.5	10.1	13.0	157.9
						
118	Caprolactone (Epsilon)	Oxepan-2-one	19.7	15.0	7.4	110.8
						
869	Carbazole (Diphenylenimine)	9H-Carbazole	21.7	6.4	6.2	152.0
						
119	Carbon Dioxide* Ro = 3.3	Methanedione	15.7	6.3	5.7	38.0
						

TABLE A.1 (CONTINUED)

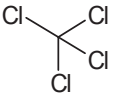
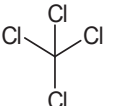
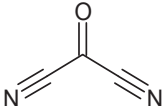
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
120	Carbon Disulfid P for 0 Dipole Moment	Methanedithione	20.5	0	0.6	60.0
						
121	Carbon Disulfid P for Group Contribution	Methanedithione	19.9	5.8	0.6	60.0
						
122	Carbon tetrachloride P for 0 Dipole Moment	Tetrachloro-methane	17.8	0	0.6	97.1
						
862	Carbon Tetrachloride P by Group Cont.	Tetrachloro-methane	16.1	8.3	0	97.1
						
124	Carbonyl Cyanide	2-Oxo-malononitrile	15.0	6.3	8.0	71.2
						
123	Carbonyl Sulfid	Thioxo-methanone	17.4	3.7	0	51.0
						
1224	Castor Oil		15.9	4.6	12.0	436.0
976	Cetyl Alcohol (1-Hexadecanol)	Hexadecan-1-ol	15.1	3.7	8.1	298.7
						

TABLE A.1 (CONTINUED)

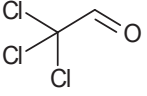
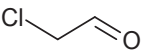
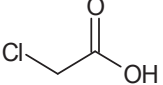
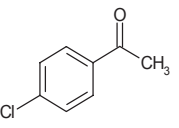
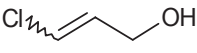
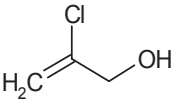
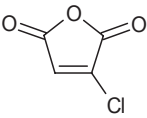
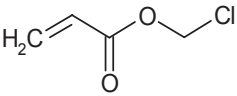
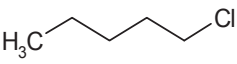
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
125	Chloral 	Trichloro-acetaldehyde	17.2	7.4	7.6	97.5
126	Chlorine	Chlorine	17.3	10.0	0	46.0
128	Chloro Acetaldehyde 	Chloro-acetaldehyde	16.2	16.1	9.0	60.4
129	Chloro Acetic Acid 	Chloro-acetic acid	17.7	10.4	12.3	68.6
797	p-Chloro Acetophenone 	1-(4-Chloro-phenyl)- ethanone	19.6	7.6	4.0	129.7
130	3-Chloro Allyl Alcohol 	3-Chloro-prop-2-en-1-ol	17.2	10.3	16.5	78.6
131	2-Chloro Allyl Alcohol 	2-Chloro-prop-2-en-1-ol	17.1	10.2	16.4	79.6
983	Chloro Maleic Anhydride 	3-Chloro-furan-2,5- dione	20.4	17.3	11.5	89.5
132	1-Chloro Methyl Acrylate 	Acrylic acid chloromethyl ester	15.9	7.3	8.5	101.4
955	1-Chloro Pentane 	1-Chloro-pentane	16.0	6.9	1.9	120.8

TABLE A.1 (CONTINUED)

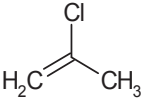
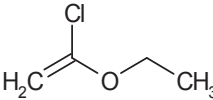
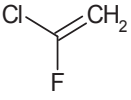
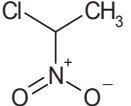
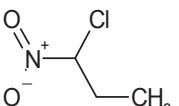
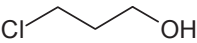
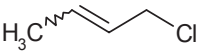
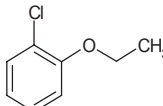
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
133	2-Chloro Propene (Isopropenyl Chloride)	2-Chloro-propene	15.5	6.7	2.2	84.9
						
135	1-Chloro Vinyl Ethyl Ether	1-Chloro-1-ethoxy-ethene	16.8	6.5	5.7	104.5
						
127	1-Chloro-1-Fluoro Ethylene	1-Chloro-1-fluoro ethene	16.0	6.9	4.0	67.6
						
136	1-Chloro-1-Nitroethane	1-Chloro-1-nitro-ethane	16.8	13.5	4.7	85.1
						
1083	1-Chloro-1-Nitropropane	1-Chloro-1-nitro-propane	16.8	13.0	4.5	102.2
						
137	3-Chloro-1-Propanol	3-Chloro-propan-1-ol	17.5	5.7	14.7	84.2
						
138	4-Chloro-1,2-Butadiene	4-Chloro-buta-1,2-diene	16.6	8.0	6.7	89.5
						
139	1-Chloro-2-Butene	1-Chloro-but-2-ene	16.2	7.7	2.0	97.4
						
817	1-Chloro-2-Ethoxy Benzene	1-Chloro-2-ethoxy-benzene	19.2	8.1	4.4	138.7
						

TABLE A.1 (CONTINUED)

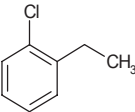
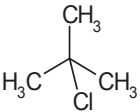
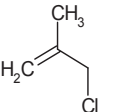
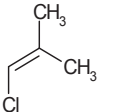
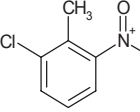
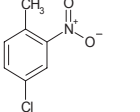
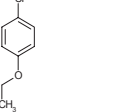
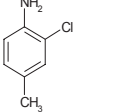
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1164	1-Chloro-2-Ethyl Benzene 	1-Chloro-2-ethyl- benzene	18.9	4.9	2.2	133.0
970	2-Chloro-2-Methyl Propane 	2-Chloro-2-methyl- propane	15.6	7.6	2.0	110.0
140	3-Chloro-2-Methyl Propene 	3-Chloro-2-methyl- propene	16.2	5.6	2.0	98.8
141	1-Chloro-2-Methyl Propene 	1-Chloro-2-methyl- propene	16.1	7.1	4.2	95.6
813	6-Chloro-2-Nitrotoluene 	1-Chloro-2-methyl-3- nitro-benzene	20.3	9.6	3.8	132.0
819	4-Chloro-2-Nitrotoluene 	4-Chloro-1-methyl-2- nitro-benzene	19.9	11.8	3.8	132.0
816	1-Chloro-4-Ethoxy Benzene 	1-Chloro-4-ethoxy- benzene	19.3	6.3	4.4	139.2
950	2-Chloro-4-Methylaniline 	2-Chloro-4-methyl- phenylamine	19.7	7.4	8.9	123.0

TABLE A.1 (CONTINUED)

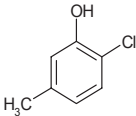
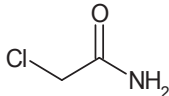
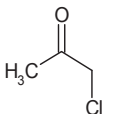
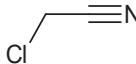
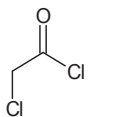
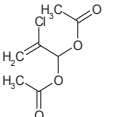
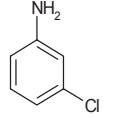
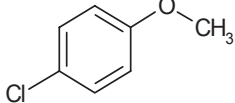
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
977	2-Chloro-5-Methyl Phenol 	2-Chloro-5-methyl- phenol	19.0	7.7	13.1	117.4
1067	2-Chloroacetamide 	2-Chloro-acetamide	17.6	12.4	12.8	93.5
142	Chloroacetone 	1-Chloro-propan-2-one	16.8	9.6	5.5	80.5
143	Chloroacetonitrile 	Chloro-acetonitrile	17.4	13.6	2.0	63.3
144	Chloroacetylchloride 	Chloro-acetyl chloride	17.5	9.2	5.5	79.5
145	Chloroacetylene $\text{HC}\equiv\text{Cl}$	Chloro-ethyne	16.2	2.1	2.5	63.7
766	2-Chloroallylidene Diacetate 	Acetic acid 1-acetoxy-2- chloro-allyl ester	16.7	7.3	8.8	160.2
1012	m-Chloroaniline 	3-Chloro-phenylamine	20.6	9.9	9.8	105.0
146	4-Chloroanisole 	1-Chloro-4-methoxy- benzene	19.6	7.8	6.7	122.5

TABLE A.1 (CONTINUED)

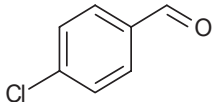
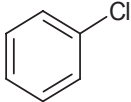
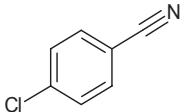
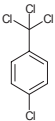
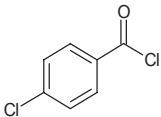
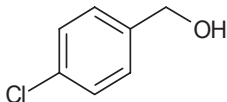
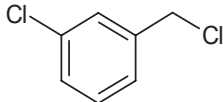
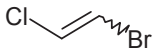
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
147	4-Chlorobenzaldehyde	4-Chloro-benzaldehyde	19.9	7.2	5.6	113.4
						
148	Chlorobenzene	Chloro-benzene	19.0	4.3	2.0	102.1
						
811	4-Chlorobenzonitrile	4-Chloro-benzonitrile	19.5	8.0	4.1	125.1
						
853	4-Chlorobenzotrichloride	1-Chloro-4-trichloromethyl-benzene	20.3	5.5	3.5	153.8
						
812	p-Chlorobenzoyl Chloride	4-Chloro-benzoyl chloride	19.9	6.7	5.1	127.1
						
149	4-Chlorobenzyl Alcohol	(4-Chloro-phenyl)-methanol	19.6	7.5	13.0	117.7
						
150	3-Chlorobenzylchloride	1-Chloro-3-chloromethyl-benzene	19.9	9.3	2.6	117.7
						
151	1,2-Chlorobromoethylene	1-Bromo-2-chloro-ethene	17.2	6.6	2.3	78.7
						

TABLE A.1 (CONTINUED)

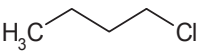
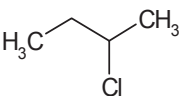
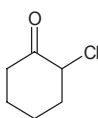
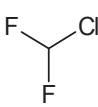
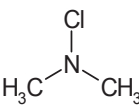
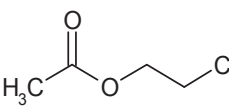
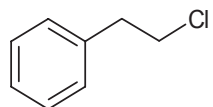
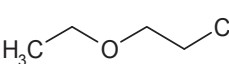
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
152	1-Chlorobutane 	1-Chloro-butane	16.2	5.5	2.0	104.5
750	2-Chlorobutane 	2-Chloro-butane	15.8	7.6	2.0	106.8
780	2-Chlorocyclohexanone 	2-Chloro-cyclohexanone	18.5	13.0	5.1	113.9
153	Chlorocyclopropane 	Chloro-cyclopropane	17.6	7.2	2.2	84.9
154	Chlorodifluoromethane (Freon 22) 	Chloro-difluoro-methan	12.3	6.3	5.7	72.9
155	N-Chlorodimethylamine 	[Chloro(methyl)amino] methane	16.0	7.8	7.9	87.4
772	2-Chloroethyl Acetate 	Acetic acid 2-chloro- ethyl ester	16.7	9.6	8.8	107.5
1165	2-Chloroethyl Benzene 	(2-Chloro-ethyl)- benzene	19.3	6.3	2.2	131.5
756	2-Chloroethyl Ethyl Ether 	1-Chloro-2-ethoxy- ethane	16.3	7.9	4.6	109.5

TABLE A.1 (CONTINUED)

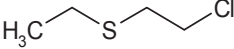
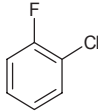
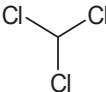
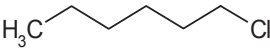
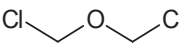
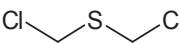
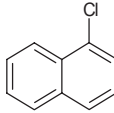
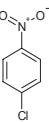
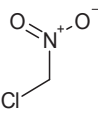
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
834	2-Chloroethyl Ethyl Sulfid 	1-Chloro-2-ethylsulfanyl-ethane	17.2	5.0	6.1	116.9
808	o-Chlorofluorobenzen 	1-Chloro-2-fluoro benzene	19.4	8.7	2.0	104.9
156	Chloroform 	Trichloro-methane	17.8	3.1	5.7	80.7
959	1-Chlorohexane 	1-Chloro-hexane	16.1	6.2	1.7	138.1
157	Bis(Chloromethyl) Ether 	Chloro-chloromethoxy-methane	17.2	4.9	6.6	86.6
158	Chloromethylsulfid 	Chloro-chloromethylsulfanyl-methane	16.6	6.4	2.0	95.0
870	1-Chloronaphthalene 	1-Chloro-naphthalene	19.9	4.9	2.5	136.2
809	p-Chloronitrobenzene 	1-Chloro-4-nitro-benzene	20.4	9.6	4.2	103.7
159	Chloronitromethane 	Chloro-nitro-methane	17.4	13.5	5.5	65.1

TABLE A.1 (CONTINUED)

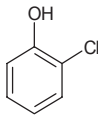
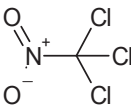
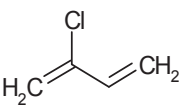
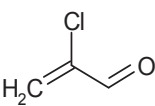
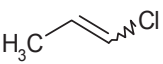
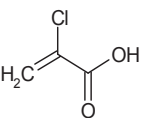
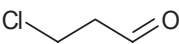
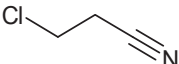
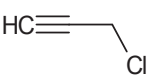
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
160	2-Chlorophenol 	2-Chloro-phenol	20.3	5.5	13.9	102.3
1081	Chloropicrin (Trichloronitromethane) 	Trichloro-nitro-methane	17.6	6.8	7.0	101.7
161	Chlorprene 	2-Chloro-buta-1,3-diene	16.1	5.4	2.1	93.2
162	2-Chloropropenal 	2-Chloro-propenal	17.1	12.9	8.1	75.5
163	1-Chloropropene 	1-Chloro-propene	15.3	6.9	2.2	82.3
164	2-Chloropropenoic Acid 	2-Chloro-acrylic acid	19.1	9.4	12.4	86.6
165	3-Chloropropionaldehyde 	3-Chloro-propionaldehyde	17.0	13.3	8.2	73.0
166	3-Chloropropionitrile 	3-Chloro-propionitrile	17.3	15.9	6.1	77.4
167	3-Chloropropyne 	3-Chloro-propyne	16.7	7.4	2.3	72.4

TABLE A.1 (CONTINUED)

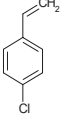
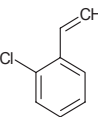
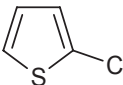
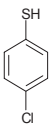
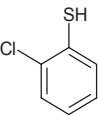
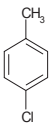
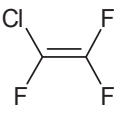
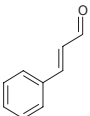
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
711	p-Chlorostyrene 	1-Chloro-4-vinyl- benzene	18.7	4.3	3.9	128.3
712	o-Chlorostyrene 	1-Chloro-2-vinyl- benzene	18.7	4.7	3.9	126.8
989	2-Chlorothiophene 	2-Chloro-thiophene	19.2	6.2	8.0	92.2
168	4-Chlorothiophenol 	4-Chloro-benzenethiol	20.8	8.6	10.6	100.0
787	o-Chlorothiophenol 	2-Chloro-benzenethiol	20.2	7.0	10.0	113.4
814	p-Chlorotoluene 	1-Chloro-4-methyl- benzene	19.1	6.2	2.6	118.3
134	Chlorotrifluoroethylene (CTFE) 	1-Chloro-1,2,2-trifluoro ethene	15.3	6.3	0	75.6
1111	trans-Cinnamaldehyde 	(E)-3-Phenyl-propenal	19.4	12.4	6.2	125.9

TABLE A.1 (CONTINUED)

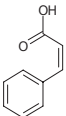
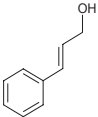
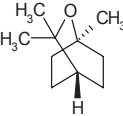
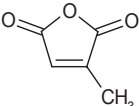
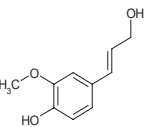
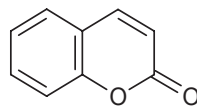
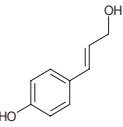
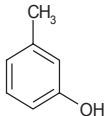
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1110	cis-Cinnamic Acid 	(Z)-3-Phenyl-acrylic acid	19.1	3.9	10.6	115.4
1107	Cinnamyl Alcohol 	(E)-3-Phenyl-prop-2-en- 1-ol	19.1	6.0	13.0	129.1
1135	Cineol (Eucalyptol) 	1,3,3-Trimethyl-2-oxa- bicyclo[2.2.2]octane	16.7	4.6	3.4	167.5
982	Citraconic Anhydride 	3-Methyl-furan-2,5- dione	19.2	17.0	11.2	89.9
1102	Coniferyl Alcohol 	4-((E)-3-Hydroxy- propenyl)-2-methoxy- phenol	19.0	7.0	16.3	171.6
974	Coumarin 	Chromen-2-one	20.0	12.5	6.7	156.3
1103	p-Coumaryl Alcohol 	4-((E)-3-Hydroxy- propenyl)-phenol	19.1	7.0	17.3	136.5
169	m-Cresol 	3-Methyl-phenol	18.0	5.1	12.9	104.7

TABLE A.1 (CONTINUED)

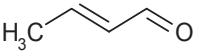
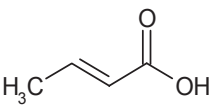
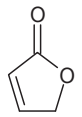
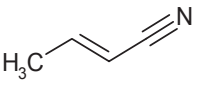
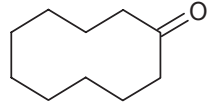
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
170	Crotonaldehyde 	(E)-But-2-enal	16.2	14.9	7.4	82.5
171	Crotonic Acid 	(E)-But-2-enoic acid	16.8	8.7	12.0	84.6
172	Crotonlactone 	5H-Furan-2-one	19.0	19.8	9.6	76.4
173	Trans-Crotononitrile 	(E)-But-2-enitrile	16.4	18.8	5.5	81.4
174	Cyanamid (Carbamonitrile) $\text{N}\equiv\text{NH}_2$	Cyanamide	15.5	27.6	16.8	32.8
175	Cyanogen $\text{N}\equiv\text{N}$	Oxalonitrile	15.1	11.8	0	54.6
176	Cyanogen Bromide $\text{N}\equiv\text{Br}$	Cyanic bromide	18.3	15.2	0	52.6
177	Cyanogen Chloride $\text{N}\equiv\text{Cl}$	Cyanic chloride	15.6	14.5	0	51.8
178	Cyclobutanone 	Cyclobutanone	18.3	11.4	5.2	73.4
179	Cyclodecanone 	Cyclodecanone	16.8	8.0	4.1	161.0

TABLE A.1 (CONTINUED)

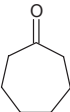
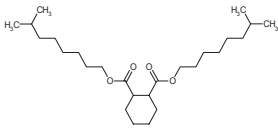
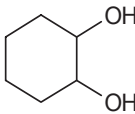
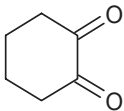
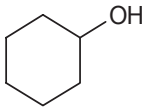
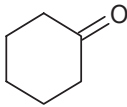
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
180	Cycloheptanone 	Cycloheptanone	17.2	10.6	4.8	118.2
181	Cyclohexane 	Cyclohexane	16.8	0	0.2	108.7
1214	Cyclohexane-1,2-Dicarboxylic Acid Di-(Isononyl) Ether 	Cyclohexane-1,2- dicarboxylic acid bis- (7-methyl-octyl) ester	16.4	2.2	5.0	422.4
1093	1,2-Cyclohexanediol 	Cyclohexane-1,2-diol	17.4	9.8	18.3	112.8
1094	1,2-Cyclohexanedione 	Cyclohexane-1,2-dione	18.6	10.3	8.0	103.8
182	Cyclohexanol 	Cyclohexanol	17.4	4.1	13.5	106.0
183	Cyclohexanone 	Cyclohexanone	17.8	6.3	5.1	104.0
871	Cyclohexene 	Cyclohexene	17.2	1.0	5.0	101.9

TABLE A.1 (CONTINUED)

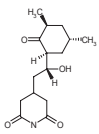
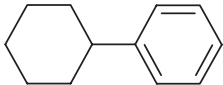
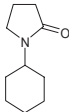
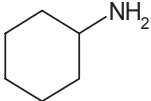
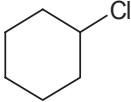
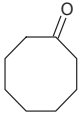
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1199	Cycloheximide 	4-[(R)-2-((1S,3S,5S)-3,5-Dimethyl-2-oxo-cyclohexyl)-2-hydroxy-ethyl]-piperidine-2,6-dione	18.3	11.0	13.8	171.0
872	Cyclohexyl Benzene 	Cyclohexyl-benzene	18.7	0	1.0	169.9
969	N-Cyclohexyl-2-Pyrrolidone 	1-Cyclohexyl-pyrrolidin-2-one	18.2	6.8	6.5	163.0
184	Cyclohexylamine 	Cyclohexylamine	17.2	3.1	6.5	113.8
185	Cyclohexylchloride 	Chloro-cyclohexane	17.3	5.5	2.0	118.6
186	Cyclooctanone 	Cyclooctanone	17.0	9.6	4.5	131.7
920	Cyclopentadiene 	Cyclopenta-1,3-diene	17.2	1.9	6.1	82.1
187	Cyclopentane 	Cyclopentane	16.4	0	1.8	94.9

TABLE A.1 (CONTINUED)

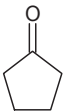
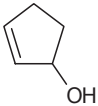
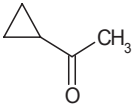
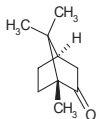
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
188	Cyclopentanone	Cyclopentanone	17.9	11.9	5.2	89.1
						
189	Cyclopentene	Cyclopentene	16.7	3.8	1.7	89.0
						
735	2-Cyclopentenyl Alcohol	Cyclopent-2-enol	18.1	7.6	15.6	86.2
						
190	Cyclopropene	Cyclopropene	17.2	2.4	2.0	50.0
						
191	Cyclopropane	Cyclopropane	17.6	0	0	58.3
						
192	Cyclopropylmethylketone	1-Cyclopropyl-ethanone	17.0	11.1	4.6	93.6
						
193	Cyclopropyl nitrile	Cyclopropane-carbonitrile	18.6	16.2	5.7	75.4
						
1223	d-Camphor	(1R,4R)-1,7,7-Trimethyl-bicyclo[2.2.1]heptan-2-one	17.8	9.4	4.7	153.7
						

TABLE A.1 (CONTINUED)

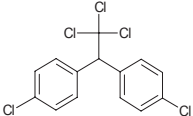
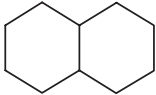
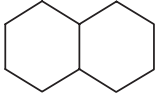
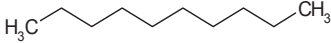
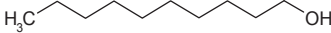
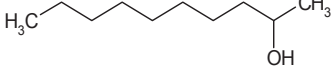
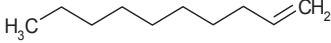
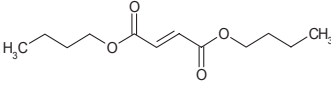
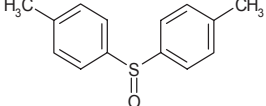
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
873	DDT	1-Chloro-4-[2,2,2-trichloro-1-(4-chlorophenyl)ethyl]benzene	20.0	5.5	3.1	268.8
						
194	Cis-Decahydronaphthalene	Decahydro-naphthalene	18.8	0	0	156.9
						
195	Trans-Decahydronaphthalene	Decahydro-naphthalene	18.0	0	0	156.9
						
196	Decane	Decane	15.7	0	0	195.9
						
197	1-Decanol*	Decan-1-ol	16.0	4.7	10.0	191.8
						
734	2-Decanol	Decan-2-ol	15.8	3.9	10.0	192.8
						
726	1-Decene	Dec-1-ene	15.8	1.0	2.2	190.6
						
759	Di-(2-Chloroethoxy) Methane	1-Chloro-2-(2-chloroethoxymethoxy)-ethane	17.1	10.2	7.1	141.1
						
773	Dibutyl Fumarate	(E)-But-2-enedioic acid dibutyl ester	16.7	3.0	6.7	232.7
						
199	Di-p-Tolyl Sulfoxide	1-Methyl-4-[(4-methylphenyl)sulfi yl] benzene	20.3	11.4	3.1	209.0
						

TABLE A.1 (CONTINUED)

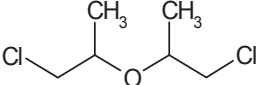
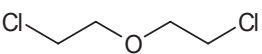
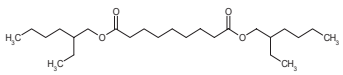
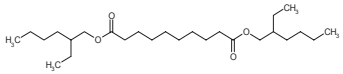
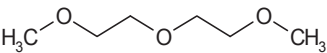
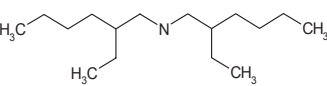
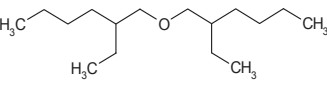
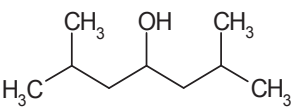
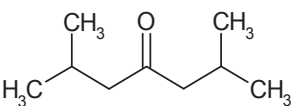
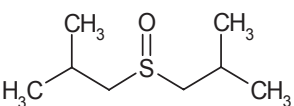
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
204	Di-(2-Chloro-Isopropyl) Ether 	1-Chloro-2-(2-chloro-1-methyl-ethoxy)-propane	19.0	8.2	5.1	146.0
235	Di-(2-Chloroethyl) Ether 	1-Chloro-2-(2-chloro-ethoxy)-ethane	18.8	9.0	5.7	117.2
1139	Di-(2-Ethyl Hexyl)azelate 	Nonanedioic acid bis-(2-ethyl-hexyl) ester	16.7	1.4	4.8	449.9
1138	Di-(2-Ethyl Hexyl)sebacate 	Decanedioic acid bis-(2-ethyl-hexyl) ester	16.8	1.0	4.7	468.7
202	Di-(2-Methoxyethyl) Ether 	1-Methoxy-2-(2-methoxy-ethoxy)-ethane	15.7	6.1	6.5	142.0
205	Di-2-Ethyl Hexyl Amine 	Bis-(2-ethyl-hexyl)-amine	15.6	0.8	3.2	301.5
736	Di-2-Ethyl Hexyl Ether 	3-(2-Ethyl-hexyloxymethyl)-heptane	15.9	2.6	2.5	300.5
208	Di-Isobutyl Carbinol 	2,6-Dimethyl-heptan-4-ol	14.9	3.1	10.8	177.8
203	Di-Isobutyl Ketone 	2,6-Dimethyl-heptan-4-one	16.0	3.7	4.1	177.1
845	Di-Isobutyl Sulfoxide 	2-Methyl-1-(2-methyl-propane-1-sulfi yl)-propane	16.3	10.5	6.1	195.0

TABLE A.1 (CONTINUED)

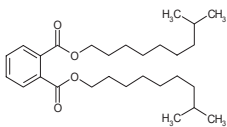
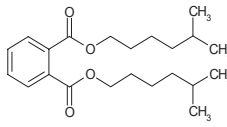
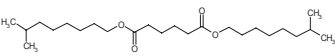
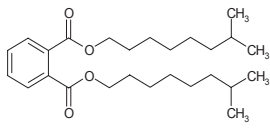
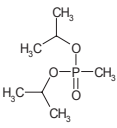
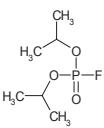
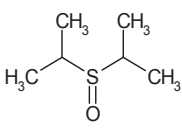
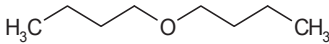
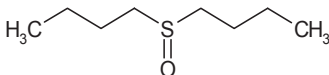
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
279	Di-Isodecyl Phthalate	Phthalic acid bis-(8-methyl-nonyl) ester	16.6	6.2	2.6	464.2
						
280	Di-Isoheptyl Phthalate	Phthalic acid bis-(5-methyl-hexyl) ester	16.8	7.2	3.4	364.9
						
281	Di-Isononyl Adipate	Hexanedioic acid bis-(7-methyl-octyl) ester	16.2	1.8	4.9	433.7
						
282	Di-Isononyl Phthalate	Phthalic acid bis-(7-methyl-octyl) ester	16.6	6.6	2.9	432.4
						
840	Di-Isopropyl Methyl Phosphonate	Methyl-phosphonic acid diisopropyl ester	16.4	10.0	5.7	184.4
						
842	Di-Isopropyl Phosphonofluoridate	Phosphorofluoridic acid diisopropyl ester	15.7	10.2	5.9	174.5
						
198	Di-Isopropyl Sulfoxide	2-(Propane-2-sulfi yl)-propane	17.0	11.5	7.4	159.9
						
219	Di-n-Butyl Ether	1-Butoxy-butane	15.2	3.4	4.2	170.3
						
200	Di-n-Butyl Sulfoxide	1-(Butane-1-sulfi yl)-butane	16.4	10.5	6.1	195.0
						

TABLE A.1 (CONTINUED)

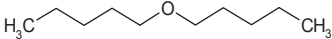
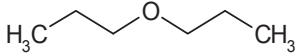
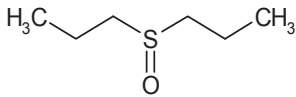
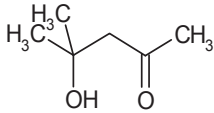
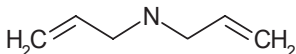
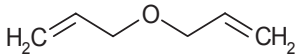
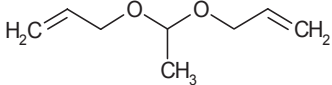
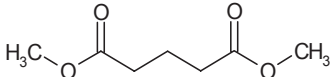
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
957	Di-n-Pentyl Ether 	1-Pentyloxy-pentane	15.6	3.1	3.0	203.2
956	Di-n-Propyl Ether 	1-Propoxy-propane	15.1	4.2	3.7	137.7
201	Di-n-Propyl Sulfoxide 	1-(Propane-1-sulfi yl)- propane	17.0	13.0	7.4	159.9
209	Diacetone Alcohol 	4-Hydroxy-4-methyl- pentan-2-one	15.8	8.2	10.8	124.2
210	Diallyl Amine 	Diallyl-amine	15.6	4.5	6.7	124.1
211	Diallyl Ether 	3-Allyloxy-propene	15.3	4.4	5.3	118.8
749	1,1-Diallyloxyethane 	3-(1-Allyloxy-ethoxy)- propene	15.4	5.1	4.8	163.3
212	Diazomethane $\text{H}_2\text{C}=\text{N}^+=\text{N}^-$	Diazomethane	14.7	6.1	11.3	78.1
1009	Dibasic Esters (dupont) Mix of Dimethyl Esters of Succinic, Glutaric, and Adipic Acids 	Pentanedioic acid dimethyl ester	16.2	4.7	8.4	159.0

TABLE A.1 (CONTINUED)

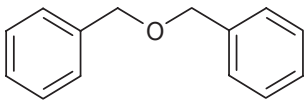
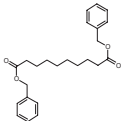
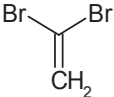
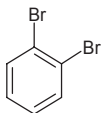
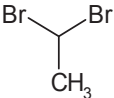
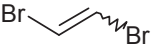
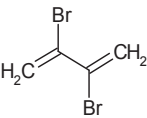
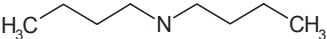
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
213	Dibenzyl Ether*	[(Benzyloxy)methyl]- benzene	19.6	3.4	5.2	197.4
						
1148	Dibenzyl Sebacate	Decanedioic acid dibenzyl ester	17.8	2.2	5.5	362.1
						
214	1,1-Dibromo Ethylene	1,1-Dibromo-ethene	15.1	4.8	7.0	85.3
						
215	Dibromo Methane	Dibromo-methane	17.8	6.4	7.0	69.8
						
804	o-Dibromobenzene	1,2-Dibromo-benzene	20.7	6.5	5.3	120.0
						
216	1,1-Dibromoethane	1,1-Dibromo-ethane	18.5	8.4	8.8	91.4
						
217	1,2-Dibromoethylene	1,2-Dibromo-ethene	18.0	4.9	3.0	82.7
						
218	2,3-Dibromoprene	2,3-Dibromo-buta-1,3- diene	17.7	11.8	6.4	103.4
						
1015	Dibutyl Amine	Dibutyl-amine	15.0	3.0	4.3	170.0
						

TABLE A.1 (CONTINUED)

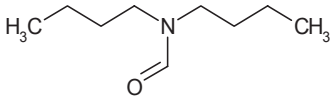
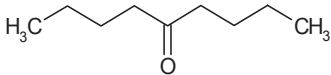
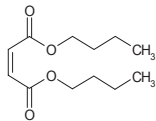
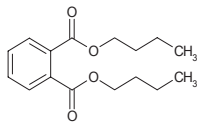
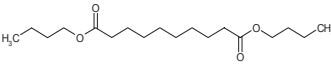
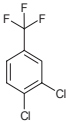
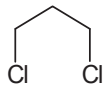
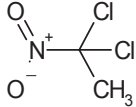
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
220	N,N-Dibutyl Formamide	N,N-Dibutyl-formamide	15.5	8.9	6.2	182.0
						
1068	Dibutyl Ketone	Nonan-5-one	16.0	7.7	4.4	173.4
						
1146	Dibutyl Maleate	(Z)-But-2-enedioic acid dibutyl ester	16.5	6.1	7.2	230.4
						
221	Dibutyl Phthalate	Phthalic acid dibutyl ester	17.8	8.6	4.1	266.0
						
222	Dibutyl Sebacate*	Decanedioic acid dibutyl ester	16.7	4.5	4.1	339.0
						
700	3,4-Dichloro α,α,α -Trifluorotoluen	1,2-Dichloro-4- trifluoromet yl- benzene	20.0	4.7	2.4	145.5
						
874	1,3-Dichloropropane	1,3-Dichloro-propane	18.6	8.2	3.0	95.1
						
1082	1,1-Dichloro-1-Nitroethane	1,1-Dichloro-1-nitro- ethane	16.8	12.1	4.3	125.2
						

TABLE A.1 (CONTINUED)

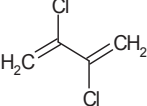
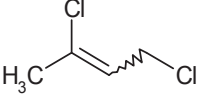
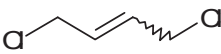
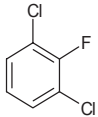
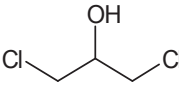
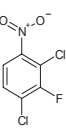
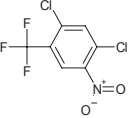
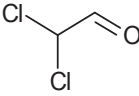
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
229	2,3-Dichloro-1,3-Butadiene	2,3-Dichloro-buta-1,3- diene	17.1	2.3	2.8	104.0
						
206	1,3-Dichloro-2-butene	1,3-Dichloro-but-2-ene	16.9	7.8	2.7	107.7
						
207	1,4-Dichloro-2-Butene	1,4-Dichloro-but-2-ene	17.8	7.6	2.0	97.8
						
825	1,3-Dichloro-2-Fluorobenzene	1,3-Dichloro-2-fluoro benzene	19.4	9.1	2.7	117.1
						
739	1,3-Dichloro-2-Propanol	1,3-Dichloro-propan-2- ol	17.5	9.9	14.6	95.5
						
824	2,4-Dichloro-3-Fluoronitrobenzene	1,3-Dichloro-2-fluoro-4 nitro-benzene	19.9	7.2	4.2	140.0
						
828	2,4-Dichloro-5-Nitrobenzotrifluorid	1,5-Dichloro-2-nitro-4- trifluoromet yl- benzene	19.9	7.6	3.7	162.0
						
230	Dichloroacetaldehyde	Dichloro-acetaldehyde	16.7	9.1	7.5	94.0
						

TABLE A.1 (CONTINUED)

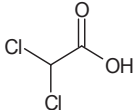
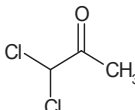
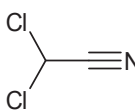
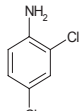
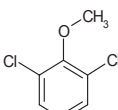
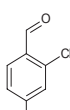
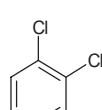
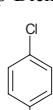
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
945	Dichloroacetic Acid	Dichloro-acetic acid	18.2	8.1	12.2	82.5
						
231	1,1-Dichloroacetone	1,1-Dichloro-propan-2-one	17.1	7.6	5.4	97.3
						
232	Dichloroacetonitrile	Dichloro-acetonitrile	17.4	9.4	6.4	80.3
						
1000	2,4-Dichloroaniline	2,4-Dichloro-phenylamine	20.9	6.2	10.0	103.4
						
233	2,6-Dichloroanisole	1,3-Dichloro-2-methoxy-benzene	19.8	8.4	6.5	137.1
						
846	2,4-Dichlorobenzaldehyde	2,4-Dichloro-benzaldehyde	19.7	8.8	5.4	135.7
						
234	o-Dichlorobenzene	1,2-Dichloro-benzene	19.2	6.3	3.3	112.8
						
715	p-Dichlorobenzene	1,4-Dichloro-benzene	19.7	5.6	2.7	118.6
						

TABLE A.1 (CONTINUED)

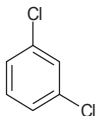
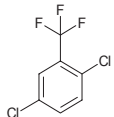
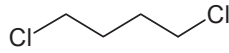
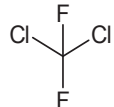
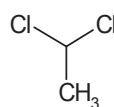
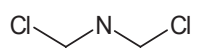
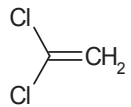
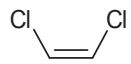
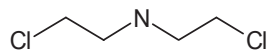
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
716	m-Dichlorobenzene 	1,3-Dichloro-benzene	19.7	5.1	2.7	114.8
826	2,5-Dichlorobenzotrifluorid 	1,4-Dichloro-2- trifluoromet yl- benzene	20.0	4.7	2.4	145.5
1019	1,4-Dichlorobutane 	1,4-Dichloro-butane	18.3	7.7	2.8	109.5
236	Dichlorodifluoromethane (Freon 12) 	Dichloro-difluoro methane	12.3	2.0	0	92.3
237	1,1-Dichloroethane 	1,1-Dichloro-ethane	16.5	7.8	3.0	84.2
225	N,N-Dichloromethyl Amine 	Bis-chloromethyl-amine	16.8	7.6	8.0	90.8
239	1,1-Dichloroethylene 	1,1-Dichloro-ethene	16.4	5.2	2.4	79.9
224	1,2-Dichloroethylene (cis) 	(Z)-1,2-Dichloro-ethene	17.0	8.0	3.2	75.5
238	N,N-Dichloroethyl Amine 	Bis-(2-chloro-ethyl)- amine	16.8	7.6	7.7	98.3

TABLE A.1 (CONTINUED)

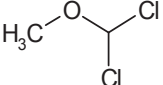
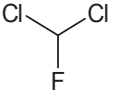
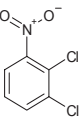
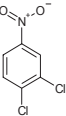
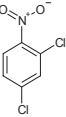
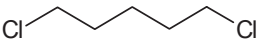
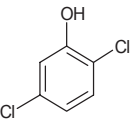
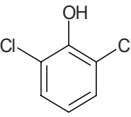
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
241	Dichloromethyl Methyl Ether	Dichloro-methoxy- methane	17.1	12.9	6.5	90.5
						
242	Dichloromonofluoromethane (Freon 21)	Dichloro-fluoro-methan	15.8	3.1	5.7	75.4
						
243	2,3-Dichloronitrobenzene	1,2-Dichloro-3-nitro- benzene	19.7	12.6	4.4	132.5
						
801	3,4-Dichloronitrobenzene	1,2-Dichloro-4-nitro- benzene	20.1	7.2	4.1	130.6
						
822	2,4-Dichloronitrobenzene	2,4-Dichloro-1-nitro- benzene	20.4	8.7	4.2	133.4
						
833	1,5-Dichloropentane	1,5-Dichloro-pentane	19.0	7.8	1.5	127.5
						
244	2,5-Dichlorophenol	2,5-Dichloro-phenol	20.0	6.3	12.1	119.0
						
852	2,6-Dichlorophenol	2,6-Dichloro-phenol	20.1	7.5	10.9	119.0
						

TABLE A.1 (CONTINUED)

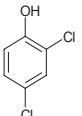
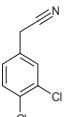
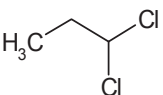
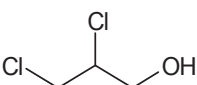
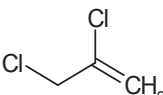
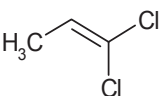
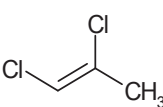
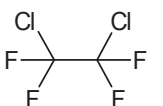
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1159	2,4-Dichlorophenol	2,4-Dichloro-phenol	20.0	7.2	13.1	119.0
						
848	3,4-Dichlorophenyl Acetonitrile	(3,4-Dichloro-phenyl)- acetonitrile	20.5	10.8	4.4	148.8
						
245	1,1-Dichloropropane	1,1-Dichloro-propane	16.1	7.8	3.5	98.3
						
738	2,3-Dichloropropanol	2,3-Dichloro-propan- 1-ol	17.5	9.2	14.6	95.2
						
226	2,3-Dichloropropene	2,3-Dichloro-propene	16.2	7.8	3.0	91.6
						
246	1,1-Dichloropropene	1,1-Dichloro-propene	16.9	6.7	2.9	93.5
						
227	1,2-Dichloropropene (cis)	(Z)-1,2-Dichloro- propene	17.0	8.5	2.9	93.9
						
247	1,2-Dichlorotetrafluoroethane (Freon 114)	1,2-Dichloro-1,1,2,2- tetrafluoro-ethan	12.6	1.8	0	117.6
						

TABLE A.1 (CONTINUED)

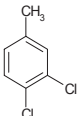
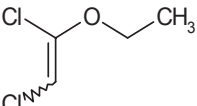
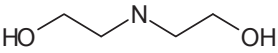
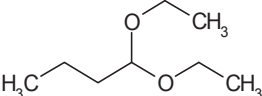
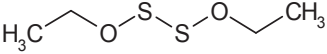
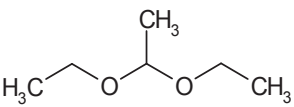
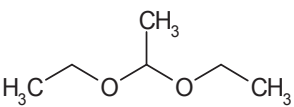
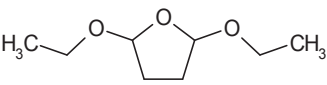
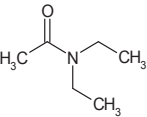
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
248	3,4-Dichlorotoluene 	1,2-Dichloro-4-methyl- benzene	19.8	9.8	2.5	128.7
228	1,2-Dichlorovinyl Ethyl Ether 	1,2-Dichloro-1-ethoxy- ethene	16.9	10.5	6.0	117.8
249	Diethanolamine 	2-(2-Hydroxy- ethylamino)-ethanol	17.2	10.8	21.2	95.9
777	1,1-Diethoxy Butane 	1,1-Diethoxy-butane	15.4	4.9	4.6	177.6
792	Diethoxy Disulfid 	[(Ethoxydithio)oxy]etha ne	15.1	8.3	7.4	141.5
250	1,1-Diethoxy Ethane 	1,1-Diethoxy-ethane	15.0	3.4	4.0	143.9
791	1,1-Diethoxy Ethane (Acetal) 	1,1-Diethoxy-ethane	15.2	5.4	5.3	142.2
1045	2,5-Diethoxy Tetrahydrofuran 	2,5-Diethoxy- tetrahydro-furan	16.6	6.4	7.3	165.7
251	N,N-Diethyl Acetamide 	N,N-Diethyl-acetamide	16.4	11.3	7.5	124.5

TABLE A.1 (CONTINUED)

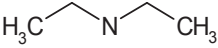
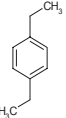
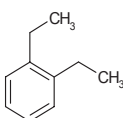
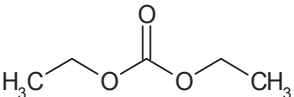
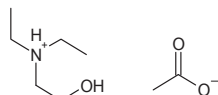
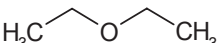
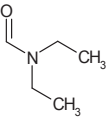
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
252	Diethyl Amine	Diethyl-amine	14.9	2.3	6.1	103.2
						
1183	Diethyl Amine/Acetic Acid	Acetate diethyl- ammonium;	16.0	20.3	18.4	
						
253	p-Diethyl Benzene	1,4-Diethyl-benzene	18.0	0	0.6	156.9
						
721	1,2-Diethyl Benzene	1,2-Diethyl-benzene	17.7	0.1	1.0	153.5
						
254	Diethyl Carbonate*	Carbonic acid diethyl ester	15.1	6.3	3.5	121.0
						
1184	Diethyl Ethanolamine/Acetic Acid	Acetate diethyl-(2- hydroxy-ethyl)- ammonium;	16.0	20.3	18.4	
						
255	Diethyl Ether	Ethoxy-ethane	14.5	2.9	5.1	104.8
						
256	N,N-Diethyl Formamide	N,N-Diethyl-formamide	16.4	11.4	9.2	111.4
						

TABLE A.1 (CONTINUED)

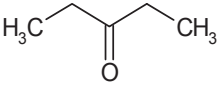
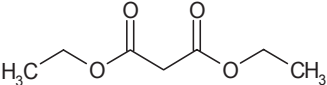
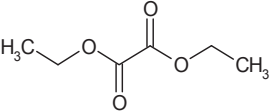
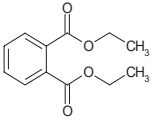
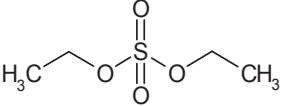
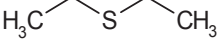
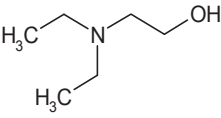
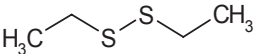
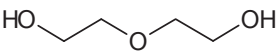
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
257	Diethyl Ketone	Pentan-3-one	15.8	7.6	4.7	106.4
						
979	Diethyl Malonate	Malonic acid diethyl ester	16.1	7.7	8.3	152.5
						
1143	Diethyl Oxalate	Oxalic acid diethyl ester	16.2	8.0	8.8	136.6
						
258	Diethyl Phthalate	Phthalic acid diethyl ester	17.6	9.6	4.5	198.0
						
259	Diethyl Sulfate	Sulfuric acid diethyl ester	15.7	14.7	7.1	131.5
						
260	Diethyl Sulfid	Ethylsulfanyl-ethane	16.8	3.1	2.0	107.4
						
261	2-(Diethylamino) Ethanol	2-Diethylamino-ethanol	14.9	5.8	12.0	133.2
						
262	Diethyldisulfid	Ethyldisulfanyl-ethane	16.7	6.7	5.7	123.1
						
263	Diethylene Glycol Commercial	2-(2-Hydroxy-ethoxy)- ethanol	16.6	12.0	20.7	94.9
						

TABLE A.1 (CONTINUED)

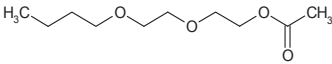
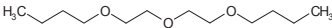
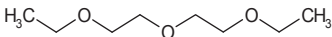
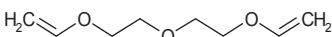
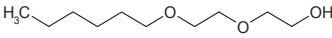
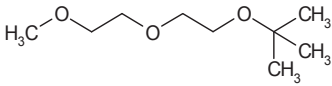
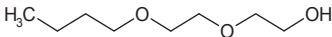
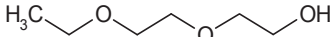
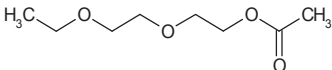
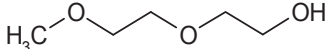
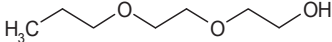
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
264	Diethylene Glycol Butyl Ether Acetate Commercial	Acetic acid 2-(2-butoxy-ethoxy)-ethyl ester	16.0	4.1	8.2	208.2
						
757	Diethylene Glycol Dibutyl Ether	1-[2-(2-Butoxy-ethoxy)-ethoxy]-butane	15.8	4.7	4.4	248.1
						
762	Diethylene Glycol Diethyl Ether	1-Ethoxy-2-(2-ethoxy-ethoxy)-ethane	15.8	5.9	5.6	179.8
						
755	Diethylene Glycol Divinyl Ether	[2-(2-Vinyloxy-ethoxy)-ethoxy]-ethene	16.0	7.3	7.9	164.1
						
265	Diethylene Glycol Hexyl Ether	2-(2-Hexyloxy-ethoxy)-ethanol	16.0	6.0	10.0	204.3
						
266	Diethylene Glycol Methyl t-Butyl Ether commercial	2-[2-(2-Methoxy-ethoxy)-ethoxy]-2-methyl-propane	16.0	7.2	7.2	193.9
						
267	Diethylene Glycol Monobutyl Ether Commercial	2-(2-Butoxy-ethoxy)-ethanol	16.0	7.0	10.6	170.6
						
268	Diethylene Glycol Monoethyl Ether Commercial	2-(2-Ethoxy-ethoxy)-ethanol	16.1	9.2	12.2	130.9
						
269	Diethylene Glycol Monoethyl Ether Acetate Commercial	Acetic acid 2-(2-ethoxy-ethoxy)-ethyl ester	16.2	5.1	9.2	175.5
						
270	Diethylene Glycol Monomethyl Ether Commercial	2-(2-Methoxy-ethoxy)-ethanol	16.2	7.8	12.6	118.0
						
271	Diethylene Glycol Monopropyl Ether	2-(2-Propoxy-ethoxy)-ethanol	16.0	7.2	11.3	153.9
						

TABLE A.1 (CONTINUED)

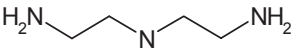
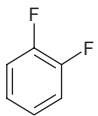
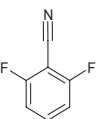
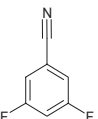
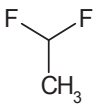
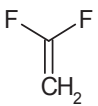
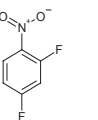
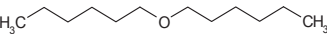
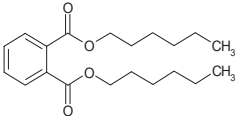
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
272	Diethylenetriamine Commercial	N*1*-(2-Amino-ethyl)- ethane-1,2-diamine	16.7	13.3	14.3	108.0
						
810	o-Difluorobenzen	1,2-Difluoro-benzen	18.0	9.0	1.0	98.5
						
829	2,6-Difluorobenzonitril	2,6-Difluoro benzonitrile	18.8	11.2	3.2	111.6
						
830	3,5-Difluorobenzonitril	3,5-Difluoro benzonitrile	18.8	11.2	3.2	111.6
						
273	1,1-Difluoroethan	1,1-Difluoro-ethan	14.9	10.2	3.0	69.5
						
274	1,1-Difluoroet ylene	1,1-Difluoro-ethen	15.0	6.8	3.6	58.2
						
823	2,4-Difluoronitrobenzen	2,4-Difluoro-1-nitro benzene	19.4	11.0	3.7	109.6
						
1069	Dihexyl Ether	1-Hexyloxy-hexane	16.0	3.0	2.8	235.8
						
275	Dihexyl Phthalate	Phthalic acid dihexyl ester	17.0	7.6	3.6	332.3
						

TABLE A.1 (CONTINUED)

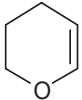
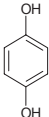
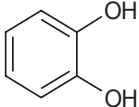
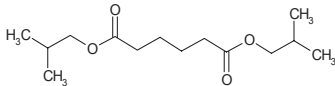
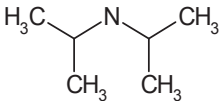
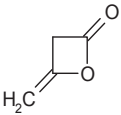
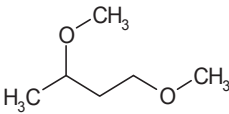
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
276	Dihydrogen Disulfid HS—SH		17.3	6.3	10.7	49.6
277	Dihhydropyran 	3,4-Dihydro-2H-pyran	17.5	5.5	5.7	91.2
917	1,4-Dihydroxybenzene (1,4-Benzenediol) 	Benzene-1,4-diol	21.0	10.2	27.2	82.7
904	1,2-Dihydroxybenzene (Catechol) 	Benzene-1,2-diol	20.0	11.3	21.8	81.9
1140	Diisobutyl Adipate 	Hexanedioic acid diisobutyl ester	16.7	2.5	6.2	269.6
1084	Diisopropylamine 	Diisopropyl-amine	14.8	1.7	3.5	141.1
283	Diketene 	4-Methylene-oxetan-2-one	16.2	15.1	8.1	75.9
763	1,3-Dimethoxy Butane 	1,3-Dimethoxy-butane	15.6	5.5	5.2	140.0

TABLE A.1 (CONTINUED)

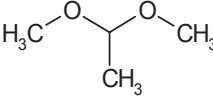
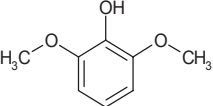
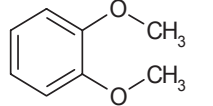
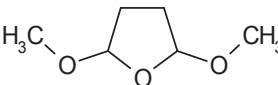
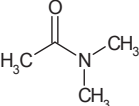
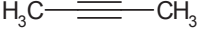
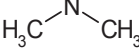
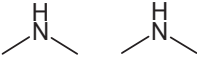
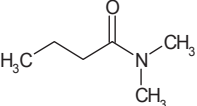
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
284	1,1-Dimethoxy Ethane	1,1-Dimethoxy-ethane	15.1	4.9	4.9	106.7
						
1108	2,6-Dimethoxy Phenol	2,6-Dimethoxy-phenol	19.3	7.6	13.7	136.4
						
889	1,2-Dimethoxybenzene	1,2-Dimethoxy-benzene	19.2	4.4	9.4	127.7
						
988	2,5-Dimethoxytetrahydrofuran	2,5-Dimethoxy-tetrahydro-furan	16.8	7.2	8.2	129.6
						
285	N,N-Dimethyl Acetamide	N,N-Dimethyl-acetamide	16.8	11.5	10.2	92.5
						
286	Dimethyl Acetylene	But-2-yne	15.1	3.4	7.6	78.9
						
287	Dimethyl Amine	Dimethyl-amine	15.3	4.8	11.2	66.2
						
288	Dimethyl Amine-Dimer	Dimethyl-amine; compound with dimethyl-amine	15.3	4.8	7.9	132.4
						
289	N,N-Dimethyl Butyramide	N,N-Dimethyl-butylamide	16.4	10.6	7.4	127.8
						

TABLE A.1 (CONTINUED)

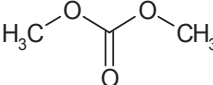
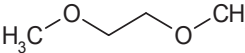
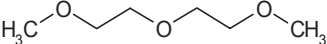
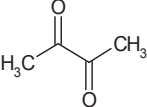
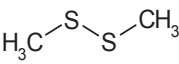
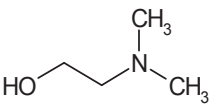
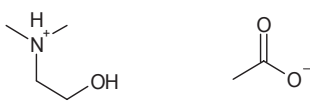
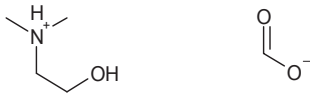
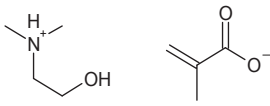
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
290	Dimethyl Carbonate	Carbonic acid dimethyl ester	15.5	3.9	9.7	84.2
						
291	Dimethyl Cellosolve	1,2-Dimethoxy-ethane	15.4	6.0	6.0	104.5
						
292	Dimethyl Diethylene Glycol	1-Methoxy-2-(2- methoxy-ethoxy)- ethane	15.8	6.1	9.2	142.0
						
293	Dimethyl Diketone	Butane-2,3-dione	15.7	5.3	11.7	88.2
						
294	Dimethyl Disulfid	Methyldisulfanyl- methane	17.3	7.8	6.5	88.6
						
295	Dimethyl Ethanolamine	2-Dimethylamino- ethanol	16.1	9.2	15.3	101.1
						
1189	Dimethyl Ethanolamine/Acetic Acid	Acetate (2-hydroxy- ethyl)-dimethyl- ammonium;	16.8	19.8	19.8	
						
1188	Dimethyl Ethanolamine/Formic Acid	Formate (2-hydroxy- ethyl)-dimethyl- ammonium;	17.2	21.5	22.5	
						
1185	Dimethyl Ethanolamine/Methacrylic Acid	2-Methyl-acrylate (2- hydroxy-ethyl)- dimethyl-ammonium;	17.2	18.8	17.6	
						

TABLE A.1 (CONTINUED)

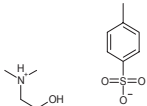
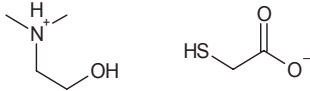
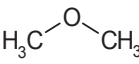
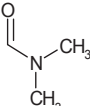
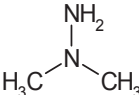
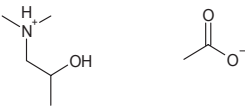
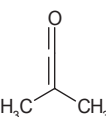
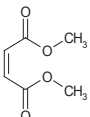
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1186	Dimethyl Ethanolamine/p-Toluene Sulfonic Acid	Toluene-4-sulfonate (2-hydroxy-ethyl)-dimethyl-ammonium;	17.2	21.5	22.5	
						
1187	Dimethyl Ethanolamine/Thioglycolic Acid	Mercapto-acetate (2-hydroxy-ethyl)-dimethyl-ammonium;	17.2	21.5	22.5	
						
296	Dimethyl Ether	Methoxymethane	15.2	6.1	5.7	63.2
						
297	Dimethyl Formamide	N,N-Dimethyl-formamide	17.4	13.7	11.3	77.0
						
298	1,1-Dimethyl Hydrazine	N,N-Dimethyl-hydrazine	15.3	5.9	11.0	76.0
						
1190	Dimethyl Isopropanol Amine/Acetic Acid	Acetate (2-hydroxy-propyl)-dimethyl-ammonium;	16.6	19.4	18.0	
						
299	Dimethyl Ketene	2-Methyl-propen-1-one	15.2	7.4	4.8	87.6
						
1145	Dimethyl Maleate	(Z)-But-2-enedioic acid dimethyl ester	16.3	8.3	9.8	125.8
						

TABLE A.1 (CONTINUED)

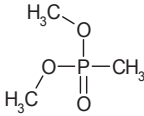
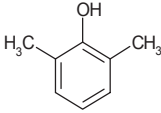
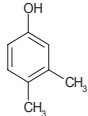
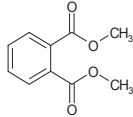
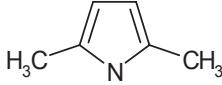
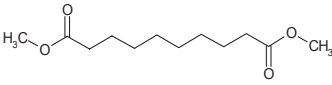
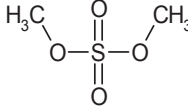
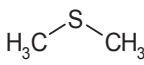
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
839	Dimethyl Methyl Phosphonate	Methyl-phosphonic acid dimethyl ester	16.7	13.1	7.5	106.9
						
849	2,6-Dimethyl Phenol	2,6-Dimethyl-phenol	19.1	4.9	12.9	116.3
						
850	3,4-Dimethyl Phenol	3,4-Dimethyl-phenol	19.2	6.0	13.4	121.0
						
300	Dimethyl Phthalate	Phthalic acid dimethyl ester	18.6	10.8	4.9	163.0
						
782	2,5-Dimethyl Pyrrole	2,5-Dimethyl-1H- pyrrole	18.3	7.6	6.8	101.8
						
1144	Dimethyl Sebacate	Decanedioic acid dimethyl ester	16.6	2.9	6.7	233.3
						
1080	Dimethyl Sulfate	Sulfuric acid dimethyl ester	17.7	17.0	9.7	94.7
						
301	Dimethyl Sulfid	Methylsulfanyl-methane	16.1	6.4	7.4	73.2
						

TABLE A.1 (CONTINUED)

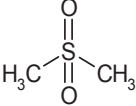
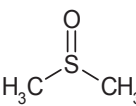
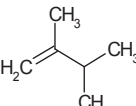
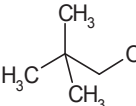
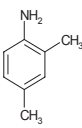
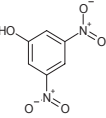
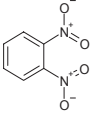
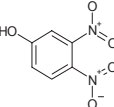
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
302	Dimethyl Sulfone	Methylsulfonyl-methane	19.0	19.4	12.3	75.0
						
303	Dimethyl Sulfoxide	Methylsulfi yl-methane	18.4	16.4	10.2	71.3
						
304	2,3-Dimethyl-1-Butene	2,3-Dimethyl-but-1-ene	14.9	1.2	2.8	125.2
						
1027	2,2-Dimethyl-1-Propanol (Neopentyl alc.)	2,2-Dimethyl-propan-1-ol	15.6	6.5	13.5	107.5
						
952	2,4-Dimethylaniline	2,4-Dimethyl-phenylamine	19.2	5.2	8.7	123.7
						
942	3,5-Dinitrophenol	3,5-Dinitro-phenol	19.5	12.9	14.4	108.2
						
940	1,2-Dinitrobenzene	1,2-Dinitro-benzene	20.6	22.7	5.4	107.4
						
941	3,4-Dinitrophenol	3,4-Dinitro-phenol	19.5	12.8	14.3	110.1
						

TABLE A.1 (CONTINUED)

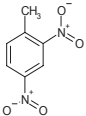
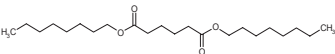
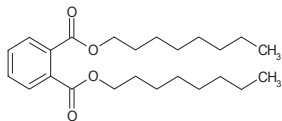
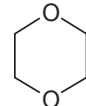
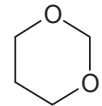
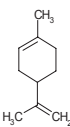
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
938	2,4-Dinitrotoluene 	1-Methyl-2,4-dinitro- benzene	20.0	13.1	4.9	137.9
1142	Diethyl Adipate Commercial 	Hexanedioic acid diethyl ester	16.7	2.0	5.1	400.0
305	Diethyl Phthalate Commercial 	Phthalic acid diethyl ester	16.6	7.0	3.1	377.0
306	1,4-Dioxane 	[1,4]Dioxane	19.0	1.8	7.4	85.7
307	1,3-Dioxolane 	[1,3]Dioxane	18.1	6.6	9.3	69.9
1197	Dipentene 	4-Isopropenyl-1-methyl- cyclohexene	17.2	1.8	4.3	162.9
1166	Diphenyl Acetylene P from 0 Dipole Moment 	(Phenylethynyl)benzene	20.1	0	3.2	184.3
1167	Diphenyl Acetylene P from Group Cont. 	(Phenylethynyl)benzene	20.1	2.0	3.2	184.3

TABLE A.1 (CONTINUED)

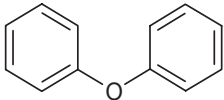
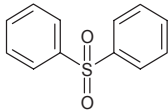
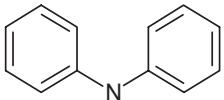
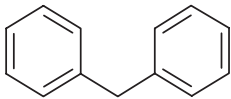
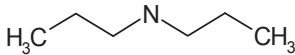
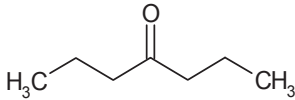
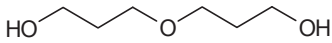
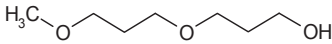
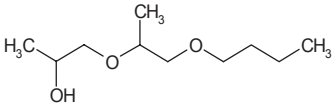
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
778	Diphenyl Ether*	Diphenyl ether	19.5	3.4	5.8	160.4
						
779	Diphenyl Sulfone	(Phenylsulfonyl)benzene	21.1	14.4	3.4	174.3
						
875	Diphenylamine	Diphenyl-amine	20.0	3.3	5.9	145.9
						
876	Diphenylmethane	Benzylbenzene	19.5	1.0	1.0	168.2
						
308	Dipropyl Amine	Dipropyl-amine	15.3	1.4	4.1	136.9
						
737	Dipropyl Ketone*	Heptan-4-one	15.8	5.7	4.9	140.8
						
309	Dipropylene Glycol Commercial	3-(3-Hydroxy-propoxy)- propan-1-ol	16.5	10.6	17.7	130.9
						
310	Dipropylene Glycol Methyl Ether Commercial	3-(3-Methoxy-propoxy)- propan-1-ol	15.5	5.7	11.2	157.4
						
1007	Dipropylene Glycol Mono n-Butyl Ether	1-(2-Butoxy-1-methyl- ethoxy)-propan-2-ol	15.7	6.5	10.0	211.2
						

TABLE A.1 (CONTINUED)

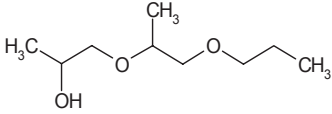
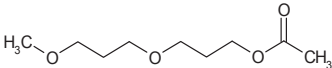
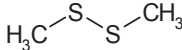
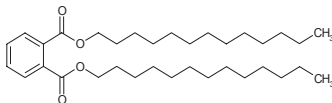
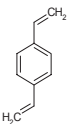
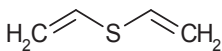
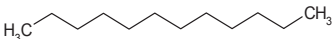
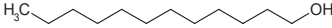
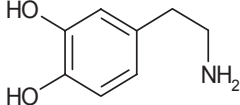
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1008	Dipropylene Glycol Mono n-Propyl Ether	1-(1-Methyl-2-propoxy-ethoxy)-propan-2-ol	15.6	6.1	11.0	185.6
						
311	Dipropylene Glycol Monomethyl Ether Acetate	Acetic acid 3-(3-methoxy-propoxy)-propyl ester	16.3	4.9	8.0	195.7
						
312	2,3-Dithiabutane	Methyldisulfanyl-methane	17.3	7.8	6.5	88.6
						
313	Ditridecyl Phthalate	Phthalic acid ditridecyl ester	16.6	5.4	1.9	558.3
						
314	p-Divinyl Benzene	1,4-Divinyl-benzene	18.6	1.0	7.0	142.8
						
315	Divinyl Sulfid	Vinylsulfanyl-ethene	16.5	4.6	5.6	93.6
						
316	Dodecane	Dodecane	16.0	0	0	228.6
						
1075	Dodecanol	Dodecan-1-ol	16.0	4.0	9.3	224.5
						
1180	Dopamine	4-(2-Amino-ethyl)-benzene-1,2-diol	18.2	10.3	19.5	183.0
						

TABLE A.1 (CONTINUED)

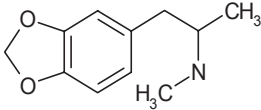
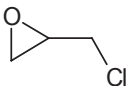
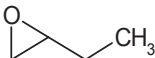
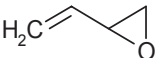
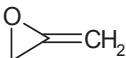
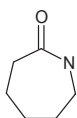
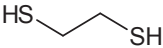
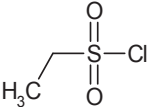
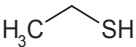
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1177	Ecstasy 	(2-Benzo[1,3]dioxol-5-yl-1-methyl-ethyl)-methyl-amine	18.0	5.1	6.1	202.9
317	Eicosane 	Icosane	16.5	0	0	359.8
318	Epichlorohydrin 	2-Chloromethyl-oxirane	18.9	7.6	6.6	78.4
987	1,2-Epoxy Butane 	2-Ethyl-oxirane	15.6	8.0	4.6	87.9
320	3,4-Epoxy-1-Butene 	2-Vinyl-oxirane	16.6	7.7	7.4	80.7
319	1,2-Epoxy-2-Propene 	2-Methylene-oxirane	16.5	8.6	6.7	70.0
321	Epsilon-Caprolactam 	Azepan-2-one	19.4	13.8	3.9	110.7
322	1,2-Ethane Dithiol 	Ethane-1,2-dithiol	17.9	7.2	8.7	83.9
323	Ethanesulfonylchloride 	Ethanesulfonyl chloride	17.7	14.9	6.8	94.7
324	Ethanethiol (Ethyl Mercaptan) 	Ethanethiol	15.7	6.5	7.1	74.3

TABLE A.1 (CONTINUED)

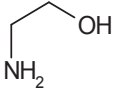
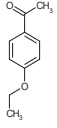
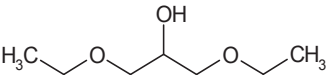
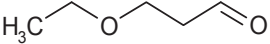
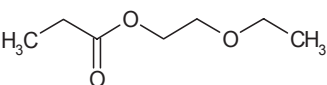
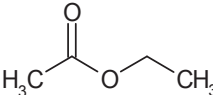
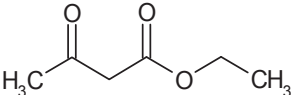
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
325	Ethanol	Ethanol	15.8	8.8	19.4	58.5
						
326	Ethanolamine	2-Amino-ethanol	17.0	15.5	21.2	59.8
						
1191	Ethanolamine/Acetic Acid	Acetate 2-hydroxy-ethyl-ammonium;	17.2	20.3	18.4	
						
794	4-Ethoxy Acetophenone	1-(4-Ethoxy-phenyl)-ethanone	18.8	10.3	6.4	162.6
						
776	1-Ethoxy Ethoxy-2-Propanol	1,3-Diethoxy-propan-2-ol	15.9	5.7	11.7	156.0
						
761	3-Ethoxy Propionaldehyde	3-Ethoxy-propionaldehyde	16.0	8.8	7.4	112.1
						
327	Ethoxyethyl Propionate	Propionic acid 2-ethoxy-ethyl ester	16.2	3.3	8.8	155.5
						
328	Ethyl Acetate	Acetic acid ethyl ester	15.8	5.3	7.2	98.5
						
1016	Ethyl Aceto Acetate (Keto)	3-Oxo-butyric acid ethyl ester	16.5	10.8	8.3	125.6
						

TABLE A.1 (CONTINUED)

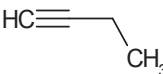
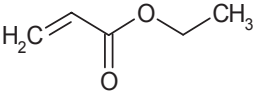
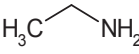
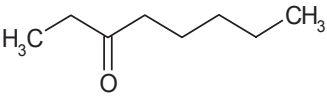
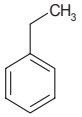
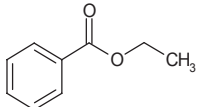
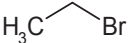
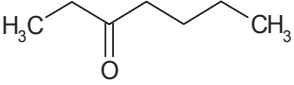
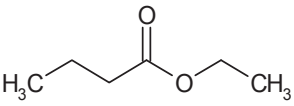
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
329	Ethyl Acetylene 	But-1-yne	15.1	3.4	5.0	81.5
330	Ethyl Acrylate 	Acrylic acid ethyl ester	15.5	7.1	5.5	108.8
331	Ethyl Amine 	Ethylamine	15.0	5.6	10.7	65.6
332	Ethyl Amyl Ketone 	Octan-3-one	16.2	4.5	4.1	156.0
333	Ethyl Benzene 	Ethyl-benzene	17.8	0.6	1.4	123.1
954	Ethyl Benzoate 	Benzoic acid ethyl ester	17.9	6.2	6.0	144.3
334	Ethyl Bromide 	Bromo-ethane	16.5	8.4	2.3	74.6
335	Ethyl Butyl Ketone 	Heptan-3-one	16.2	5.0	4.1	139.0
928	Ethyl Butyrate 	Butyric acid ethyl ester	15.5	5.6	5.0	132.9

TABLE A.1 (CONTINUED)

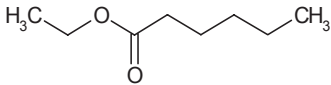
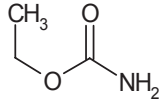
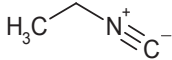
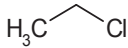
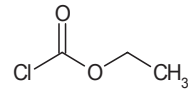
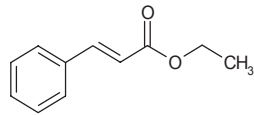
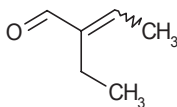
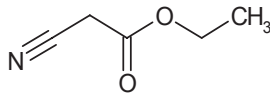
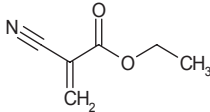
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
929	Ethyl Caproate 	Hexanoic acid ethyl ester	15.5	3.2	5.9	149.6
336	Ethyl Carbamate 	Carbamic acid ethyl ester	16.8	10.1	13.0	91.2
337	Ethyl Carbylamine 	Isocyano-ethane	15.6	15.2	5.8	74.4
338	Ethyl Chloride 	Chloro-ethane	15.7	6.1	2.9	70.0
339	Ethyl Chloroformate 	Ethyl chloridocarbonate	15.5	10.0	6.7	95.6
340	Ethyl Cinnamate 	(E)-3-Phenyl-acrylic acid ethyl ester	18.4	8.2	4.1	166.8
341	2-Ethyl Croton Aldehyde 	2-Ethyl-but-2-enal	16.1	8.0	5.5	115.2
1042	Ethyl Cyanoacetate 	Cyano-acetic acid ethyl ester	16.7	7.9	8.3	107.1
342	Ethyl Cyanoacrylate 	2-Cyano-acrylic acid ethyl ester	15.2	10.3	9.0	117.1

TABLE A.1 (CONTINUED)

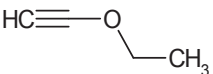
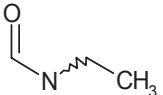
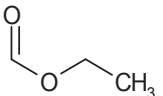
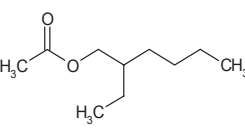
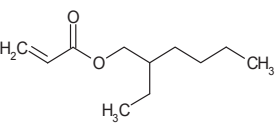
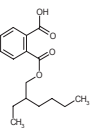
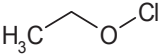
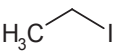
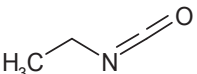
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
343	Ethyl Ethynylether 	Ethoxy-ethyne	15.4	7.9	5.9	87.6
966	N-Ethyl Formamide 	N-Ethyl-formamide	17.2	10.0	14.0	76.5
344	Ethyl Formate 	Formic acid ethyl ester	15.5	8.4	8.4	80.2
346	2-Ethyl Hexyl Acetate 	Acetic acid 2-ethyl-hexyl ester	15.8	2.9	5.1	196.0
347	2-Ethyl Hexyl Acrylate 	Acrylic acid 2-ethyl-hexyl ester	14.8	4.7	3.4	208.2
1153	Mono 2-Ethyl-Hexyl Phthalate (MEHP) 	Phthalic acid mono-(2-ethyl-hexyl) ester	17.3	6.2	6.8	265.0
348	Ethyl Hypochlorite 	Ethyl hypochlorite	15.7	8.6	6.5	79.5
349	Ethyl Iodide 	Iodo-ethane	17.3	7.9	7.2	81.2
350	Ethyl Isocyanate 	Isocyanato-ethane	15.4	12.0	2.5	78.7

TABLE A.1 (CONTINUED)

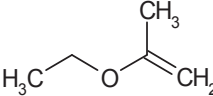
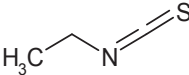
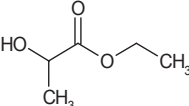
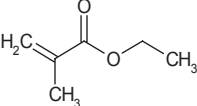
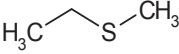
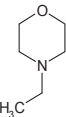
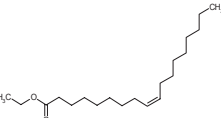
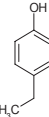
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
351	Ethyl Isopropenyl Ether	2-Ethoxy-propene	14.8	3.5	5.1	113.3
						
352	Ethyl Isothiocyanate	Isothiocyanato-ethane	17.2	14.7	9.0	87.1
						
353	Ethyl Lactate	2-Hydroxy-propionic acid ethyl ester	16.0	7.6	12.5	115.0
						
354	Ethyl Methacrylate	2-Methyl-acrylic acid ethyl ester	15.8	7.2	7.5	125.8
						
355	Ethyl Methyl Sulfid	Methylsulfanyl-ethane	17.1	4.8	2.5	91.2
						
1085	4-Ethyl Morpholine	4-Ethyl-morpholine	17.7	5.0	6.6	116.5
						
1117	Ethyl Oleate	(Z)-Octadec-9-enoic acid ethyl ester	14.5	3.8	3.7	357.3
						
1071	4-Ethyl Phenol	4-Ethyl-phenol	19.2	5.3	12.8	118.6
						

TABLE A.1 (CONTINUED)

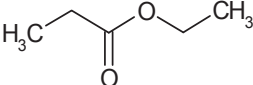
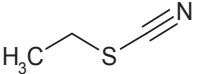
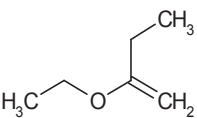
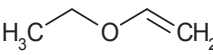
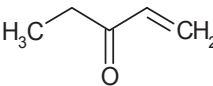
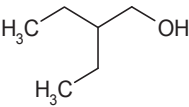
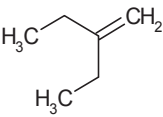
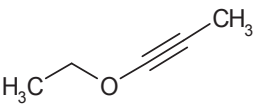
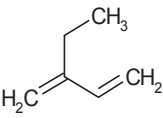
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1010	Ethyl Propionate	Propionic acid ethyl ester	15.5	6.1	4.9	115.5
						
356	Ethyl Thiocyanate	Thiocyanato-ethane	15.4	13.4	5.4	87.1
						
357	1-Ethyl Vinyl Ethyl Ether	2-Ethoxy-but-1-ene	15.3	4.2	4.6	125.6
						
358	Ethyl Vinyl ether	Ethoxy-ethene	14.9	4.9	5.6	94.9
						
359	Ethyl Vinylketone	Pent-1-en-3-one	15.8	11.3	4.5	99.3
						
360	2-Ethyl-1-Butanol	2-Ethyl-butan-1-ol	15.8	4.3	13.5	123.2
						
727	2-Ethyl-1-Butene	3-Methylene-pentane	14.9	1.7	3.5	123.1
						
361	Ethyl-1-Propynylether	1-Ethoxy-propyne	15.9	6.4	5.4	101.6
						
362	2-Ethyl-1,3-Butadiene	3-Methylene-pent-1-ene	15.3	1.6	3.8	115.2
						

TABLE A.1 (CONTINUED)

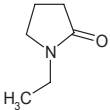
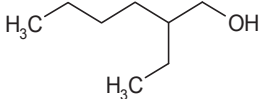
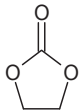
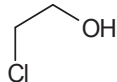
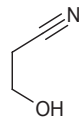
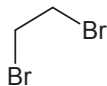
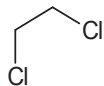
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
961	N-Ethyl-2-Pyrrolidone 	1-Ethyl-pyrrolidin-2-one	18.0	12.0	7.0	113.9
345	2-Ethyl-hexanol 	2-Ethyl-hexan-1-ol	15.9	3.3	11.8	156.6
1058	Ethylene $\text{H}_2\text{C}=\text{CH}_2$	Ethene	15.0	2.0	3.8	63.0
363	Ethylene Carbonate 	[1,3]Dioxolan-2-one	19.4	21.7	5.1	66.0
364	Ethylene Chlorohydrin 	2-Chloro-ethanol	16.9	8.8	17.2	67.3
365	Ethylene Cyanohydrin 	3-Hydroxy-propionitrile	17.2	18.8	17.6	68.3
366	Ethylene Dibromide 	1,2-Dibromo-ethane	19.2	3.5	8.6	87.0
367	Ethylene Dichloride 	1,2-Dichloro-ethane	19.0	7.4	4.1	79.4

TABLE A.1 (CONTINUED)

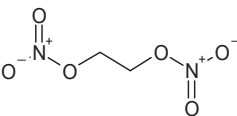
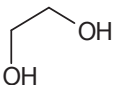
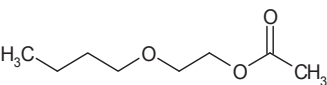
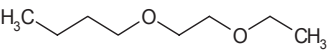
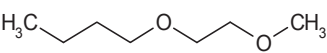
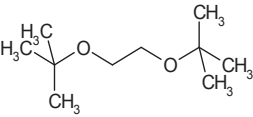
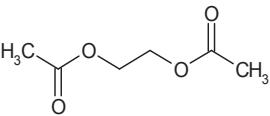
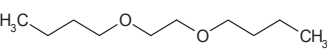
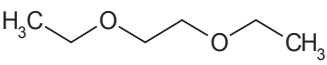
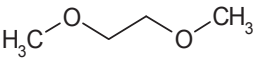
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
937	Ethylene Dinitrate	1,2-Bis-nitrooxy-ethane	16.2	13.0	4.5	101.7
						
368	Ethylene Glycol	Ethane-1,2-diol	17.0	11.0	26.0	55.8
						
369	Ethylene Glycol Butyl Ether Acetate	Acetic acid 2-butoxy-ethyl ester	15.3	4.5	8.8	171.2
						
752	Ethylene Glycol Butyl Ethyl Ether	1-(2-Ethoxy-ethoxy)-butane	15.3	4.9	4.6	175.5
						
753	Ethylene Glycol Butyl Methyl Ether	1-(2-Methoxy-ethoxy)-butane	15.5	5.2	4.9	157.2
						
370	Ethylene Glycol Di-tert-Butyl Ether	2-(2-tert-Butoxy-ethoxy)-2-methylpropane	14.7	4.1	8.2	210.0
						
371	Ethylene Glycol Diacetate	Acetic acid 2-acetoxy-ethyl ester	16.2	4.7	9.8	132.8
						
758	Ethylene Glycol Dibutyl Ether	1-(2-Butoxy-ethoxy)-butane	15.7	4.5	4.2	209.5
						
760	Ethylene Glycol Diethyl Ether	1,2-Diethoxy-ethane	15.4	5.4	5.2	141.6
						
775	Ethylene Glycol Dimethyl Ether	1,2-Dimethoxy-ethane	15.4	6.3	6.0	103.9
						

TABLE A.1 (CONTINUED)

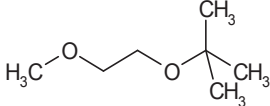
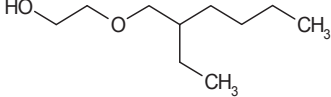
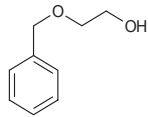
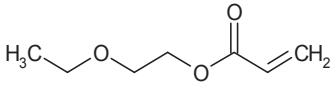
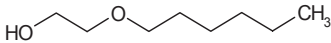
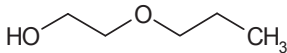
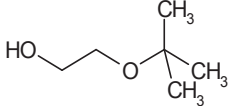
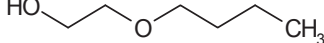
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
372	Ethylene Glycol Methyl t-Butyl Ether*	2-(2-Methoxy-ethoxy)- 2-methyl-propane	15.3	5.1	8.2	157.4
						
373	Ethylene Glycol Mono-2-Ethyl Hexyl Ether*	2-(2-Ethyl-hexyloxy)- ethanol	16.0	4.1	10.5	194.7
						
731	Ethylene Glycol Monobenzyl Ether	2-Benzyloxy-ethanol	17.8	5.9	12.2	143.0
						
771	Ethylene Glycol Monoethyl Ether Acrylate	Acrylic acid 2-ethoxy- ethyl ester	15.9	5.1	9.3	147.7
						
1056	Ethylene Glycol Mono n-Hexyl Ether	2-Hexyloxy-ethanol	16.0	5.0	11.4	164.5
						
1004	Ethylene Glycol Mono n-Propyl Ether	2-Propoxy-ethanol	16.1	8.7	13.5	114.1
						
374	Ethylene Glycol Mono-t-Butyl Ether Commercial	2-tert-Butoxy-ethanol	15.3	6.1	10.8	131.0
						
375	Ethylene Glycol Monobutyl Ether Commercial	2-Butoxy-ethanol	16.0	5.1	12.3	131.6
						

TABLE A.1 (CONTINUED)

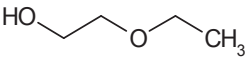
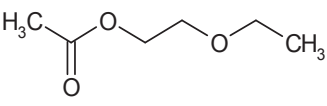
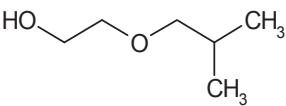
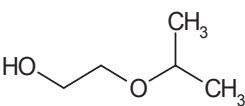
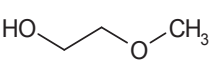
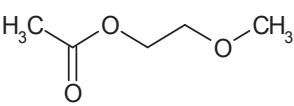
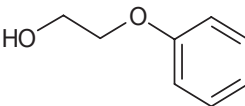
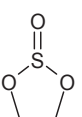
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
376	Ethylene Glycol Monoethyl Ether Commercial	2-Ethoxy-ethanol	16.2	9.2	14.3	97.8
						
377	Ethylene Glycol Monoethyl Ether Acetate Commercial	Acetic acid 2-ethoxy-ethyl ester	15.9	4.7	10.6	136.1
						
378	Ethylene Glycol Monoisobutyl Ether Commercial	2-Isobutoxy-ethanol	15.2	4.9	9.6	132.5
						
379	Ethylene Glycol Monoisopropyl Ether	2-Isopropoxy-ethanol	16.0	8.2	13.1	115.8
						
380	Ethylene Glycol Monomethyl Ether Commercial	2-Methoxy-ethanol	16.2	9.2	16.4	79.1
						
381	Ethylene Glycol Monomethyl Ether Acetate	Acetic acid 2-methoxy-ethyl ester	15.9	5.5	11.6	121.6
						
924	Ethylene Glycol Monophenyl Ether	2-Phenoxy-ethanol	17.0	7.2	12.3	125.3
						
789	Ethylene Glycol Sulfit	[1,3,2]Dioxathiolane 2-oxide	20.0	15.9	5.1	75.1
						

TABLE A.1 (CONTINUED)

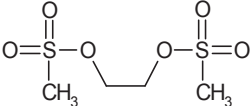
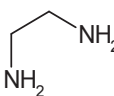
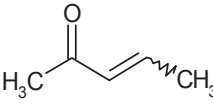
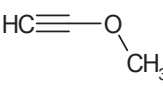
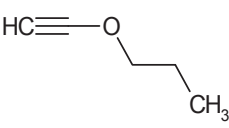
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
382	Ethylene Methyl Sulfonate	Methanesulfonic acid 2-methanesulfonyloxy-ethyl ester	16.9	9.3	9.6	97.9
						
383	Ethylene Oxide	Oxirane	15.6	10.0	11.0	49.9
						
384	Ethylene Sulfid	Thiirane	19.3	9.0	6.5	59.5
						
385	Ethylenediamine	Ethane-1,2-diamine	16.6	8.8	17.0	67.3
						
386	Ethyleneimine	Aziridine	18.6	9.8	7.7	51.8
						
946	2-Ethylhexylamine	Ethyl-hexyl-amine	15.7	4.2	6.1	163.3
						
387	Ethylidene Acetone	Pent-3-en-2-one	16.2	12.1	4.5	99.0
						
389	Ethynyl Methyl Ether	Methoxy-ethyne	15.8	8.1	6.5	70.1
						
388	Ethynyl Propyl Ether	1-Ethynyloxy-propane	15.4	3.8	5.4	104.1
						

TABLE A.1 (CONTINUED)

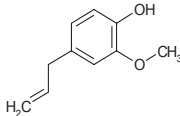
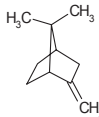
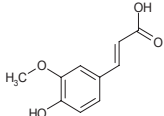
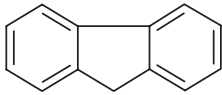
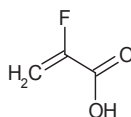
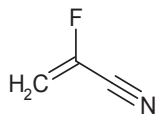
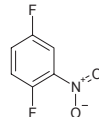
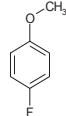
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1109	Eugenol 	4-Allyl-2-methoxy- phenol	19.0	7.5	13.0	154.0
898	Fenchene(Alfa) 	7,7-Dimethyl-2- methylene- bicyclo[2.2.1]heptane	16.9	1.5	3.1	157.3
1104	Ferulic Acid 	(E)-3-(4-Hydroxy-3- methoxy-phenyl)- acrylic acid	19.0	6.6	15.1	155.5
877	Fluorene 	9H-Fluorene	20.0	1.7	1.7	138.2
390	1-Fluoro Acrylic Acid 	2-Fluoro-acrylic acid	16.0	8.7	13.0	90.0
391	1-Fluoro Acrylonitrile 	2-Fluoro-acrylonitrile	14.1	15.4	5.7	88.8
827	4-Fluoro-3-Nitrobenzofluorid 	1,4-Difluoro-2-nitro benzene	19.4	7.2	3.7	108.4
799	p-Fluoroanisole 	1-Fluoro-4-methoxy- benzene	18.7	7.3	6.7	113.2

TABLE A.1 (CONTINUED)

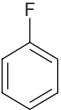
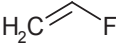
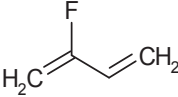
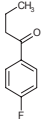
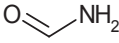
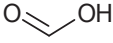
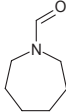
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
392	Fluorobenzene	Fluoro-benzene	18.7	6.1	2.0	94.7
						
393	Fluoroethylene	Fluoro-ethene	15.2	7.0	1.0	57.5
						
394	Fluoromethane	Fluoromethane	13.4	10.6	9.5	40.7
						
395	Fluoroprene	2-Fluoro-buta-1,3-diene	14.2	5.8	1.0	85.5
						
820	4-Fluoropropiophenone	1-(4-Fluoro-phenyl)- butan-1-one	19.6	7.1	3.5	138.8
						
396	Formaldehyde	Formaldehyde	12.8	14.4	15.4	36.8
						
397	Formamide	Formamide	17.2	26.2	19.0	39.8
						
398	Formic Acid	Formic acid	14.3	11.9	16.6	37.8
						
741	Formyl Fluoride	Formyl fluorid	15.0	10.1	8.6	56.5
						
400	N-Formyl Hexamethylene Imine	Azepane-1-carbaldehyde	18.5	10.4	7.6	127.0
						

TABLE A.1 (CONTINUED)

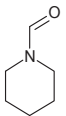
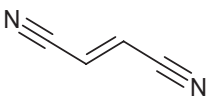
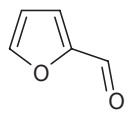
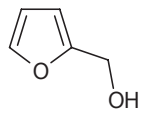
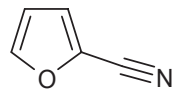
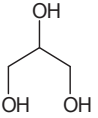
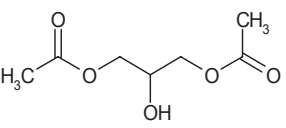
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
401	N-Formyl Piperidine 	Piperidine-1-carbaldehyde	18.7	10.6	7.8	111.5
402	Fumaronitrile 	(E)-But-2-enedinitrile	16.7	13.6	7.8	83.0
403	Furan 	Furan	17.8	1.8	5.3	72.5
404	Furfural 	Furan-2-carbaldehyde	18.6	14.9	5.1	83.2
405	Furfuryl Alcohol 	Furan-2-yl-methanol	17.4	7.6	15.1	86.5
994	2-Furonitrile 	Furan-2-carbonitrile	18.4	15.0	8.2	87.5
406	Glycerol 	Propane-1,2,3-triol	17.4	12.1	29.3	73.3
1222	Glycerol Diacetate (Isomer Mix) 	Acetic acid 3-acetoxy-2-hydroxy-propyl ester	16.4	8.9	14.2	149.4

TABLE A.1 (CONTINUED)

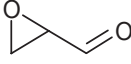
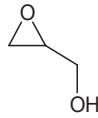
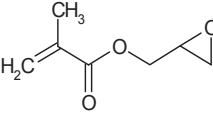
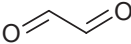
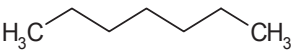
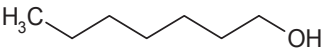
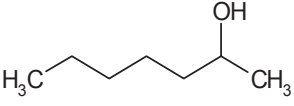
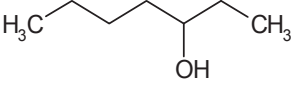
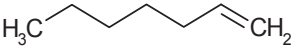
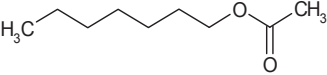
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
990	Glycidaldehyde*	Oxirane-2-carbaldehyde	17.5	13.4	9.8	63.2
						
1087	Glycidol	Oxiranyl-methanol	18.2	9.0	17.9	66.5
						
407	Glycidyl Methacrylate	2-Methyl-acrylic acid oxiranylmethyl ester	16.3	8.5	5.7	136.4
						
408	Glyoxal (Ethandial)	Ethanedial	15.0	17.0	13.3	50.9
						
409	Heptane	Heptane	15.3	0	0	147.4
						
931	1-Heptanol	Heptan-1-ol	16.0	5.3	11.7	141.4
						
1044	2-Heptanol	Heptan-2-ol	15.7	5.4	11.7	142.9
						
1013	3-Heptanol	Heptan-3-ol	15.9	5.4	11.7	141.2
						
410	1-Heptene	Hept-1-ene	15.0	1.1	2.6	141.9
						
411	n-Heptyl Acetate	Acetic acid heptyl ester	15.8	2.9	5.5	181.1
						

TABLE A.1 (CONTINUED)

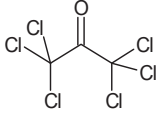
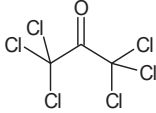
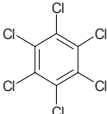
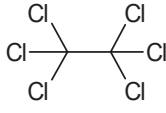
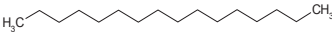
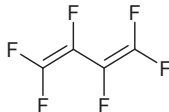
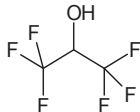
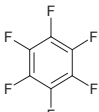
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
706	Hexachloroacetone P Based on 0 Dipole Moment	1,1,1,3,3,3-Hexachloro- propan-2-one	18.3	0	0	151.9
						
707	Hexachloroacetone P based on group contributions	1,1,1,3,3,3-Hexachloro- propan-2-one	18.3	6.6	6.4	151.9
						
1128	Hexachlorobenzene (Lindane)	1,2,3,4,5,6-Hexachloro- benzene	21.9	2.1	0	139.3
						
1121	Hexachloroethane	1,1,1,2,2,2-Hexachloro- ethane	22.0	4.7	0	113.2
						
412	Hexadecane	Hexadecane	16.3	0	0	294.1
						
413	Hexafluoro 1,3-Butadiene	1,1,2,3,4,4-Hexafluoro buta-1,3-diene	13.8	0	0	104.3
						
414	Hexafluoro Isopropanol	1,1,1,3,3,3-Hexafluoro propan-2-ol	17.2	4.5	14.7	105.3
						
1119	Hexafluorobenzene	1,2,3,4,5,6-Hexafluoro benzene	16.9	0	0	115.8
						

TABLE A.1 (CONTINUED)

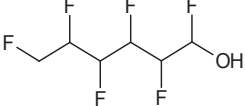
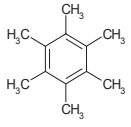
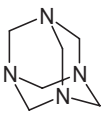
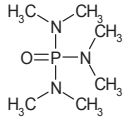
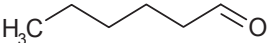
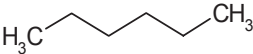
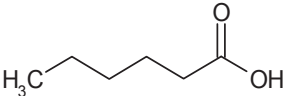
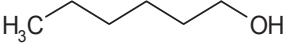
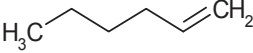
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
415	Hexafluoro hexanol	1,2,3,4,5,6-Hexafluoro hexan-1-ol	15.1	4.4	9.9	123.3
						
1123	Hexamethyl Benzene	1,2,3,4,5,6-Hexamethyl- benzene	19.2	1.6	0	152.7
						
948	Hexamethylenetetramine	1,3,5,7-Tetraaza- tricyclo[3.3.1.1*3,7*] decane	19.4	11.6	13.8	105.3
						
416	Hexamethylphosphoramide		18.5	8.6	11.3	175.7
						
1097	Hexanal	Hexanal	15.8	8.5	5.4	120.2
						
417	n-Hexane	n-Hexane	14.9	0	0	131.6
						
1022	Hexanoic Acid	Hexanoic acid	15.0	4.1	9.4	125.9
						
930	1-Hexanol	Hexan-1-ol	15.9	5.8	12.5	124.9
						
418	1-Hexene*	Hex-1-ene	14.7	1.1	0	126.1
						

TABLE A.1 (CONTINUED)

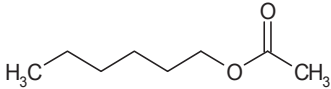
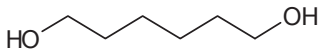
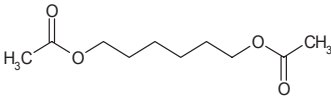
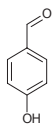
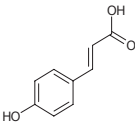
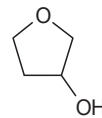
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
419	Hexyl Acetate	Acetic acid hexyl ester	15.8	2.9	5.9	165.0
						
420	Hexylene Glycol	Hexane-1,6-diol	15.7	8.4	17.8	123.0
						
421	Hexylene Glycol Diacetate	Acetic acid 6-acetoxy- hexyl ester	15.3	4.5	7.2	204.3
						
422	Hydrazine	Hydrazine	14.2	8.3	8.9	32.1
	$\text{H}_2\text{N}-\text{NH}_2$					
423	Hydrogen Cyanide	Nitrilomethane	12.3	17.6	9.0	39.3
	$\text{HC}\equiv\text{N}$					
1173	Hydrogen Peroxide	Hydrogen peroxide	15.5	12.2	42.7	23.2
	$\text{HO}-\text{OH}$					
424	Hydrogen Sulfid		17.9	6.0	10.2	36.1
	SH_2					
905	4-Hydroxy Benzaldehyde	4-Hydroxy- benzaldehyde	19.4	15.2	14.5	108.2
						
1105	4-Hydroxy Cinnamic Acid	(E)-3-(4-Hydroxy- phenyl)-acrylic acid	19.1	6.7	15.9	128.3
						
991	3-Hydroxy Tetrahydrofuran	Tetrahydro-furan-3-ol	18.9	9.4	16.3	80.0
						

TABLE A.1 (CONTINUED)

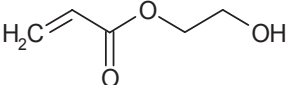
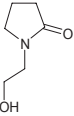
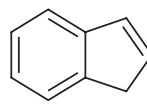
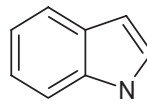
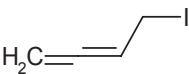
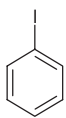
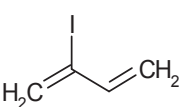
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
425	Hydroxyethyl Acrylate 	Acrylic acid 2-hydroxy-ethyl ester	16.0	13.2	13.4	114.9
960	N-(2-Hydroxyethyl)-2-Pyrrolidone 	1-(2-Hydroxy-ethyl)-pyrrolidin-2-one	18.0	9.2	15.7	113.4
1078	Indene 	1H-Indene	18.7	2.6	9.0	116.5
831	Indole 	1H-Indole	19.8	7.5	6.5	110.1
426	4-Iodo-1,2-Butadiene 	4-Iodo-buta-1,2-diene	17.4	6.3	6.2	105.1
427	Iodobenzene* 	Iodo-benzene	19.5	6.0	6.1	114.4
1126	Iodoform 	Triiodo-methane	20.2	3.6	10.6	98.2
428	Iodoprene 	2-Iodo-buta-1,3-diene	17.2	2.5	6.2	104.2

TABLE A.1 (CONTINUED)

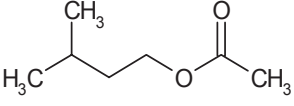
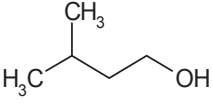
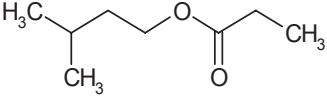
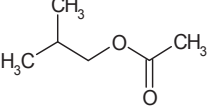
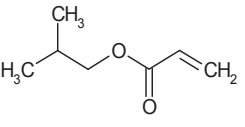
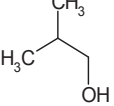
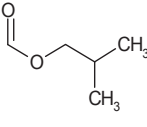
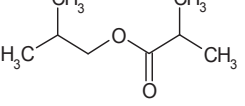
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
429	Isoamyl Acetate	Acetic acid 3-methyl- butyl ester	15.3	3.1	7.0	148.8
						
754	Isoamyl Alcohol (3-Methyl-1-Butanol)	3-Methyl-butan-1-ol	15.8	5.2	13.3	109.4
						
925	Isoamyl Propionate	Propionic acid 3-methyl- butyl ester	15.7	5.2	5.6	165.7
						
430	Isobutyl Acetate	Acetic acid isobutyl ester	15.1	3.7	6.3	133.5
						
770	Isobutyl Acrylate	Acrylic acid isobutyl ester	15.5	6.2	5.0	145.0
						
431	Isobutyl Alcohol	2-Methyl-propan-1-ol	15.1	5.7	15.9	92.8
						
1035	Isobutyl Formate	Formic acid isobutyl ester	15.5	6.5	6.7	117.4
						
432	Isobutyl Isobutyrate	Isobutyric acid isobutyl ester	15.1	2.9	5.9	163.0
						

TABLE A.1 (CONTINUED)

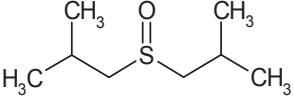
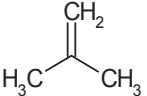
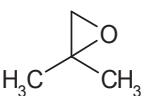
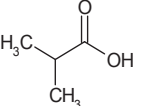
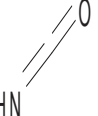
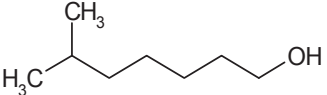
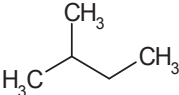
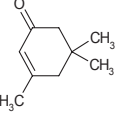
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
786	Isobutyl Sulfoxide 	2-Methyl-1-(2-methyl- propane-1-sulfi yl)- propane	16.3	10.5	6.1	195.0
433	Isobutylene 	Isobutene	14.5	2.0	1.5	89.4
434	Isobutyleneoxide 	2,2-Dimethyl-oxirane	16.1	4.8	5.8	90.0
1039	Isobutyric Acid 	Isobutyric acid	16.5	5.4	11.1	93.4
435	Isocyanic Acid 	Imino-methanone	15.8	10.5	13.6	37.7
436	Isooctyl Alcohol 	6-Methyl-heptan-1-ol	14.4	7.3	12.9	156.6
437	Isopentane 	2-Methyl-butane	13.7	0	0	117.4
438	Isophorone 	3,5,5-Trimethyl- cyclohex-2-enone	16.6	8.2	7.4	150.5

TABLE A.1 (CONTINUED)

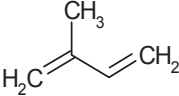
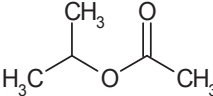
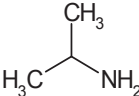
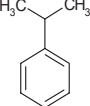
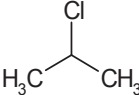
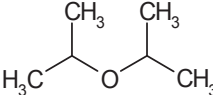
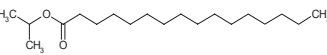
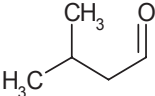
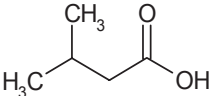
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
439	Isoprene (2-Methyl-1,3-Butadiene)	2-Methyl-buta-1,3-diene	14.7	1.4	4.1	100.9
						
440	Isopropyl Acetate	Acetic acid isopropyl ester	14.9	4.5	8.2	117.1
						
441	Isopropyl Amine (2-Propan Amine)	Isopropylamine	14.8	4.4	6.6	86.8
						
878	Isopropyl Benzene (Cumene)	Isopropyl-benzene	18.1	1.2	1.2	139.1
						
442	Isopropyl Chloride (2-Chloro Propane)	2-Chloro-propane	15.9	8.3	2.1	91.7
						
443	Isopropyl Ether	2-Isopropoxy-propane	13.7	3.9	2.3	140.9
						
444	Isopropyl Palmitate	Hexadecanoic acid isopropyl ester	14.3	3.9	3.7	330.0
						
445	Isovaleraldehyde	3-Methyl-butylaldehyde	14.7	9.5	5.0	106.0
						
1040	Isovaleric Acid	3-Methyl-butyric acid	16.4	4.1	10.7	111.0
						

TABLE A.1 (CONTINUED)

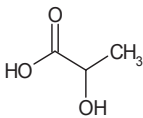
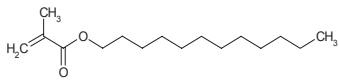
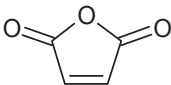
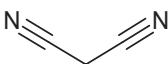
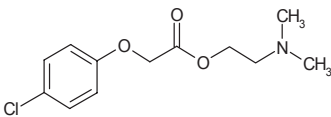
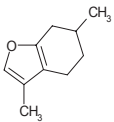
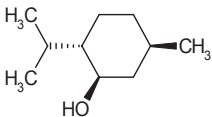
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
446	Isoxazole 	Isoxazole	18.8	13.4	11.2	64.1
447	Ketene $\text{H}_2\text{C}=\text{C}=\text{O}$	Ethenone	15.4	7.3	5.8	53.0
709	Lactic Acid (DL) 	2-Hydroxy-propionic acid	17.0	8.3	28.4	72.1
448	Lauryl Methacrylate 	2-Methyl-acrylic acid dodecyl ester	14.4	2.2	5.1	293.1
980	Maleic Anhydride 	Furan-2,5-dione	20.2	18.1	12.6	66.3
449	Malononitrile 	Malononitrile	17.7	18.4	6.7	55.5
1208	Meclofenoxate (Base Only) 	(4-Chloro-phenoxy)- acetic acid 2- dimethylamino-ethyl ester	16.0	6.2	9.0	198.3
1129	Menthofuran 	3,6-Dimethyl-4,5,6,7- tetrahydro-benzofuran	17.2	4.6	5.6	158.0
1132	L-Menthol 	(1R,2S,5R)-2-Isopropyl- 5-methyl-cyclohexanol	16.6	4.7	10.6	175.6

TABLE A.1 (CONTINUED)

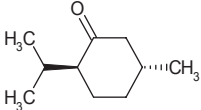
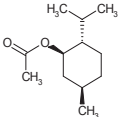
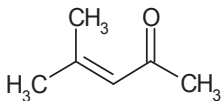
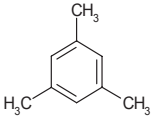
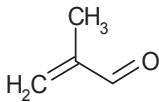
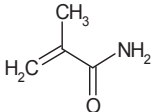
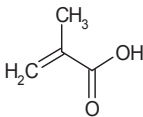
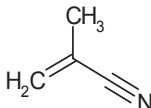
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1130	L-Menthone	(2S,5R)-2-Isopropyl-5-methyl-cyclohexanone	17.0	8.1	4.4	172.3
						
1131	L-Menthyl Acetate	Acetic acid (1R,2S,5R)-2-isopropyl-5-methyl-cyclohexyl ester	16.8	4.7	4.9	215.8
						
450	Mesityl Oxide	4-Methyl-pent-3-en-2-one	16.4	6.1	6.1	115.6
						
451	Mesitylene	Mesitylene	18.0	0	0.6	139.8
						
452	Methacrylaldehyde	2-Methyl-propenal	15.7	11.1	7.4	83.3
						
453	Methacrylamide	2-Methyl-acrylamide	15.8	11.0	11.6	76.0
						
454	Methacrylic Acid	2-Methyl-acrylic acid	15.8	2.8	10.2	84.8
						
455	Methacrylonitrile	2-Methyl-acrylonitrile	15.8	15.1	5.4	83.9
						

TABLE A.1 (CONTINUED)

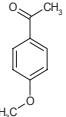
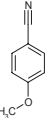
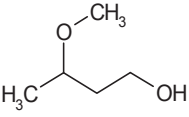
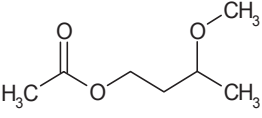
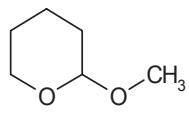
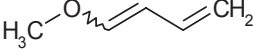
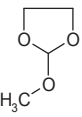
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
456	Methanol 	Methanol	15.1	12.3	22.3	40.7
793	4-Methoxy Acetophenone 	1-(4-Methoxy-phenyl)- ethanone	18.9	11.2	7.0	137.8
781	4-Methoxy Benzonitrile 	4-Methoxy-benzonitrile	19.4	16.7	5.4	121.0
1003	3-Methoxy Butanol 	3-Methoxy-butan-1-ol	15.3	5.4	13.6	113.2
1001	3-Methoxy Butyl Acetate 	Acetic acid 3-methoxy- butyl ester	15.3	4.1	8.1	153.9
996	2-Methoxy Tetrahydropyran 	2-Methoxy-tetrahydro- pyran	17.2	6.6	6.0	105.1
457	1-Methoxy-1,3-Butadiene 	1-Methoxy-buta-1,3- diene	15.5	8.3	5.4	101.8
992	2-Methoxy-1,3-Dioxolane 	2-Methoxy- [1,3]dioxolane	17.8	8.4	7.7	95.3

TABLE A.1 (CONTINUED)

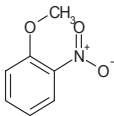
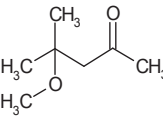
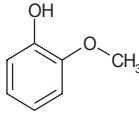
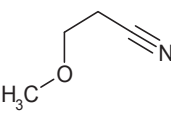
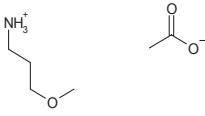
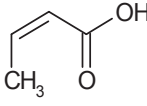
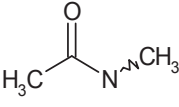
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1021	1-Methoxy-2-Nitrobenzene 	1-Methoxy-2-nitro- benzene	19.6	16.3	5.5	123.5
458	Methoxyhexanone (Pentoxone) 	4-Methoxy-4-methyl- pentan-2-one	15.3	6.0	5.9	143.5
459	o-Methoxyphenol (Guaiacol) 	2-Methoxy-phenol	18.0	8.2	13.3	109.5
460	3-Methoxypropionitrile 	3-Methoxy-propionitrile	16.6	14.4	7.8	91.1
1192	3-Methoxypropyl Amine/Acetic Acid 	Acetate 3-methoxy- propyl-ammonium;	17.2	22.5	23.5	
461	2-Methyl (cis) Acrylic Acid 	(Z)-But-2-enoic acid	16.8	5.2	12.4	83.9
463	Methyl 1-Propenyl Ether 	1-Methoxy-propene	15.0	4.3	5.7	93.5
958	N-Methyl Acetamide 	N-Methyl-acetamide	16.9	18.7	13.9	76.9

TABLE A.1 (CONTINUED)

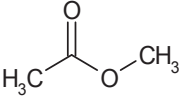
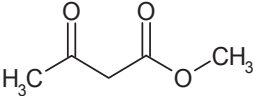
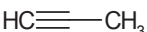
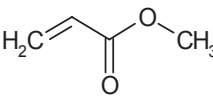
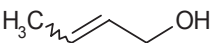
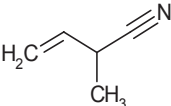
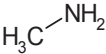
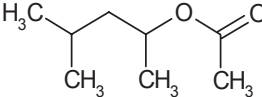
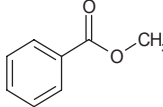
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
464	Methyl Acetate	Acetic acid methyl ester	15.5	7.2	7.6	79.7
						
1037	Methyl Aceto Acetate	3-Oxo-butyric acid methyl ester	16.4	8.6	8.9	108.3
						
465	Methyl Acetylene	Propyne	15.1	3.8	9.2	59.6
						
467	Methyl Acrylate	Acrylic acid methyl ester	15.3	6.7	9.4	90.3
						
468	3-Methyl Allyl Alcohol	But-2-en-1-ol	16.0	6.0	15.5	84.4
						
469	Methyl Allyl Cyanide	2-Methyl-but-3-enenitrile	16.4	11.3	5.1	97.7
						
470	Methyl Amine	Methylamine	13.0	7.3	17.3	44.4
						
471	Methyl Amyl Acetate	Acetic acid 1,3-dimethyl-butyl ester	15.2	3.1	6.8	167.4
						
472	Methyl Benzoate*	Benzoic acid methyl ester	18.9	8.2	4.7	124.9
						

TABLE A.1 (CONTINUED)

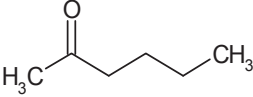
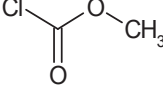
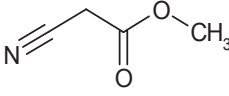
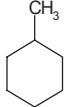
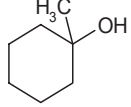
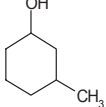
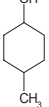
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
473	Methyl Bromide 	Bromomethane	17.0	8.8	2.6	56.8
474	Methyl Butyl Ketone 	Hexan-2-one	15.3	6.1	4.1	123.6
475	Methyl Chloride 	Chloromethane	15.3	6.1	3.9	55.4
476	Methyl Chloroformate 	Methyl chloridocarbonate	16.3	9.5	8.5	77.3
1041	Methyl Cyanoacetate 	Cyano-acetic acid methyl ester	16.8	14.8	9.1	88.3
477	Methyl Cyclohexane 	Methyl-cyclohexane	16.0	0	1.0	128.3
1030	1-Methyl Cyclohexanol 	1-Methyl-cyclohexanol	17.1	6.4	12.5	123.4
1031	3-Methyl Cyclohexanol (mix) 	3-Methyl-cyclohexanol	17.2	6.4	12.5	124.8
1032	4-Methyl Cyclohexanol (mix) 	4-Methyl-cyclohexanol	17.2	6.3	12.5	125.5

TABLE A.1 (CONTINUED)

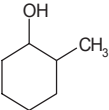
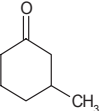
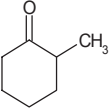
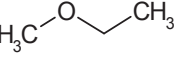
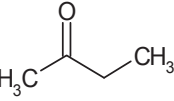
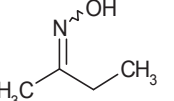
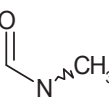
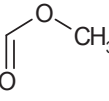
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1033	2-Methyl Cyclohexanol (mix)	2-Methyl-cyclohexanol	17.1	6.5	12.5	124.1
						
478	3-Methyl Cyclohexanone	3-Methyl- cyclohexanone	17.7	6.3	4.7	122.5
						
479	2-Methyl Cyclohexanone	2-Methyl- cyclohexanone	17.6	6.3	4.7	121.3
						
480	Methyl Ethyl Ether	Methoxy-ethane	14.7	4.9	6.2	84.1
						
481	Methyl Ethyl Ketone	Butan-2-one	16.0	9.0	5.1	90.1
						
482	Methyl Ethyl Ketoxime	Butan-2-one oxime	14.7	4.9	7.8	94.8
						
965	N-Methyl Formamide	N-Methyl-formamide	17.4	18.8	15.9	59.1
						
483	Methyl Formate	Formic acid methyl ester	15.3	8.4	10.2	62.2
						

TABLE A.1 (CONTINUED)

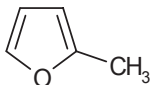
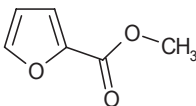
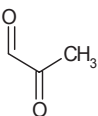
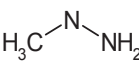
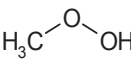
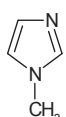
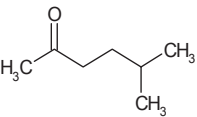
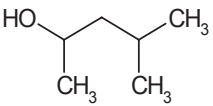
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
995	2-Methylfuran	2-Methyl-furan	17.3	2.8	7.4	89.7
						
998	Methyl Furoate	Furan-2-carboxylic acid methyl ester	17.4	6.9	9.7	107.0
						
484	Methyl Glyoxal	2-Oxo-propionaldehyde	15.5	16.1	9.7	68.9
						
485	Methyl Hydrazine	Methyl-hydrazine	16.2	8.7	14.8	52.7
						
486	Methyl Hydroperoxide	Methyl-hydroperoxide	15.0	15.0	30.0	24.1
						
487	1-Methyl Imidazole	1-Methyl-1H-imidazole	19.7	15.6	11.2	79.5
						
488	Methyl Iodide	Iodomethane	17.5	7.7	5.3	62.3
						
489	Methyl Isoamyl Ketone	5-Methyl-hexan-2-one	16.0	5.7	4.1	142.8
						
490	Methyl Isobutyl Carbinol	4-Methyl-pentan-2-ol	15.4	3.3	12.3	127.2
						

TABLE A.1 (CONTINUED)

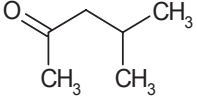
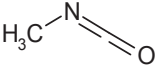
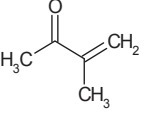
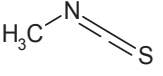
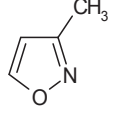
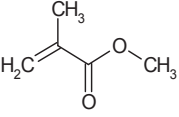
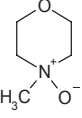
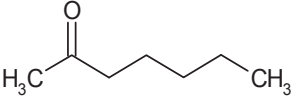
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
491	Methyl Isobutyl Ketone 	4-Methyl-pentan-2-one	15.3	6.1	4.1	125.8
492	Methyl Isocyanate 	Isocyanatomethane	15.6	7.3	2.5	61.8
493	Methyl Isopropenyl Ketone 	3-Methyl-but-3-en-2-one	15.9	12.1	4.5	99.3
494	Methyl Isothiocyanate 	Isothiocyanatomethane	17.3	16.2	10.1	68.4
495	3-Methyl Isoxazole 	3-Methyl-isoxazole	19.4	14.8	11.8	57.7
496	Methyl Mercaptan 	Methanethiol	16.6	7.7	8.6	54.1
497	Methyl Methacrylate 	2-Methyl-acrylic acid methyl ester	15.8	6.5	5.4	106.1
1106	N-Methyl Morpholine-N-Oxide 	4-Methyl-morpholine 4-oxide	19.0	16.1	10.2	97.6
498	Methyl n-Amyl Ketone 	Heptan-2-one	16.2	5.7	4.1	139.8

TABLE A.1 (CONTINUED)

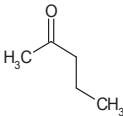
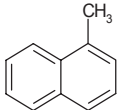
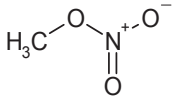
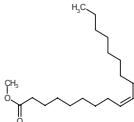
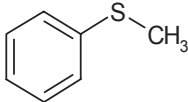
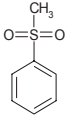
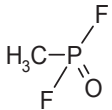
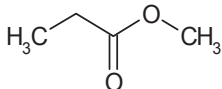
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
499	Methyl n-Propyl Ketone	Pentan-2-one	16.0	7.6	4.7	106.7
						
500	1-Methyl Naphthalene	1-Methyl-naphthalene	20.6	0.8	4.7	138.8
						
501	Methyl Nitrate	Nitrooxymethane	15.8	14.0	4.8	63.8
						
502	Methyl Oleate	(Z)-Octadec-9-enoic acid methyl ester	14.5	3.9	3.7	340.0
						
879	Methyl Phenyl Sulfid	Methylsulfanyl-benzene	19.6	4.8	4.7	117.4
						
1076	Methyl Phenyl Sulfone	Methanesulfonyl-benzene	20.0	16.9	7.8	124.6
						
841	Methyl Phosphonic Difluorid	Methylphosphonic difluorid	14.0	14.0	8.4	73.9
						
503	Methyl Propionate	Propionic acid methyl ester	15.5	6.5	7.7	96.8
						

TABLE A.1 (CONTINUED)

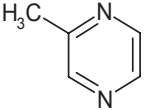
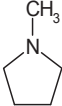
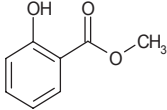
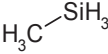
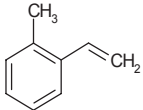
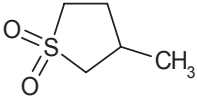
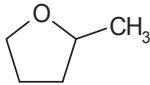
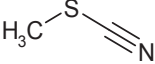
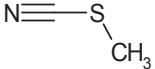
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
902	2-Methyl Pyrazine 	2-Methyl-pyrazine	18.3	12.3	10.5	91.4
986	N-Methyl Pyrrolidine 	1-Methyl-pyrrolidine	17.0	2.8	6.9	104.0
504	Methyl Salicylate 	2-Hydroxy-benzoic acid methyl ester	18.1	8.0	13.9	129.6
740	Methyl Silane 	Methyl-silane	15.5	3.3	0	71.0
886	alfa-Methyl Styrene 	1-Methyl-2-vinyl- benzene	18.5	2.4	2.4	130.0
506	Methyl Sulfolane 	3-Methyl-tetrahydro- thiophene 1,1-dioxide	19.4	17.4	5.3	112.7
997	2-Methyl Tetrahydrofuran 	2-Methyl-tetrahydro- furan	16.9	5.0	4.3	100.2
507	Methyl Thiocyanate 	Thiocyanatomethane	17.3	15.0	6.0	68.5
968	Methyl Thiocyanate 	Thiocyanatomethane	17.4	15.0	8.7	68.5

TABLE A.1 (CONTINUED)

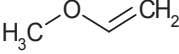
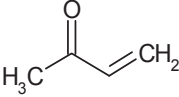
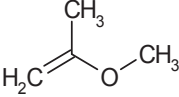
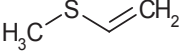
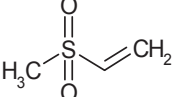
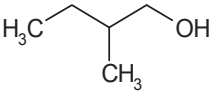
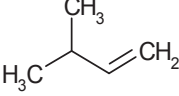
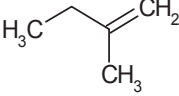
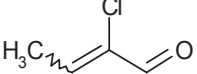
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
508	Methyl Vinyl Ether 	Methoxy-ethene	14.9	5.3	6.3	75.2
509	Methyl Vinyl Ketone 	But-3-en-2-one	15.6	12.5	5.0	81.2
510	1-Methyl Vinyl Methyl Ether 	2-Methoxy-propene	14.8	4.2	5.6	96.1
511	Methyl Vinyl Sulfid 	Methylsulfanyl-ethene	16.4	4.9	6.0	82.1
512	Methyl Vinyl Sulfone 	Methylsulfonyl-ethene	16.8	19.6	4.8	87.6
462	2-Methyl-1-Butanol 	2-Methyl-butan-1-ol	16.0	5.1	14.3	109.5
513	3-Methyl-1-Butene 	3-Methyl-but-1-ene	14.0	1.4	3.8	112.9
514	2-Methyl-1-Butene 	2-Methyl-but-1-ene	14.2	1.8	2.3	108.7
515	2-Methyl-1-Chloro Acrolein 	2-Chloro-but-2-enal	17.1	10.6	7.3	91.7

TABLE A.1 (CONTINUED)

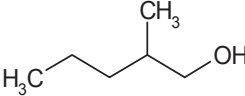
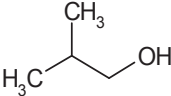
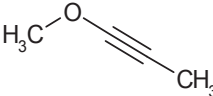
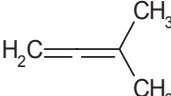
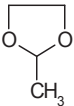
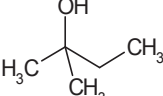
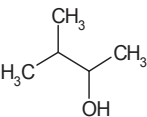
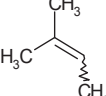
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1028	2-Methyl-1-Pentanol	2-Methyl-pentan-1-ol	15.8	4.9	13.5	124.0
						
516	2-Methyl-1-Propanol	2-Methyl-propan-1-ol	15.1	5.7	15.9	92.8
						
517	Methyl-1-Propynyl Ether	1-Methoxy-propyne	15.7	6.3	5.9	85.4
						
518	3-Methyl-1,2-Butadiene	3-Methyl-buta-1,2-diene	15.1	2.5	4.5	99.7
						
519	2-Methyl-1,3-Dioxolane	2-Methyl-[1,3]dioxolane	17.3	4.8	5.8	89.8
						
732	2-Methyl-2-Butanol	2-Methyl-butan-2-ol	15.3	6.1	13.3	109.6
						
1029	3-Methyl-2-Butanol	3-Methyl-butan-2-ol	15.6	5.2	13.4	107.7
						
520	2-Methyl-2-Butene	2-Methyl-but-2-ene	14.3	2.0	3.9	106.7
						

TABLE A.1 (CONTINUED)

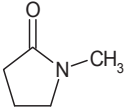
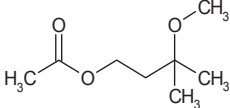
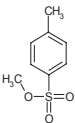
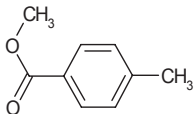
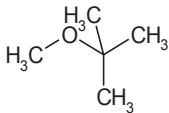
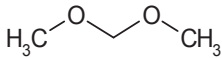
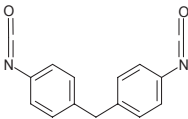
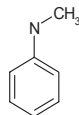
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
521	N-Methyl-2-Pyrrolidone 	1-Methyl-pyrrolidin-2-one	18.0	12.3	7.2	96.5
1002	3-Methyl-3-Methoxy Butyl Acetate 	Acetic acid 3-methoxy-3-methyl-butyl ester	15.3	3.8	7.7	168.6
972	Methyl-4-Toluenesulfonate 	Toluene-4-sulfonic acid methyl ester	19.6	15.3	3.8	152.6
796	Methyl-p-Toluate 	4-Methyl-benzoic acid methyl ester	19.0	6.5	3.8	140.4
522	Methyl-t-Butyl Ether 	2-Methoxy-2-methyl-propane	14.8	4.3	5.0	119.8
523	Methylal (Dimethoxymethane) 	Dimethoxy-methane	15.0	1.8	8.6	169.4
1217	4,4'-Methylenebis(Phenylisocyanate) 	1-Isocyanato-4-(4-isocyanatobenzyl) benzene	19.5	4.1	1.7	212.1
951	N-Methylaniline 	Methyl-phenyl-amine	19.5	6.0	11.5	108.4

TABLE A.1 (CONTINUED)

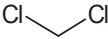
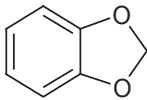
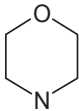
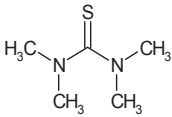
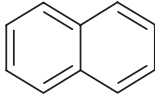
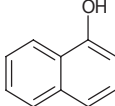
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
524	Methylene Dichloride 	Dichloro-methane	18.2	6.3	6.1	63.9
525	Methylene Diiodide 	Diiodo-methane	17.8	3.9	5.5	80.5
888	1,2-Methylenedioxybenzene 	Benzo[1,3]dioxole	19.0	6.7	5.9	114.8
526	Morpholine 	Morpholine	18.8	4.9	9.2	87.1
1193	Morpholine/Acetic Acid 	Acetate morpholin-4-ium;	17.2	20.3	18.4	
844	N,N,N,N-Tetramethylthiourea 	Tetramethyl-thiourea	17.3	6.0	10.5	132.2
529	Naphtha.High-Flash		17.9	0.7	1.8	181.8
530	Naphthalene 	Naphthalene	19.2	2.0	5.9	111.5
896	1-Naphthol 	Naphthalen-1-ol	19.7	6.3	12.3	131.7

TABLE A.1 (CONTINUED)

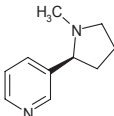
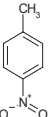
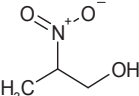
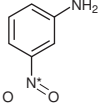
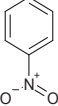
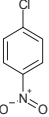
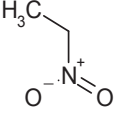
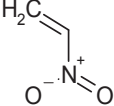
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1198	Nicotine 	3-((S)-1-Methyl- pyrrolidin-2-yl)- pyridine	18.5	7.8	6.5	160.7
784	p-Nitro Toluene 	1-Methyl-4-nitro- benzene	20.1	9.6	3.9	98.5
1092	2-Nitro-1-Propanol 	2-Nitro-propan-1-ol	16.1	13.7	15.4	88.8
939	3-Nitroaniline 	3-Nitro-phenylamine	21.2	18.7	10.3	96.1
531	Nitrobenzene 	Nitro-benzene	20.0	8.6	4.1	102.7
973	4-Nitrochlorobenzene 	1-Chloro-4-nitro- benzene	20.0	8.8	3.9	121.2
532	Nitroethane 	Nitro-ethane	16.0	15.5	4.5	71.5
533	Nitroethylene 	Nitro-ethene	16.3	16.6	5.0	59.9

TABLE A.1 (CONTINUED)

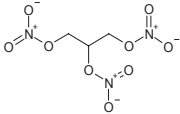
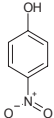
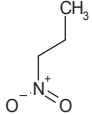
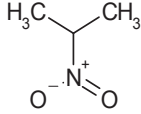
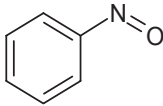
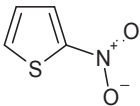
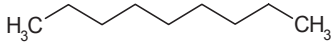
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1112	Nitroglycerin (Glyceryl Trinitrate)	1,2,3-Tris-nitrooxy- propane	16.2	17.8	5.9	142.5
						
534	Nitromethane	Nitromethane	15.8	18.8	5.1	54.3
						
936	4-Nitrophenol	4-Nitro-phenol	20.4	20.9	15.1	93.1
						
535	1-Nitropropane	1-Nitro-propane	16.6	12.3	5.5	88.4
						
536	2-Nitropropane	2-Nitro-propane	16.2	12.1	4.1	86.9
						
1073	Nitrosobenzene	Nitroso-benzene	20.0	12.7	4.0	89.3
						
984	2-Nitrothiophene	2-Nitro-thiophene	19.7	16.2	8.2	94.6
						
537	Nonane	Nonane	15.7	0	0	179.7
						
1095	1,9-Nonanediol	Nonane-1,9-diol	15.7	7.0	15.1	170.5
						

TABLE A.1 (CONTINUED)

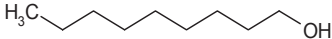
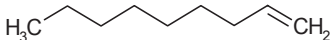
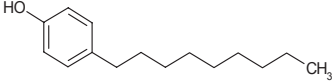
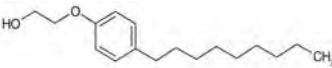
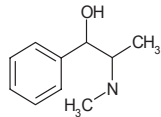
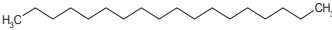
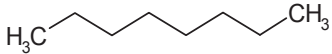
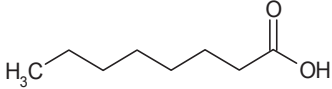
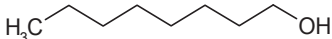
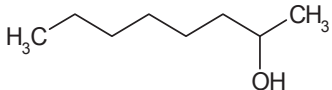
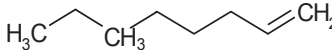
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
932	1-Nonanol	Nonan-1-ol	16.0	4.8	10.6	174.4
						
1070	1-Nonene	Non-1-ene	15.4	1.0	2.2	170.5
						
538	Nonyl Phenol	4-Nonyl-phenol	16.5	4.1	9.2	231.0
						
539	Nonyl Phenoxy Ethanol	2-(4-Nonyl-phenoxy)- ethanol	16.7	10.2	8.4	275.0
						
1209	Norephedrin	2-Methylamino-1- phenyl-propan-1-ol	18.0	10.7	24.1	141.9
						
923	Octadecane	Octadecane	16.4	0	0	326.9
						
540	Octane	Octane	15.5	0	0	163.5
						
541	Octanoic Acid	Octanoic acid	15.1	3.3	8.2	159.0
						
542	1-Octanol*	Octan-1-ol	16.0	5.0	11.9	157.7
						
543	2-Octanol	Octan-2-ol	16.1	4.9	11.0	159.1
						
544	1-Octene	Oct-1-ene	15.3	1.0	2.4	158.0
						

TABLE A.1 (CONTINUED)

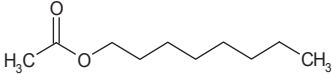
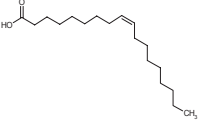
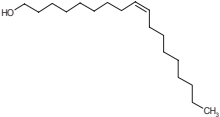
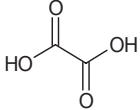
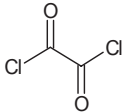
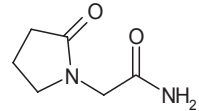
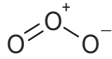
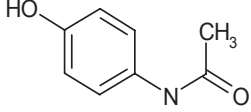
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1065	Octyl Acetate	Acetic acid octyl ester	15.8	2.9	5.1	196.0
						
545	Oleic Acid*	(Z)-Octadec-9-enoic acid	16.0	2.8	6.2	317.0
						
546	Oleyl Alcohol	(Z)-Octadec-9-en-1-ol	14.3	2.6	8.0	316.0
						
1156	Oxalic Acid	Oxalic acid	17.0	14.3	22.0	47.4
						
547	Oxalylchloride	Oxalyl dichloride	16.1	3.8	7.5	85.8
						
1026	Oxetane (Trimethylene Oxide)	Oxetane	18.0	9.1	5.4	65.0
						
1210	2-Oxopyrrolidinacetamid	2-(2-Oxo-pyrrolidin-1-yl)-acetamide	17.5	15.6	11.2	116.6
						
1169	Ozone		19.8	4.2	0	24.0
						
1211	Paracetamol	N-(4-Hydroxy-phenyl)-acetamide	17.8	10.5	13.9	151.2
						

TABLE A.1 (CONTINUED)

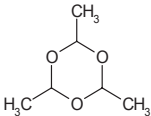
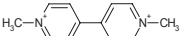
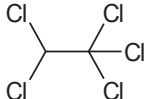
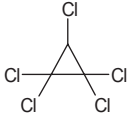
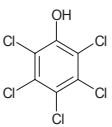
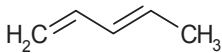
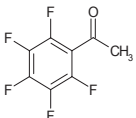
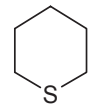
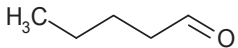
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
978	Paraldehyde 	2,4,6-Trimethyl- [1,3,5]trioxane	16.6	7.5	7.3	132.4
1114	Paraquat 1,1'-Dimethyl-4,4'- 	bipyridinium dichloride	19.5	8.8	5.9	205.7
1161	Pentachloro Ethane 	1,1,1,2,2-Pentachloro- ethane	18.2	3.2	2.4	120.9
818	Pentachlorocyclopropane 	1,1,2,2,3-Pentachloro- cyclopropane	18.5	10.5	3.7	128.5
788	Pentachlorophenol 	2,3,4,5,6-Pentachloro- phenol	21.5	6.9	12.8	134.7
548	1,3-Pentadiene (Trans) 	(E)-Penta-1,3-diene	14.7	2.5	5.0	101.7
821	Pentafluorobenzophenon 	1-Pentafluorophenyl- ethanone	19.3	8.1	5.4	185.5
549	Pentamethylene Sulfid 	Tetrahydro-thiopyran	18.5	6.3	8.9	103.6
1061	Pentanal (Valeraldehyde) 	Pentanal	15.7	9.4	5.8	106.4

TABLE A.1 (CONTINUED)

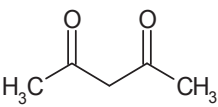
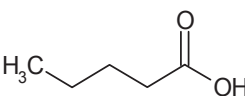
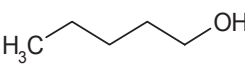
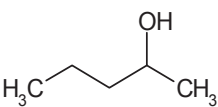
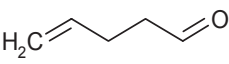
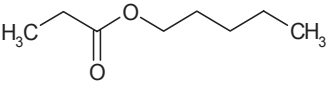
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
550	Pentane	Pentane	14.5	0	0	116.2
						
551	2,4-Pentanedione	Pentane-2,4-dione	17.1	9.0	4.1	103.1
						
1023	Pentanoic Acid	Pentanoic acid	15.0	4.1	10.3	109.2
						
552	1-Pentanol*	Pentan-1-ol	15.9	5.9	13.9	108.6
						
733	2-Pentanol	Pentan-2-ol	15.6	6.4	13.3	109.6
						
553	4-Pentenal	Pent-4-enal	15.5	8.1	6.8	98.7
						
554	1-Pentene	Pent-1-ene	13.9	1.4	3.8	110.4
						
1005	n-Pentyl Propionate	Propionic acid pentyl ester	15.8	5.2	5.7	165.3
						
556	Perfluoro Dimethylcyclohexane	1,1,2,2,3,3,4,4,5,5,6,6-Decafluoro-3,6-bis(trifluoromethyl)cyclohexane	12.4	0	0	217.4
						

TABLE A.1 (CONTINUED)

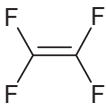
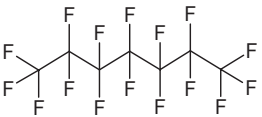
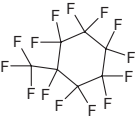
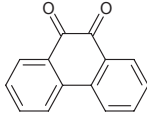
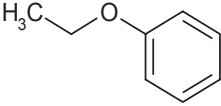
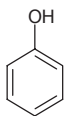
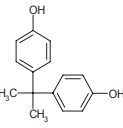
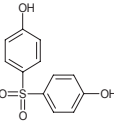
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
555	Perfluoro Ethylene (Tetrafluoro Ethylene)	1,1,2,2-Tetrafluoro ethene	15.1	0	0	65.8
						
557	Perfluoroheptan	1,1,1,2,2,3,3,4,4,5,5,6,6,7,7,7-Hexadecafluoro heptane	12.0	0	0	227.3
						
558	Perfluoromethylcyclohexane	1,1,2,2,3,3,4,4,5,5,6,6-Undecafluoro-6-trifluoromethylcyclohexane	12.4	0	0	196.0
						
975	9,10-Phenanthrenequinone	Phenanthrene-9,10-dione	20.3	17.1	4.8	148.2
						
880	Phenetole (Ethyl Phenyl Ether)	Ethoxy-benzene	18.4	4.5	4.0	127.3
						
559	Phenol	Phenol	18.0	5.9	14.9	87.5
						
1100	Bisphenol A	4-[1-(4-Hydroxyphenyl)-1-methylethyl]phenol	19.2	5.9	13.8	207.5
						
1155	Bisphenol-S	4-[(4-Hydroxyphenyl)sulfonyl]phenol	20.0	14.6	16.3	183.2
						

TABLE A.1 (CONTINUED)

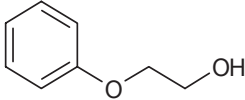
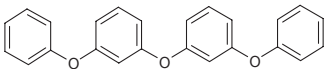
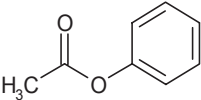
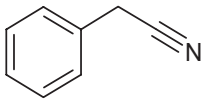
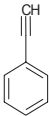
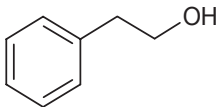
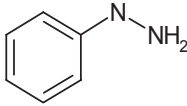
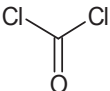
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
560	2-Phenoxy Ethanol	2-Phenoxy-ethanol	17.8	5.7	14.3	124.7
						
561	Bis-(m-Phenoxyphenyl) Ether	1-Phenoxy-3-(3-phenoxyphenoxy)benzene	19.6	3.1	5.1	373.0
						
881	Phenyl Acetate	Acetic acid phenyl ester	19.8	5.2	6.4	126.9
						
851	Phenyl Acetonitrile	Phenyl-acetonitrile	19.5	12.3	3.8	114.9
						
1168	Phenyl Acetylene	Ethynyl-benzene	18.8	2.8	4.0	109.1
						
1074	2-Phenyl Ethanol	2-Phenyl-ethanol	19.0	5.8	12.8	120.0
						
1047	Phenylhydrazine	Phenyl-hydrazine	20.4	6.5	14.0	98.5
						
562	Phosgene	Carbonyl dichloride	16.4	5.3	5.3	71.7
						

TABLE A.1 (CONTINUED)

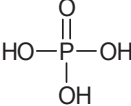
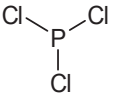
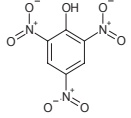
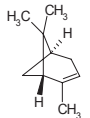
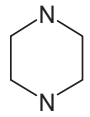
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1216	Phosphoric Acid	Phosphoric acid	14.7	18.6	26.8	52.8
						
1171	Phosphorous Oxychloride (Phosphoryl Trichloride)	Phosphoric trichloride	18.1	9.3	0	93.2
						
563	Phosphorus Trichloride	Phosphorous trichloride	18.4	3.6	0	87.3
						
564	Phthalic Anhydride	Isobenzofuran-1,3-dione	20.6	20.1	10.1	96.8
						
1089	Picric Acid (2,4,6-Trinitrophenol)	Picric acid	19.2	7.0	6.0	130.0
						
565	Pine Oil		15.6	3.0	9.8	155.0
900	2-Pinene (dl)	(1S,5S)-2,6,6-Trimethyl- bicyclo[3.1.1]hept-2- ene	16.9	1.8	3.1	159.5
						
1052	Piperazine	Piperazine	18.1	5.6	8.0	97.9
						

TABLE A.1 (CONTINUED)

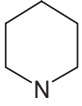
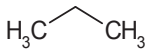
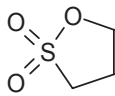
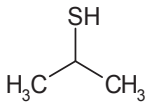
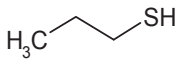
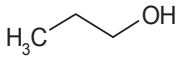
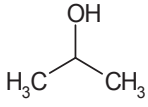
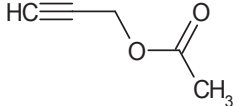
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1050	Piperidine 	Piperidine	17.6	4.5	8.9	98.9
566	1,2-Propadiene (Allene) $\text{H}_2\text{C}=\text{CH}_2$	Propa-1,2-diene	15.3	3.0	6.8	60.1
1064	Propane 	Propane	13.4	0	0	89.5
943	1,3-Propane Sultone 	[1,2]Oxathiolane 2,2-dioxide	18.4	16.0	9.0	87.7
1038	1,3-Propanediol (Trimethyleneglycol) 	Propane-1,3-diol	16.8	13.5	23.2	72.5
567	2-Propanethiol 	Propane-2-thiol	16.3	6.8	6.5	94.1
568	1-Propanethiol 	Propane-1-thiol	16.1	5.8	5.7	90.5
569	1-Propanol 	Propan-1-ol	16.0	6.8	17.4	75.2
570	2-Propanol 	Propan-2-ol	15.8	6.1	16.4	76.8
1036	Propargyl Acetate 	Acetic acid prop-2-ynyl ester	16.3	5.2	8.3	98.8

TABLE A.1 (CONTINUED)

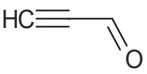
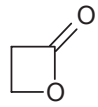
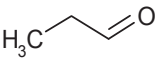
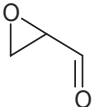
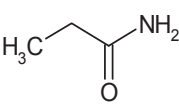
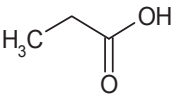
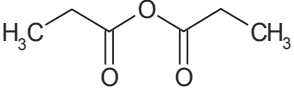
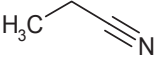
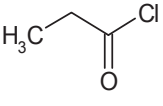
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
571	Propargylaldehyde 	Propynal	16.2	11.9	8.7	60.0
572	beta-Propiolactone 	Oxetan-2-one	19.7	18.2	10.3	65.5
1062	Propionaldehyde* 	Propionaldehyde	15.3	11.1	6.9	73.4
574	Propionaldehyde-2,3-Epoxy* 	Oxirane-2-carbaldehyde	17.5	13.4	9.8	63.2
575	Propionamide 	Propionamide	16.7	9.8	11.5	78.9
576	Propionic Acid 	Propionic acid	14.7	5.3	12.4	75.0
1043	Propionic Anhydride 	Propanoic anhydride	16.2	10.0	8.7	129.4
577	Propionitrile 	Propionitrile	15.3	14.3	5.5	70.9
578	Propionylchloride 	Propionyl chloride	16.1	10.3	5.3	86.9

TABLE A.1 (CONTINUED)

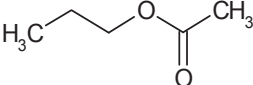
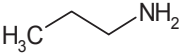
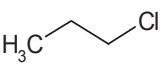
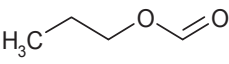
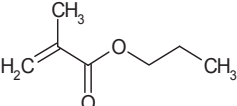
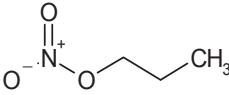
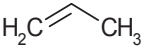
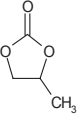
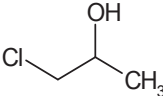
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
579	n-Propyl Acetate 	Acetic acid propyl ester	15.3	4.3	7.6	115.3
580	Propyl Amine 	Propylamine	16.9	4.9	8.6	83.0
581	Propyl Chloride 	1-Chloro-propane	16.0	7.8	2.0	88.1
926	Propyl Formate 	Formic acid propyl ester	15.5	7.1	8.6	97.9
582	Propyl Methacrylate 	2-Methyl-acrylic acid propyl ester	15.5	6.3	6.6	158.8
935	n-Propyl Nitrate 	1-Nitrooxy-propane	15.8	11.3	4.1	99.7
583	Propylene 	Propene	15.1	1.6	1.5	68.8
584	Propylene Carbonate 	4-Methyl-[1,3]dioxolan- 2-one	20.0	18.0	4.1	85.0
746	Propylene Chlorohydrin 	1-Chloro-propan-2-ol	16.8	9.8	15.3	85.4

TABLE A.1 (CONTINUED)

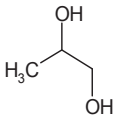
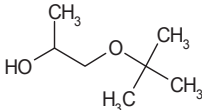
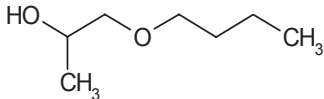
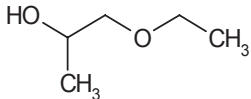
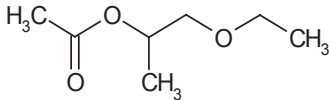
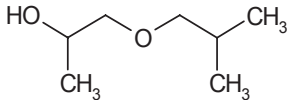
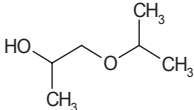
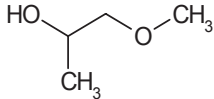
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
585	Propylene Glycol	Propane-1,2-diol	16.8	9.4	23.3	73.6
						
586	Propylene Glycol Mono-t-Butyl Ether	1-tert-Butoxy-propan-2-ol	15.3	6.1	10.8	151.6
						
587	Propylene Glycol Monobutyl Ether	1-Butoxy-propan-2-ol	15.3	4.5	9.2	132.0
						
588	Propylene Glycol Monoethyl Ether	1-Ethoxy-propan-2-ol	15.7	6.5	10.5	115.6
						
589	Propylene Glycol Monoethyl Ether Acetate	Acetic acid 2-ethoxy-1-methyl-ethyl ester	15.6	4.3	9.0	155.1
						
590	Propylene Glycol Monoisobutyl Ether	1-Isobutoxy-propan-2-ol	15.1	4.7	9.8	132.2
						
857	Propylene Glycol Monoisopropyl Ether	1-Isopropoxy-propan-2-ol	15.5	6.1	11.0	134.6
						
591	Propylene Glycol Monomethyl Ether	1-Methoxy-propan-2-ol	15.6	6.3	11.6	93.8
						

TABLE A.1 (CONTINUED)

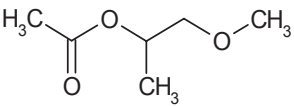
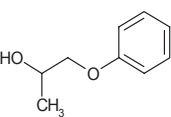
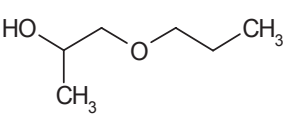
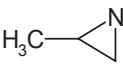
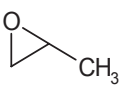
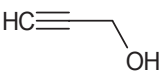
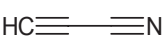
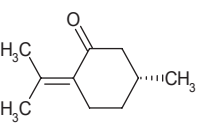
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
592	Propylene Glycol Monomethyl Ether Acetate	Acetic acid 2-methoxy- 1-methyl-ethyl ester	15.6	5.6	9.8	137.1
						
593	Propylene Glycol Monophenyl Ether	1-Phenoxy-propan-2-ol	17.4	5.3	11.5	143.2
						
594	Propylene Glycol Monopropyl Ether	1-Propoxy-propan-2-ol	15.8	7.0	9.2	130.3
						
1086	Propylene Imine (2-Methyl Aziridine)	2-Methyl-aziridine	18.1	8.4	6.5	71.2
						
595	Propylene Oxide	2-Methyl-oxirane	15.2	8.6	6.7	67.6
						
1091	2-Propyn-1-ol	Prop-2-yn-1-ol	16.1	8.8	19.1	57.7
						
596	Propynonitrile	Propynenitrile	15.5	17.0	6.3	62.5
						
1133	Pulegone	(R)-2-Isopropylidene-5- methyl-cyclohexanone	17.5	8.9	5.5	162.9
						
985	Pyrazole	1H-Pyrazole	20.2	10.4	12.4	68.1
						

TABLE A.1 (CONTINUED)

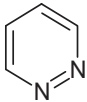
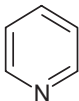
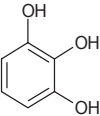
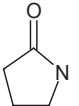
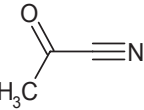
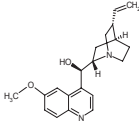
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
597	Pyridazine	Pyridazine	20.2	17.4	11.7	72.6
						
598	Pyridine	Pyridine	19.0	8.8	5.9	80.9
						
901	Pyrogallol (1,2,3-Trihydroxybenzene)	Benzene-1,2,3-triol	20.7	10.7	21.1	87.0
						
600	Pyrrole	1H-Pyrrole	19.2	7.4	6.7	69.2
						
1051	Pyrrolidine	Pyrrolidine	17.9	6.5	7.4	83.5
						
599	2-Pyrrolidone	Pyrrolidin-2-one	19.4	17.4	11.3	76.4
						
601	Pyruvitrile	2-Oxo-propionitrile	15.9	18.9	8.0	70.9
						
1202	Quinine	(R)-(6-Methoxy-quinolin-4-yl)- ((2S,4S,5R)-5-vinyl-1- aza-bicyclo[2.2.2]oct- 2-yl)-methanol	19.0	6.6	11.0	310.7
						

TABLE A.1 (CONTINUED)

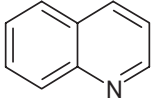
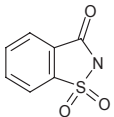
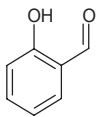
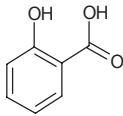
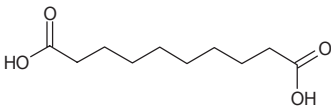
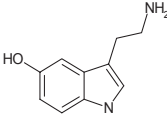
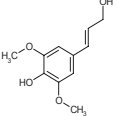
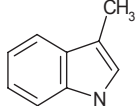
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
602	Quinoline	Quinoline	19.8	5.6	5.7	118.0
						
1196	Saccharin	1,1-Dioxo-1,2-dihydro- 1lambda*6*- benzo[d]isothiazol-3- one	21.0	13.9	8.8	206.8
						
704	Salicylaldehyde	2-Hydroxy- benzaldehyde	19.4	10.7	14.7	104.6
						
1204	Salicylic Acid	2-Hydroxy-benzoic acid	19.4	10.1	17.4	95.7
						
1225	Sebacic Acid	Decanedioic acid	17.1	7.1	11.7	167.6
						
1178	Serotonin	3-(2-Amino-ethyl)-1H- indol-5-ol	18.0	8.2	14.4	144.4
						
1101	Sinapyl Alcohol	4-((E)-3-Hydroxy- propenyl)-2,6- dimethoxy-phenol	19.2	7.3	16.1	194.6
						
832	Skatole	3-Methyl-1H-indole	20.0	7.1	6.2	122.6
						

TABLE A.1 (CONTINUED)

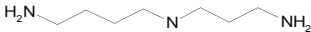
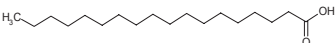
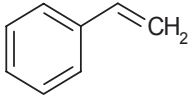
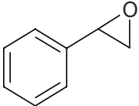
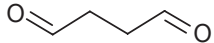
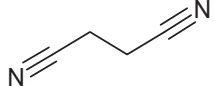
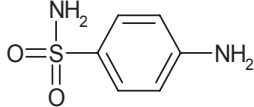
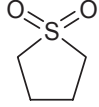
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1207	Spermidin 	N*1*-(3-Amino-propyl)-butane-1,4-diamine	16.7	11.2	12.0	155.6
603	Stearic Acid 	Octadecanoic acid	16.3	3.3	5.5	326.0
604	Styrene 	Vinyl-benzene	18.6	1.0	4.1	115.6
1014	Styrene Oxide (Phenyl Oxirane) 	2-Phenyl-oxirane	19.4	5.8	6.6	114.2
605	Succinaldehyde (Butanedial) 	Succinaldehyde	16.8	9.8	10.5	81.2
606	Succinic Anhydride 	Dihydro-furan-2,5-dione	18.6	19.2	16.6	66.8
607	Succinonitrile 	Succinonitrile	17.9	16.2	7.9	81.2
1077	Sulfanilamide 	4-Amino-benzenesulfonamide	20.0	19.5	10.7	159.5
608	Sulfolane 	Tetrahydro-thiophene 1,1-dioxide	20.3	18.2	10.9	95.7

TABLE A.1 (CONTINUED)

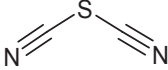
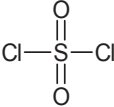
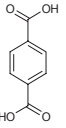
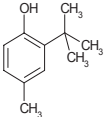
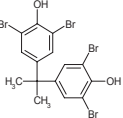
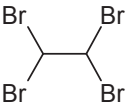
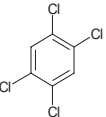
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
609	Sulfur Dicyanide	Cyanic thiocyanate	18.1	13.5	0	60.0
						
610	Sulfur Dioxide	Oxosulfane oxide	15.8	8.4	10.0	44.0
						
1098	Sulfuryl Chloride	Sulfuryl dichloride	17.6	7.2	0	81.0
						
897	Terephthalic Acid	Terephthalic acid	18.8	6.1	10.7	166.0
						
1079	2-tert-Butyl-4-Methyl Phenol	2-tert-Butyl-4-methyl-phenol	17.3	3.7	10.5	177.6
						
1127	Tetrabromo Bisphenol A	2,6-Dibromo-4-[1-(3,5-dibromo-4-hydroxyphenyl)-1-methylethyl]phenol	20.2	9.1	13.8	249.5
						
612	1,1,2,2-Tetrabromoethane	1,1,2,2-Tetrabromoethane	22.6	5.1	8.2	116.8
						
708	1,2,4,5-Tetrachlorobenzene 1,2,4,5-T	tetrachlorobenzene	21.2	10.7	3.4	116.2
						

TABLE A.1 (CONTINUED)

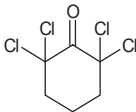
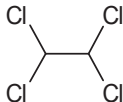
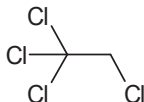
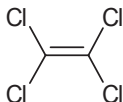
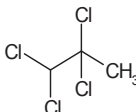
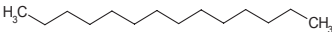
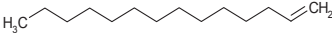
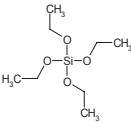
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
783	2,2,6,6-Tetrachlorocyclohexanone	2,2,6,6-Tetrachloro- cyclohexanone	19.5	14.0	6.3	138.8
						
614	1,1,2,2-Tetrachloroethane	1,1,2,2-Tetrachloro- ethane	18.8	5.1	5.3	105.2
						
882	1,1,1,2-Tetrachloroethane	1,1,1,2-Tetrachloro- ethane	18.0	4.4	4.2	105.0
						
615	Tetrachloroethylene	1,1,2,2-Tetrachloro- ethene	18.3	5.7	0	101.2
						
613	1,1,2,2-Tetrachloropropane	1,1,2,2-Tetrachloro- propane	17.9	6.7	3.3	123.7
						
922	n-Tetradecane	Tetradecane	16.2	0	0	261.3
						
1011	1-Tetradecene	Tetradec-1-ene	16.1	0.5	1.9	253.4
						
616	Tetraethylorthosilicate		13.9	4.3	0.6	224.0
						
617	Tetrahydrofuran	Tetrahydro-furan	16.8	5.7	8.0	81.7
						

TABLE A.1 (CONTINUED)

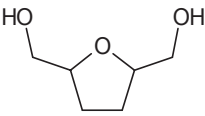
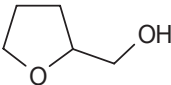
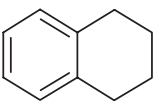
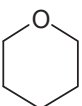
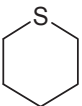
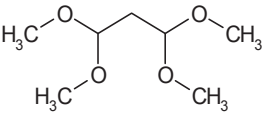
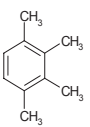
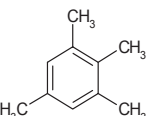
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
999	2,5-Tetrahydrofuran Dimethanol	(5-Hydroxymethyl- tetrahydro-furan-2-yl)- methanol	17.9	8.5	18.9	114.5
						
1096	Tetrahydrofurfuryl Alcohol	(Tetrahydro-furan-2-yl)- methanol	17.8	8.2	10.2	97.0
						
618	Tetrahydronaphthalene	1,2,3,4-Tetrahydro- naphthalene	19.6	2.0	2.9	136.0
						
619	Tetrahydropyran	Tetrahydro-pyran	16.4	6.3	6.0	97.8
						
620	Tetrahydrothiapyran	Tetrahydro-thiopyran	18.5	6.3	8.9	103.6
						
895	1,1,3,3-Tetramethoxypropane	1,1,3,3-Tetramethoxy- propane	15.0	7.1	6.8	164.7
						
884	1,2,3,4-Tetramethylbenzene	1,2,3,4-Tetramethyl- benzene	18.8	0.5	0.5	148.3
						
885	1,2,3,5-Tetramethylbenzene	1,2,3,5-Tetramethyl- benzene	18.6	0.5	0.5	150.8
						

TABLE A.1 (CONTINUED)

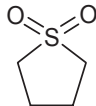
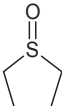
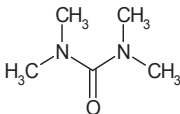
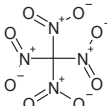
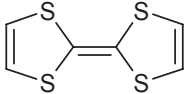
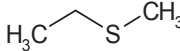
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
622	Tetramethylene Sulfid	Tetrahydro-thiophene	18.6	6.7	9.1	88.3
						
790	Tetramethylene Sulfone (Sulfolane)	Tetrahydro-thiophene 1,1-dioxide	20.3	18.2	10.9	95.7
						
623	Tetramethylene Sulfoxide	Tetrahydro-thiophene 1- oxide	18.2	11.0	9.1	90.0
						
624	Tetramethylurea	Tetramethyl-urea	16.7	8.2	11.0	120.4
						
933	Tetranitromethane	Tetranitro-methane	15.5	9.9	7.5	119.7
						
890	2,2',5,5'-Tetrathiafulvalen	[2,2']Bi[[1,3]dithiolyli- dene]	21.0	8.2	8.4	204.0
						
625	2-Thiabutane	Methylsulfanyl-ethane	16.2	5.9	5.3	90.4
						
626	Thiacyclopropane	Thiirane	19.3	9.1	5.0	58.0
						
627	Thiazole*	Thiazole	20.5	18.8	10.8	70.9
						

TABLE A.1 (CONTINUED)

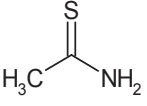
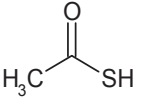
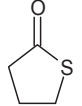
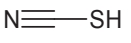
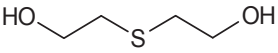
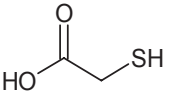
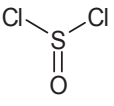
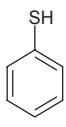
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
628	Thioacetamide	Thioacetamide	17.5	20.6	20.2	75.0
						
629	Thioacetic Acid	Thioacetic acid	17.0	6.7	8.9	71.5
						
630	Gamma-Thiobutyrolactone	Dihydro-thiophen-2-one	19.0	6.9	6.2	86.6
						
631	Thiocyanic Acid	Thiocyanic acid	16.8	8.9	10.9	51.7
						
1034	Thiodiethylglycol	2-(2-Hydroxy-ethylsulfanyl)-ethanol	17.3	8.8	19.8	103.6
						
1017	Thioglycolic Acid (Mercapto Acetic Acid)	Mercapto-acetic acid	16.0	8.6	14.8	69.5
						
632	Thionyl Chloride	Thionyl dichloride	16.9	6.2	5.9	79.0
						
633	Thiophene	Thiophene	18.9	2.4	7.8	79.0
						
703	Thiophenol	Benzenethiol	20.0	4.5	10.3	102.4
						

TABLE A.1 (CONTINUED)

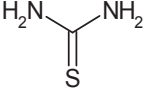
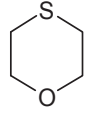
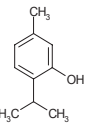
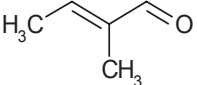
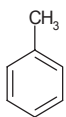
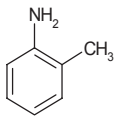
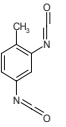
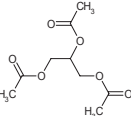
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
634	Thiourea 	Thiourea	20.0	21.7	14.8	72.8
635	1,4-Thioxane 	[1,4]Oxathiane	19.0	6.6	7.7	93.5
1203	Thymol 	2-Isopropyl-5-methyl-phenol	19.0	4.5	10.8	166.9
636	Tigaldehyde 	(E)-2-Methyl-but-2-enal	16.2	12.9	6.8	96.6
637	Toluene 	Toluene	18.0	1.4	2.0	106.8
953	2-Toluidine 	o-Tolylamine	19.4	5.8	9.4	107.8
638	Tolyene Diisocyanate 	2,4-Diisocyanato-1-methyl-benzene	19.3	7.9	6.1	143.5
1147	Triacetin (1,2,3-Propanetriol Triacetate) 	Acetic acid 2-acetoxy-1-acetoxymethyl-ethyl ester	16.5	4.5	9.1	188.2

TABLE A.1 (CONTINUED)

No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1137	Tri-n-Butyl Acetyl Citrate	3-Acetoxy-3-butoxycarbonyl-pentanedioic acid dibutyl ester	16.7	2.5	7.4	384.3
1020	Tri-n-Butyl Borate	Tri-n-butyl borate	16.7	1.8	4.6	269.7
1220	Tri-n-Butyl Citrate	3-Butoxycarbonyl-3-hydroxy-pentanedioic acid dibutyl ester	16.6	3.8	10.1	345.5
641	Tri Butyl Phosphate*	Phosphoric acid tributyl ester	16.3	6.3	4.3	274.0
639	1,2,3-Triazole	1H-[1,2,3]Triazole	20.7	8.8	15.0	58.2
1118	2,4,6-Tribromo Phenol	2,4,6-Tribromo-phenol	20.6	10.2	14.1	129.7
640	Tribromo Ethylene*	1,1,2-Tribromo-ethene	18.3	9.4	8.0	97.8
642	3,3,3-Trichloro Propene	3,3,3-Trichloro-propene	17.7	15.5	3.4	106.2

TABLE A.1 (CONTINUED)

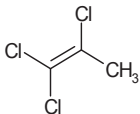
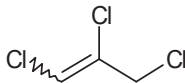
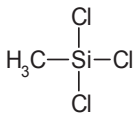
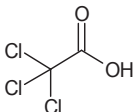
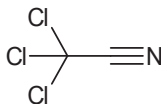
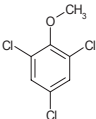
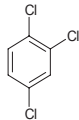
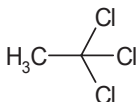
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
643	1,1,2-Trichloro Propene	1,1,2-Trichloro-propene	17.7	15.7	3.4	104.8
						
644	1,2,3-Trichloro Propene	1,2,3-Trichloro-propene	17.8	15.7	3.4	105.0
						
981	Trichloro-Methyl-Silane	Trichloro-methyl-silane	16.5	6.6	3.5	117.4
						
861	Trichloroacetic Acid	Trichloro-acetic acid	18.3	5.8	11.4	100.2
						
645	Trichloroacetonitrile	Trichloro-acetonitrile	16.4	7.4	6.1	100.0
						
1115	2,4,6-Trichloroanisole	1,3,5-Trichloro-2-methoxy-benzene	21.0	3.9	7.0	129.0
						
701	1,2,4-Trichlorobenzene*	1,2,4-Trichloro-benzene	20.2	6.0	3.2	125.5
						
647	1,1,1-Trichloroethane	1,1,1-Trichloro-ethane	16.8	4.3	2.0	99.3
						

TABLE A.1 (CONTINUED)

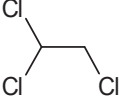
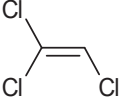
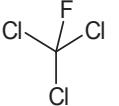
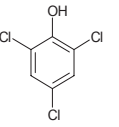
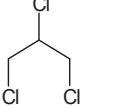
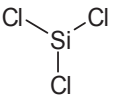
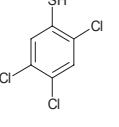
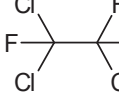
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
648	1,1,2-Trichloroethane	1,1,2-Trichloro-ethane	18.2	5.3	6.8	92.9
						
649	Trichloroethylene	1,1,2-Trichloro-ethene	18.0	3.1	5.3	90.2
						
650	Trichlorofluoromethane (Freon 11)	Trichloro-fluoro methane	15.3	2.0	0	92.8
						
702	2,4,6-Trichlorophenol	2,4,6-Trichloro-phenol	20.3	5.1	10.8	132.5
						
883	1,2,3-Trichloropropane	1,2,3-Trichloro-propane	17.8	12.3	3.4	106.1
						
1174	Trichlorosilane		14.2	3.6	3.8	101.1
						
651	2,4,5-Trichlorothiophenol	2,4,5-Trichloro- benzenethiol	21.0	4.5	9.1	145.0
						
652	1,1,2-Trichlorotrifluoroethane (Freon 113)	1,1,2-Trichloro-1,2,2- trifluoro-ethan	14.7	1.6	0	119.2
						

TABLE A.1 (CONTINUED)

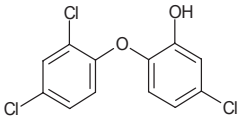
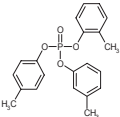
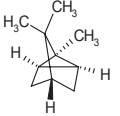
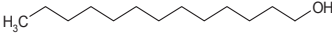
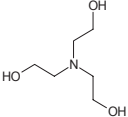
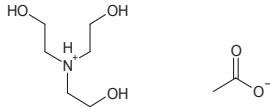
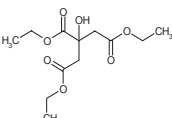
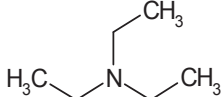
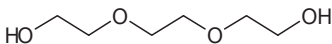
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1160	Triclosan 	5-Chloro-2-(2,4-dichloro-phenoxy)-phenol	20.0	7.7	10.0	263.0
653	Tricresyl Phosphate 	Phosphoric acid m-tolyl ester o-tolyl ester p-tolyl ester	19.0	12.3	4.5	316.0
899	Tricyclene 	(1S,2S,4R,6R)-1,7,7-Trimethyl-tricyclo[2.2.1.0*2,6*]heptane	16.9	0	0	161.4
654	Tridecyl Alcohol* 	Tridecan-1-ol	16.2	3.1	9.0	242.0
655	Triethanolamine 	2-[Bis-(2-hydroxy-ethyl)-amino]-ethanol	17.3	22.4	23.3	133.2
1194	Triethanolamine/Acetic Acid 	Acetate tris-(2-hydroxy-ethyl)-ammonium;	17.2	20.3	18.4	
1218	Triethyl Citrate 	3-Ethoxycarbonyl-3-hydroxy-pentanedioic acid diethyl ester	16.5	4.9	12.0	243.0
656	Triethylamine 	Triethyl-amine	17.8	0.4	1.0	138.6
657	Triethylene Glycol 	2-[2-(2-Hydroxy-ethoxy)-ethoxy]-ethanol	16.0	12.5	18.6	114.0

TABLE A.1 (CONTINUED)

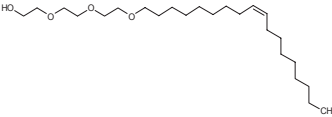
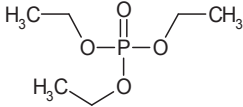
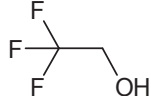
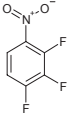
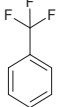
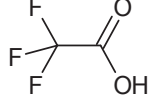
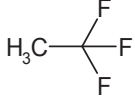
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
856	Triethylene Glycol Monomethyl Ether 	2-[2-(2-Methoxy-ethoxy)-ethoxy]-ethanol	16.2	7.6	12.5	160.0
658	Triethylene Glycol Monooleyl Ether* 	2-(2-{2-[(Z)-Octadec-9-enyl]oxy}-ethoxy)-ethoxy)-ethanol	16.0	3.1	8.4	418.5
659	Triethylphosphate 	Phosphoric acid triethyl ester	16.7	11.4	9.2	171.0
944	2,2,2-Trifluoro Ethanol 	2,2,2-Trifluoro-ethanol	15.4	8.3	16.4	72.3
795	2,3,4-Trifluoro Nitrobenzen 	1,2,3-Trifluoro-4-nitro benzene	19.5	7.7	3.5	122.0
1158	alpha,alpha,alpha Trifluoro toluene 	Trifluoromethyl-benzen	17.5	8.8	0	122.9
660	Trifluoroacetic Acid 	Trifluoro-acetic acid	15.6	9.9	11.6	74.2
661	1,1,1-Trifluoroethanol 	1,1,1-Trifluoro-ethanol	14.6	10.7	0	64.6

TABLE A.1 (CONTINUED)

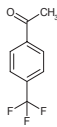
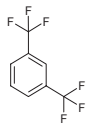
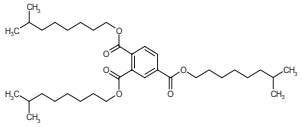
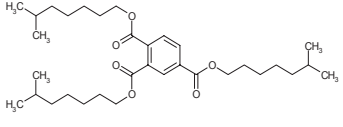
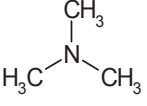
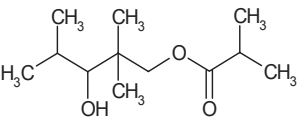
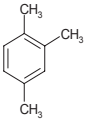
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
662	Trifluoromethane (Freon 23)	Trifluoro-methan	14.4	8.9	6.5	46.1
						
847	4-(Trifluoromethyl) Acetophenone	1-(4-Trifluoromethyl-phenyl)-ethanone	18.8	6.1	3.5	151.7
						
1157	1,3-Bis(Trifluoromethyl)Benzen	1,3-Bis-trifluoromethyl benzene	17.0	6.8	0	155.2
						
663	Triisononyl Trimellitate	Benzene-1,2,4-tricarboxylic acid tris-(7-methyl-octyl) ester	16.6	5.7	2.2	602.9
						
664	Triisooctyl Trimellitate	Benzene-1,2,4-tricarboxylic acid tris-(6-methyl-heptyl) ester	16.6	6.0	2.5	553.1
						
665	Trimethyl Amine	Trimethyl-amine	14.6	3.4	1.8	90.3
						
666	2,2,4-Trimethyl-1,3-Pentanediol Monoisobutyrate	Isobutyric acid 3-hydroxy-2,2,4-trimethyl-pentyl ester	15.1	6.1	9.8	227.4
						
667	Trimethylbenzene*	1,2,4-Trimethyl-benzene	18.0	1.0	1.0	137.3
						

TABLE A.1 (CONTINUED)

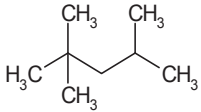
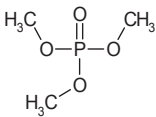
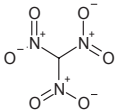
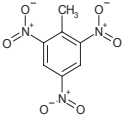
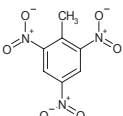
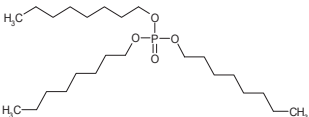
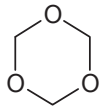
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
669	Trimethylenesulfid	Thietane	18.8	7.8	9.4	72.8
						
670	2,2,4-Trimethylpentane	2,2,4-Trimethyl-pentane	14.1	0	0	166.1
						
671	Trimethylphosphate	Phosphoric acid trimethyl ester	16.7	15.9	10.2	115.8
						
934	Trinitomethane	Trinitro-methane	15.5	10.3	7.3	94.6
						
918	Trinitrotoluene (TNT) P by Dipole Moment	2-Methyl-1,3,5-trinitro- benzene	19.5	3.7	4.5	137.6
						
919	Trinitrotoluene (TNT) P by Group Cont.	2-Methyl-1,3,5-trinitro- benzene	19.5	10.0	4.5	137.6
						
1151	Trioctylphosphate	Phosphoric acid trioctyl ester	16.2	5.9	4.2	469.8
						
855	1,3,5-Trioxane	[1,3,5]Trioxane	18.7	9.2	8.6	77.0
						

TABLE A.1 (CONTINUED)

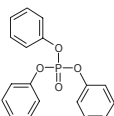
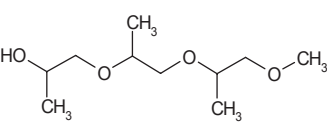
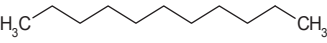
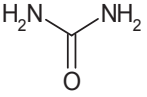
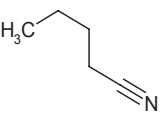
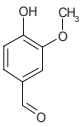
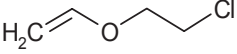
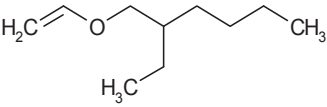
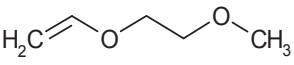
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
1150	Triphenyl Phosphate	Phosphoric acid triphenyl ester	20.1	6.4	6.8	271.9
						
672	Tripropylene Glycol Monomethyl Ether*	1-[2-(2-Methoxy-1-methyl-ethoxy)-1-methyl-ethoxy]-propan-2-ol	15.3	5.5	10.4	214.0
						
1059	Undecane	Undecane	16.0	0	0	212.7
						
860	Urea (Ro = 19.4)	Urea	20.9	18.7	26.4	45.8
						
673	Valeronitrile	Pentanenitrile	15.3	11.0	4.8	103.8
						
1162	Vanillin (4-Hydroxy-3-Methoxy Benzaldehyde)	4-Hydroxy-3-methoxy-benzaldehyde	19.4	9.8	11.2	144.1
						
674	Vinyl 2-Chloro Ethyl Ether	(2-Chloro-ethoxy)-ethene	16.3	6.7	5.8	101.7
						
908	Vinyl 2-Ethyl-Hexyl Ether	3-Vinyloxymethyl-heptane	15.6	3.4	4.2	194.2
						
675	Vinyl 2-Methoxy Ethyl Ether	(2-Methoxy-ethoxy)-ethene	15.9	6.7	6.8	96.2
						

TABLE A.1 (CONTINUED)

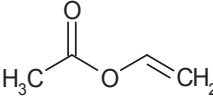
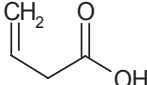
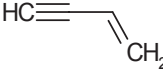
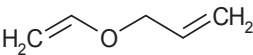
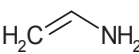
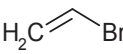
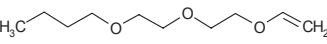
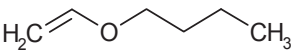
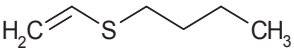
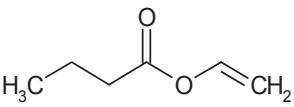
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
676	Vinyl Acetate	Acetic acid vinyl ester	16.0	7.2	5.9	92.6
						
677	Vinyl Acetic Acid	But-3-enoic acid	16.8	5.2	12.3	85.3
						
678	Vinyl Acetylene	But-1-en-3-yne	15.1	1.7	12.0	74.3
						
679	Vinyl Allyl Ether	3-Vinyloxy-propene	14.9	6.5	5.3	105.4
						
742	Vinyl Amine	Vinylamine	15.7	7.2	11.8	51.8
						
680	Vinyl Bromide	Bromo-ethene	15.9	6.3	5.4	71.6
						
914	Vinyl Butyl Carbitol	1-[2-(2-Vinyloxy-ethoxy)-ethoxy]-butane	16.0	5.0	6.0	206.2
						
964	Vinyl Butyl Ether	1-Vinyloxy-butane	15.2	4.1	5.1	129.4
						
681	Vinyl Butyl Sulfid	1-Vinylsulfanyl-butane	16.0	5.0	5.4	136.7
						
682	Vinyl Butyrate	Butyric acid vinyl ester	15.6	3.9	6.9	126.5
						

TABLE A.1 (CONTINUED)

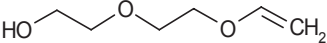
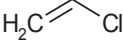
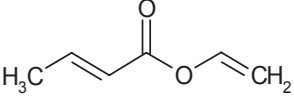
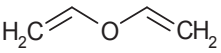
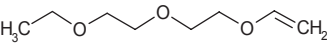
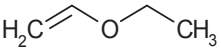
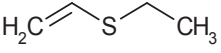
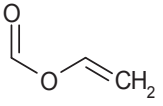
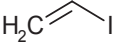
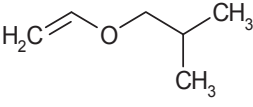
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
906	Vinyl Carbitol 	2-(2-Vinyloxy-ethoxy)- ethanol	16.3	7.5	13.8	129.2
683	Vinyl Chloride 	Chloro-ethene	16.0	6.5	2.4	68.7
684	Vinyl Crotonate 	(E)-But-2-enoic acid vinyl ester	15.9	5.0	9.0	118.8
685	Vinyl Ether 	Vinyloxy-ethene	14.8	4.2	5.8	90.7
907	Vinyl Ethyl Carbitol 	[2-(2-Ethoxy-ethoxy)- ethoxy]-ethene	15.9	6.0	6.6	171.9
1024	Vinyl Ethyl Ether 	Ethoxy-ethene	14.5	4.9	6.0	95.0
686	Vinyl Ethyl Sulfid 	Ethylsulfanyl-ethene	16.4	5.8	6.3	101.3
687	Vinyl Formate 	Formic acid vinyl ester	15.3	6.5	9.7	74.7
688	Vinyl Iodide (Iodoethene) 	Iodo-ethene	17.1	5.5	7.3	75.6
909	Vinyl Isobutyl Ether 	2-Methyl-1-vinyloxy- propane	15.1	4.1	5.1	131.3

TABLE A.1 (CONTINUED)

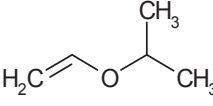
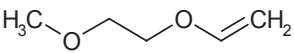
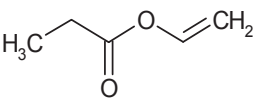
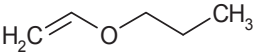
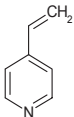
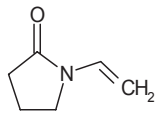
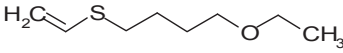
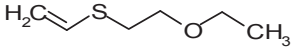
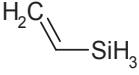
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
910	Vinyl Isopropyl Ether	2-Vinyloxy-propane	14.7	3.5	5.2	115.3
						
911	Vinyl Methyl Cellosolve	(2-Methoxy-ethoxy)- ethene	15.6	6.2	6.8	114.7
						
689	Vinyl Propionate	Propionic acid vinyl ester	15.6	8.0	4.7	110.1
						
690	Vinyl Propyl Ether	1-Vinyloxy-propane	14.9	3.5	5.2	113.0
						
843	4-Vinyl Pyridine	4-Vinyl-pyridine	18.1	7.2	6.8	107.3
						
691	Vinyl Pyrrolidone	1-Vinyl-pyrrolidin-2-one	16.4	9.3	5.9	106.9
						
913	Vinyl S-Butyl Mercapto Butyl Ether	1-Ethoxy-4- vinylsulfanyl-butane	16.5	5.0	5.8	174.0
						
912	Vinyl S-Ethyl Mercapto Ethyl Ether	(2-Ethoxy- ethylsulfanyl)-ethene	16.4	7.0	6.0	139.7
						
692	Vinyl Silane	Vinyl-silane	15.5	2.6	4.0	89.4
						

TABLE A.1 (CONTINUED)

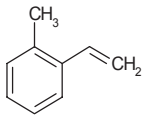
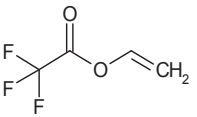
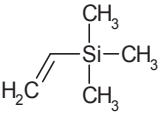
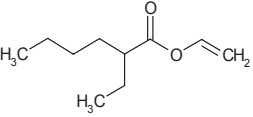
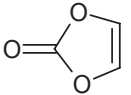
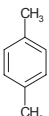
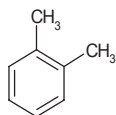
No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
916	2-Vinyl Toluene 	1-Methyl-2-vinyl- benzene	18.6	1.0	3.8	131.8
693	Vinyl Trifluoro Acetate 	Trifluoro-acetic acid vinyl ester	13.9	4.3	7.6	116.4
694	Vinyl Trimethyl Silane 	Trimethyl-vinyl-silane	14.5	1.0	2.5	145.3
915	Vinyl-2-Ethyl Hexanoate 	2-Ethyl-hexanoic acid vinyl ester	15.6	2.7	5.8	196.0
695	Vinylencarbonate 	[1,3]Dioxol-2-one	17.3	18.1	9.6	86.0
696	Water OH ₂	Water	15.5	16.0	42.3	18.0
859	Water - 1% in Ro = 18.1 OH ₂	Water	15.1	20.4	16.5	18.0
858	Water - Complete Miscibility Ro = 13.0 OH ₂	Water	18.1	17.1	16.9	18.0
697	Xylene 	p-Xylene	17.6	1.0	3.1	123.3

TABLE A.1 (CONTINUED)

No.	Solvent Name	Autonom/ ACD Name	Dispersion	Polarity	Hydrogen Bonding	Molar Volume
698	o-Xylene	o-Xylene	17.8	1.0	3.1	121.2



Appendix A: Table A.2

COMMENTS TO TABLE A.2

Documentation for the sources of data used for the HSP correlations is given in the following. The quality of the entries is sometimes less than desired because of the data being too few, too limited in scope and range of HSP, or for other reasons discussed in the text, such as the influence of molecular weight (molecular volume) of the test solvents in the given study. All entries have been included (with some apologies) as they have some value in terms of estimating, however.

POLYMERS 1–109

These polymers are listed in Reference 1 with suppliers. This report from the Scandinavian Paint and Printing Ink Research Institute updates an earlier one from 1982. The institute no longer exists. See also Reference 2.

POLYMER 110

This is an intermediate value for the permeation of chemicals through Challenge materials [3]. See also Table 13.1 and Figure 13.2. Improved values are found below in 141 and 142. This correlation was based on few data to help locate additional solvents for testing. Results from tests with these then resulted in the correlations below.

POLYMERS 111–112

These are correlations of true solubilities for the DO W epoxy Novolacs 438 and 444.

POLYMERS 113–114

These are correlations of the chemical resistance of coatings based on inorganic zinc silicate and a two component epoxy produced by Hempel's Marine Paints. Data taken from resistance tables.

POLYMER 115

The data are solubilities determined for PVDF with the correlation being previously published in [4].

POLYMER 116

Data for coal tar pitch generated for the solubility of the solids not dissolved in some cases where the solution was darkened with only partial solution.

POLYMERS 117–140

Permeation correlations for chemical protective clothing described in detail in Reference 5. See also Chapter 13, Table 13.1.

POLYMERS 141–142

Final permeation correlations for Challenge 5100 and 5200 materials. Data from Reference 3 where there is considerable discussion. See also Chapter 13, Table 13.1, and Figure 13.2.

POLYMERS 143–144

These correlations are based on which solvents dissolve PVDC at elevated temperatures and use data from Wessling [6]. These were additionally used to check new calculations for solubility parameters of the solvents where these were lacking.

POLYMERS 145–148

These chemical resistance data for PES (ICI-Victrex) and PPS (Philips-Ryton) were based on supplier data sheets and are reported in Reference 7.

Polymers 149–160

These correlations for many common plastics types are based on the resistance tables reported in the PLASTGUIDE (1989) published by the Danish company Dukadan, which no longer exists. A single correlation for the solubility of PAA 6,6 is based on its solubility only with data from Reference 8.

POLYMER 161

Beerbower treated several sets of data and made correlations of swelling and solubility (and other phenomena). This one is for polyvinyl silane.

POLYMERS 162–163

These correlations for swelling of cellophane and solubility of ethylene vinyl alcohol copolymer are based on data generated at NIF (Scandinavian Paint and Printing Ink Research Institute).

POLYMERS 164–167

These are supplementary breakthrough time correlations for Saranex, Safety 44H, and polyvinylalcohol protective gloves. See also Reference 5 and Chapter 13. Elimination of plasticizer data for the 4H gloves improved predictability for lower molecular weight materials.

POLYMERS 168–181

These correlations for common polymer types are based on data in resistance tables in the *Modern Plastics Encyclopedia* in the 1984/1985 issue [9]. Such data are not always sufficiently encompassing to allow good correlations.

POLYMER 182

Correlation based on high temperature solvents for ECTFE.

POLYMER 183

Data for this correlation of solubility of polyacrylonitrile were taken from the *Polymer Handbook* [10], Table of solvents and nonsolvents, p. VII/385-VII/386. See also Chapter 5, Table 5.3.

POLYMERS 184–186

Data for this correlation are the tendency of Polyethylene imide (PEI) (GE Ultem) to environmental stress crack (ESC) at different stress/strain levels. These data were generated by General Electric as published in the *Modern Plastics Encyclopedia* 1984/1985 [9].

POLYMERS 187–224

The *Handbook of Solubility Parameters and Other Cohesion Parameters* [11] as well as the *Polymer Handbook* [12] included so-called “solvent range” data. Solvents were divided into groups of poor, moderate, and strong hydrogen bonding, and many experiments were run. The correlations show that not all the data were well taken, but a reasonable indication is possible. The full Hansen solubility parameter system is not covered very well by this limited solubility data. These polymers are included in Reference 11, Table 1, on page 280. Heating samples to speed up the solution process was also done. This can easily lead to errors.

POLYMERS 225–346

These entries have the same problem as those in 187–224 in that the data are sometimes questionable and not sufficient enough to do what has been done, i.e. convert solvent range data to Hansen solubility parameter spheres. These entries cover the acrylics, polyesters, polystyrenes, vinyls, and miscellaneous categories. Some categories are not yet included. Data on page 281-289 (Table 2) in Reference 11.

Polymer 347

These values for VYHH (Union Carbide) were taken from Reference 1.

POLYMER 348

This questionable correlation for PVF includes only one solvent as being good [13].

POLYMER 349

Data on PES true solubility taken by author. See Chapter 5 and Table 5.4.

POLYMERS 350–358

These entries are not all polymers but mostly biological materials with the source of data being [14].

POLYMER 359

The solubility of cholesterol, data collected by the author. See Chapter 15.

POLYMER 360

Solubility data generated by high school students as part of project. Included in Reference 4. Source of chlorophyll was crushed leaves.

Polymer 361

Correlation on strength of paper immersed in different solvents reported in Reference 4. Data was taken from Reference 15.

POLYMER 362

Solubility of ULTRASON PES has been reported by BASF in their product data. These data were combined with supplementary solubility data for this correlation. Also reported in Reference 16. See Chapter 5.

POLYMERS 363–364

Chemical resistance of BAREX 210 from data in BP Chemicals datasheet. Styrene is an outlier in the first, whereas its removal from consideration gives a perfect fit and presumably a more useful correlation.

POLYMERS 365–367

These data were generated in connection with a lecture to the Nordic Conservation Congress in Copenhagen [17]. All give perfect fits, partly because of too few data, but the correlations can be useful. Paraloid B72 and Dammar are used as protective lacquers.

POLYMERS 368–369

These correlations divide the permeation coefficients given in Reference 18 into >80 and >0.8 , respectively. The units are $(\text{g} \times \text{mm})/(\text{m}^2 \times \text{d})$. The fits are good. See Chapter 13.

POLYMERS 370–371

These are correlations of experimental solubility data for the Rhône-Poulenc reactive isocyanates Tolonate HDT (which gave the same result as Tolonate HDT-LV) and Tolonate HDB (which gave the same results as Tolonate HDB-LV). The fits were perfect and the numbers reasonable. The data could not include alcohol or amine solvents because of reactions.

POLYMERS 372–389

The data correlated for these 18 rubbers are from a RAPRA database [19]. The information used was satisfactory or unsatisfactory, all other information such as limited suitability was neglected. No precise weight gain or other information is available, just the general suitability or not.

The values in parentheses are (data fit/number of solvents).

ACM	acrylate rubbers (.981/55)
ECO	epichlorohydrin rubbers (.988/37)
CSM	chlorosulphonated polyethylene rubber (.906/53)
E	ebonite (.722/41)
EPM	ethylene-propylene copolymer (.987/47)
EPDM	ethylene-propylene terpolymer (.968/51)
FQ	fluorosilicone rubber (.844/40)
FKM	hexafluoroprop.-vinylidene fluoride copol. (viton) (.769/50)
NR	natural rubber (1.000/59)
NBR	nitrile rubber (.990/65)
FFKM Kalrez	(Du Pont) (too resistant to correlate)
CR	polychloroprene (.877/54)
AU	polyester polyurethane (.959/63)
EU	polyether polyurethane (.959/63)
T	polysulphide rubber (.799/48)
Q silicone	(.748/53)

SBR	styrene butadiene rubber (.942/54)
TFP	tetrafluoroethylene-propylene copolymer (.744/26)

POLYMERS 390–412

These correlations use data from the RAPRA collection of data on chemical resistance for plastics [20]. Approach same as for RAPRA rubber data just above.

POLYMERS 413–450

These data are from the collected report of the EC project on self-stratifying coatings reported in a full issue of *Progress in Organic Coatings*. The specific reference is Reference 21. The evaluations were made at different concentrations in many cases. Some alkyls were omitted here.

POLYMERS 451–452

These data are for strong swelling of two different film samples of brominated butyl rubber.

POLYMER 453

The correlation is based on strong swelling of a film of polyisoprene

POLYMERS 454–458

These correlations are based on chemical resistance data from Reference 22.

POLYMER 459

Correlation based on solubility of Ethylene Vinylacetate adhesive EVA 4055.

POLYMER 460

Correlation based on solubility of Topas 6013 from Ticona GmbH (Hoechst AG).

POLYMER 461

Correlation based on solubility of CZ Resin from the West Company.

POLYMER 462

An older correlation for the solubility of Kauri Gum, used in the Kauri-Butanol test, was made with a data fit of 0.95 for the standard solvents.

POLYMER 463

The data for the solubility of polyvinylpyrrolidone used in this correlation are found in Reference 23. The data fit was 0.992, but as with many water soluble polymers, there is a considerable extrapolation into the “unknown” where there are no liquids.

ENTRY 464

The data fit for the correlation of solubility of palm oil with the standard set of solvents was 0.992.

ENTRY 465

This is a correlation of the solubility of a fungicide and algicide called Bethoxazin using solubility data in 19 liquids from Reference 24. The data fit as 0.976.

ENTRY 466

This is a correlation for the solubility of carbon-60 at a given small level as reported in Reference 25; 15 of the 87 liquids were considered as "good" giving a data fit of 0.972

REFERENCES

1. Saarnak, A., Hansen C.M., and Wallström E., Solubility Parameters — Characterization of Paints and Polymers, Report from Scandinavian Paint and Printing Ink Research Institute, January 1990, Hoer - sholm, Denmark
2. Hansen, C.M., Solubility Parameters, in *Paint Testing Manual*, Manual 17, J.V. Koleske, Ed., American Society for Testing and Materials, Philadelphia, 1995, pp. 38–404.
3. Hansen, C.M., Billing, C.B., and Bentz, A.P., Selection and Use of Molecular Parameters to Predict Permeation Through Fluoropolymer-Based Protective Clothing Materials, *The Performance of Protective Clothing*; Fourth Volume, ASTM STP 1133, J.P. McBriarty and N.W. Henry, Eds., American Society for Testing and Materials, Philadelphia, 1992, pp. 894–907.
4. Hansen, C.M., 25 Years with Solubility Parameters (in Danish: 25 År med Opløselighedsparametrene), *Dansk Kemi*, 73(8), 18–22, 1992.
5. Hansen, C.M. and Hansen, K.M., Solubility Parameter Prediction of the Barrier Properties of Chemical Protective Clothing, *Performance of Protective Clothing: Second Symposium*. ASTM STP 989, S.Z. Mansdorf, R. Sager, and A.P. Nielsen, Eds., American Society for Testing and Materials, Philadelphia, 1988, pp. 197–208.
6. Wessling, R.A., The Solubility of Poly(vinylidene Chloride), *Journal of Applied Polymer Science*, 14, 1531–1545, 1970.
7. Hansen, C.M., *Solubility Parameters for Polyphenylene Sulfide (PPS) and Polyether Sulphone (PES)*, Centre for Polymer Composites (Denmark), Danish Technological Institute, Taastrup, 1991, 89 pages. ISBN 87-7756-139-2
8. Wyzgoski, M.G., The Role of Solubility in Stress Cracking of Nylon 6,6, in *Macromolecular Solutions — Solvent Property Relationships in Polymers*, R.B. Seymour and G.A. Stahl, Eds. Pergamon, New York, 1982, pp. 41–60.
9. Anonymous, Modern Plastics Encyclopedia 1984/1985, McGraw-Hill, New York, pp. 482–455.
10. Fuchs, O., Tables of Solvents and Non-solvents, *Polymer Handbook*, 3rd Ed., J. Branderup and E.H. Immergut, Eds., Wiley, New York, 1989, pp. VII/379–VII/407.
11. Barton, A.F.M., *Handbook of Solubility Parameters and Other Cohesion Parameters*, CRC Press Inc., Boca Raton, FL. 1983, pp. 280–289.
12. Grulke, E.A., Table 3.4, Solubility Parameter Ranges of Commercial Polymers, *Polymer Handbook*, 3rd Ed., J. Branderup and E.H. Immergut, Eds., Wiley, New York, 1989, pp. VII/544–VII/550.
13. Fuchs, O., Tables of Solvents and Non-solvents, *Polymer Handbook*, 3rd Ed., J. Branderup and E.H. Immergut, Eds., Wiley, New York, 1989, p. VII/385.
14. Hansen, C.M. and Andersen, B.H., The Affinities of Organic Solvents in Biological Systems, *American Industrial Hygiene Association Journal*, 49(6), 301–308, 1988.
15. Robertson, A.A., Cellulose-Liquid Interactions, *Pulp and Paper Magazine of Canada*, 65(4), T-171–T-178, 1964.
16. Hansen, C.M., *Solvent Resistance of Polymer Composites — Glass Fibre Reinforced Polyether Sulfone (PES)*, Centre for Polymer Composites (Denmark), Danish Technological Institute, Taastrup, 1994.
17. Hansen, C.M., *Conservation and Solubility Parameters*, Nordic Conservation Congress Preprints, Copenhagen, 1994, pp. 1–13.
18. Pauly, S., Permeability and Diffusion Data, *Polymer Handbook*, 3rd Ed., J. Branderup and E.H. Immergut, Eds., Wiley, New York, 1989, pp. VI/435–VI/449.

19. Anonymous, Chemical Resistance Data Sheets, Volume 2. Rubbers, New Edition — 1993, *Rapra Technology*, Shawbury, Shrewsbury, Shropshire, 1993.
20. Anonymous, Chemical Resistance Data Sheets, Volume 1. Plastics, New Edition — 1993, *Rapra Technology*, Shawbury, Shrewsbury, Shropshire, 1993.
21. Benjamin, S., Carr, C., and Walbridge, D.J., Self-stratifying Coatings for Metallic Substrates, *Progress in Organic Coatings*, 28, 197-207, 1996.
22. Anonymous, Engineering Guide to Du Pont Elastomers, *The Du Pont Company*, Switzerland, 1987.
23. Hansen, C.M., The Universality of the Solubility Parameter, *Ind. Eng. Chem. Prod. Res. Dev.*, 8(1), 2-11, 1969.
24. Bosselaers, J., Blancquaert, P., Gors, J., Heylen, I., Lauwaerts, A., Nys, J., Van der Flaas, M., and Valcke Janssen, A., A New Fungicide and Algicide, *Färg och Lack Scandinavia*, 49(1), 5-11 2003.
25. Hansen, C.M., and Smith, A.L., Using Hansen Solubility Parameters to Correlate Solubility of C60 Fullerene in Organic Solvents and in Polymers, *Carbon*, 42(8-9), 1591-1597, 2004.

LIST OF TRADE NAMES AND SUPPLIERS

PAINTS AND BINDERS:

Bayer (D): Cellit, Desmophen, Desmolac, Pergut, Cellidora, Desmodur, Baysilon, Alkydal
 Hercules (US): Piccolyte, Cellolyn, Pentolyn, Ester Gum, Parlon
 Ciba-Geigy (CH): Araldite
 Shell (D): Epikote, Cariflex
 Union Carbide (US): Vinylite, Phenoxyl
 Hoechst (D): Macrynal, Phenodur, Alpex, Mowithal, Alftal, Mowilith
 Reichhold (CH): Super Beckasite, Uformite
 Polymer Corp. (CAN): Polysar
 Goodrich (US): Hycar
 Hüls (D): Vilit, Vesturit, Buna Hüls, Lutonal, Laroflex, Plastopal, Polystren
 Monsanto (US): Modaflex, Multiflex, Butvar
 Montecatini Edison (I): Vipla
 ICI (GB): Cereclor, Allophen, Suprasec
 Du Pont (US): Lucite
 Hagedorn (D): 1/2 sec. Nitrocellulose H 23
 Röhm (D): Plexigum
 Rohm and Haas (U.S.): Paraloid
 Dynamit Nobel (D): Dynapol
 SOAB (S): Soamin
 BIP Chemicals (GB): Beetle
 Dyno Cyanamid (N): Dynomin
 DSM Resins (NL): Uracron
 Wacker (D): Wacker
 Dow Chemical (CH): Ethocel
 Cray Valley (GB): Versamid
 W. Biesterfeld (D): Chlorparaffin
 Synres (NL): Synresin
 American Cyanamide (US): Cymel
 Polyplex (DK): Plexal
 Pennsylvania Industrial Chemical Corp. (US): Piccopale, Piccoumarone

OTHERS:

Chemical Fabrics Corporation: Challenge

Chevron Phillips: Ryton
ICI (Vitrex plc): Vitrex
Saranex: Dow
Safety 4, 4H: North
General Electric: Ultem
BASF: Ultrason
BP Chemicals: Barex
Rhône-Poulenc: Tolonate
Ticona (Celanese): Topas
West Company (DAIKYO): CZ Resin

The capital letters in parenthesis are the international symbols for the respective countries:

D	Germany
US	United States of America
CH	Switzerland
CAN	Canada
I	Italy
GB	Great Britain
S	Sweden
N	Norway
NL	Netherlands
DK	Denmark

TABLE A.2
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
Cellulose Acetobutyrate					
1	CELLIT BP-300	16.60	12.00	6.70	10.20
Cellulose Acetate					
2	CELLIDORA A	18.20	12.40	10.80	7.40
Ethyl Cellulose					
3	ETHOCEL HE10	17.90	4.30	3.90	5.90
4	ETHOCEL STD 20	20.10	6.90	5.90	9.90
Epoxy					
5	ARALDITE DY O25	14.00	7.40	9.40	13.70
6	EPIKOTE 828	23.10	14.60	5.00	20.50
7	EPIKOTE 1001	20.00	10.32	10.11	10.02
8	EPIKOTE 1004	17.40	10.50	9.00	7.90
9	EPIKOTE 1007	21.00	11.10	13.40	11.70
10	EPIKOTE 1009	19.30	9.37	10.95	8.26
11	PKHH	23.40	7.20	14.80	14.90
Epoxy Curing Agents					
12	VERSAMID 100	23.80	5.30	16.20	16.10
13	VERSAMID 115	20.30	6.60	14.10	9.60
14	VERSAMID 125	24.90	3.10	18.70	20.30
15	VERSAMID 140	26.90	2.40	18.50	24.00
Polyurethane					
16	DESMOPHEN 651	17.70	10.60	11.60	9.50
17	DESMOPHEN 800	19.10	12.20	9.90	8.00
18	DESMOPHEN 850	21.54	14.94	12.28	16.78
19	DESMOPHEN 1100	16.00	13.10	9.20	11.40
20	DESMOPHEN 1150	20.60	7.80	11.60	13.10
21	DESMOPHEN 1200	19.40	7.40	6.00	9.80
22	DESMOPHEN 1700	17.90	9.60	5.90	8.20
23	DESMOLAC 4200	18.70	9.60	9.90	8.20
24	MACRYNAL SM 510N	19.90	8.10	6.00	9.80
Phenolic Resins					
25	SUPER BECKACITE 1001	23.26	6.55	8.35	19.85
26	PHENODUR 373U	19.74	11.62	14.59	12.69
Hydrocarbon Resins					
27	PLIOLYTE S-100	16.47	0.37	2.84	8.59
28	PICCOPALE 110	17.55	1.19	3.60	6.55
29	PICCORONE 450L	19.42	5.48	5.77	9.62
Styrene-Butadiene (SBR)					
30	POLYSAR 5630	17.55	3.35	2.70	6.55
Acrylonitrile-Butadiene					
31	HYCAR 1052	18.62	8.78	4.17	9.62

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
Polybutadiene					
32	BUNA HULS B10	17.53	2.25	3.42	6.55
Polyisoprene					
33	CARIFLEX IR 305	16.57	1.41	-0.82	9.62
Polyisobutylene					
34	LUTONAL IC/1203	14.20	2.50	4.60	12.40
35	LUTANAL I60	16.90	2.50	4.00	7.20
36	POLYVINYL BUTYL ETHER	17.40	4.30	8.40	7.40
Special					
37	LIGNIN	20.17	14.61	15.04	11.66
38	MODAFLOW	16.10	3.70	7.90	8.90
Polyvinylchloride					
39	VIPLA KR (PVC)	18.40	6.60	8.00	3.00
Chloroparaffin					
40	CERECLOR 70	20.00	8.30	6.80	9.80
41	CHLOROPAR 40	17.00	7.60	7.90	11.90
Chlorinated Rubber					
42	PERGUT S 5	17.40	9.50	3.80	10.00
43	ALLOPREN R10	17.40	4.30	3.90	6.10
Chlorinated Polypropylene					
44	PARLON P 10	20.26	6.32	5.40	10.64
Chlorosulfonated PE					
45	HYPALON 20	18.10	3.40	4.90	3.60
46	HYPALON 30	18.20	4.70	2.00	5.00
Cyclized Rubber					
47	ALPEX	19.90	0.00	0.00	9.40
Nitrocellulose					
48	1/2-sec.-NITRO CELLULOSE H 23	15.41	14.73	8.84	11.46
Rosin Derivatives					
49	CELLOLYN 102	21.73	0.94	8.53	15.75
50	PENTALYN 255	17.55	9.37	14.32	10.64
51	PENTALYN 830	20.03	5.81	10.93	11.66
52	ESTER GUM BL	19.64	4.73	7.77	10.64
Polyamide					
53	VERSAMID 930	17.43	-1.92	14.89	9.62
54	VERSAMID 961	18.90	9.60	11.10	6.20
55	VERSAMID 965	20.15	6.04	12.90	9.20

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
Isocyanate					
56	DESMODUR L	17.50	11.30	5.90	8.50
57	DESMODUR N	17.60	10.00	3.70	9.30
58	SUPRASEC F-5100	19.70	12.90	12.80	11.40
Polyvinylbutyral					
59	MOWITAL B 30 H	18.60	12.90	10.30	8.30
60	MOWITAL B 60 H	20.20	11.20	13.30	11.20
61	BUTVAR B 76	18.60	4.36	13.03	10.64
Polyacrylate					
62	LUCITE 2042 PEMA	17.60	9.66	3.97	10.64
63	LUCITE 2044 PMMA	16.20	6.80	5.70	9.10
64	PLEXIGUM MB319	18.60	10.80	4.10	11.50
65	PLEXIGUM M527	18.40	9.40	6.50	10.70
66	PMMA	18.64	10.52	7.51	8.59
Polyvinylacetate					
67	MOWILITH 50 PVAC	20.93	11.27	9.66	13.71
Polystyrene					
68	POLYSTYRENE LG	22.28	5.75	4.30	12.68
Vinyl Chloride Copolymers					
69	LAROFLEX MP 45	18.40	8.40	5.80	9.00
70	VILIT MB 30	20.00	8.30	6.70	9.40
71	VILIT MC 31	20.00	8.30	6.70	9.40
72	VILIT MC 39	18.40	7.60	6.70	6.80
73	VINYLITE VAGD	17.10	10.40	6.50	7.50
74	VINYLITE VAGH	16.50	10.90	6.40	7.70
75	VINYLITE VMCA	17.70	11.10	6.90	8.70
76	VINYLITE VMCC	17.60	11.10	6.80	8.80
77	VINYLITE VMCH	17.60	11.10	6.40	8.60
78	VINYLITE VYHH	17.40	10.20	5.90	7.80
79	VINYLITE VYLF	18.10	10.30	4.20	8.30
Binders in Solution: Alkyds and Polyesters					
80	ALFTALAT AC 366	18.60	10.00	5.00	10.40
81	ALFTALAT AM 756	23.00	2.20	4.20	16.90
82	ALFTALAT AN 896	22.90	15.20	7.60	18.10
83	ALFTALAT AN 950	22.60	13.80	8.10	17.10
84	ALFTALAT AT 316	20.50	9.30	9.10	12.40
85	ALFTALAT AT 576	19.20	5.30	6.30	11.90
86	ALKYDAL F 261 HS	23.60	1.00	7.60	19.00
87	ALKYDAL F 41	20.60	4.60	5.50	12.60
88	DUROFTAL T 354	17.30	4.20	7.90	9.30
89	DYNAPOL L 812	22.60	13.10	5.80	16.80

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
90	DYNAPOL L 850	20.00	6.20	7.00	9.50
91	PLEXAL C-34	18.50	9.21	4.91	10.64
92	SOALKYD 1935-EGAX	18.00	11.60	8.50	9.00
93	VESTURIT BL 908	18.80	12.00	6.00	11.50
94	VESTURIT BL 915	17.70	13.00	7.60	11.50
Amino Resins					
95	BE 370	20.70	6.10	12.70	14.80
96	BEETLE 681	22.20	-0.40	10.10	18.40
97	CYMEL 300	19.35	12.83	12.87	9.82
98	CYMEL 325	25.50	15.20	9.50	22.20
99	DYNOMIN MM 9	18.80	14.00	12.30	10.50
100	DYNOMIN UM 15	19.90	15.80	13.40	11.70
101	SOAMIN M 60	15.90	8.10	6.50	10.60
102	SYNRESIN A 560	22.10	5.00	11.30	15.50
103	PLASTOPAL H	20.81	8.29	14.96	12.69
104	UFORMITE MX-61	22.70	2.80	5.40	16.20
Acrylate Resins					
105	URACRON 15	19.20	7.70	5.70	10.60
106	PARALOID P 400	19.20	9.60	9.30	12.20
107	PARALOID P 410	19.60	9.10	6.80	12.20
108	PARALOID EXPER. RES. QR 954	18.40	9.80	10.00	12.40
Silicone Resins					
109	BAYSILON UD 125	19.40	9.90	10.10	6.90
110	TEFLON (SL2-)	17.10	8.10	1.30	4.70
Special Data					
111	DOW EPOXY NOVOLAC 438	20.30	15.40	5.30	15.10
112	DOW EPOXY NOVOLAC 444	19.50	11.60	9.30	10.00
113	ZINK SILICATE - CHEMICAL RES.	23.50	17.50	16.80	15.60
114	2-COMP EPOXY CHEMICAL RES.	18.40	9.40	10.10	7.00
115	POLYVINYLIDINE FLUORIDE SOL.	17.00	12.10	10.20	4.10
116	COAL TAR PITCH SOL.	18.70	7.50	8.90	5.80
Breakthrough Time (Bt) Correlations for Common Types of Chemical Protective Films at Practical Film Thickness					
117	NITRILE 20 MIN	17.50	7.30	6.50	5.10
118	NITRILE 1 HR	16.60	9.10	4.40	10.00
119	NITRILE 4 HR	19.00	12.60	3.80	13.30
120	BUTYL 20 MIN	16.50	1.00	5.10	5.00
121	BUTYL 1 HR	15.80	-2.10	4.00	8.20
122	BUTYL 4 HR (2)	17.60	2.10	2.10	7.00
123	NATURAL RUBBER 20 MIN	14.50	7.30	4.50	11.00
124	NATURAL RUBBER 1 HR	15.60	3.40	9.10	14.00
125	NATURAL RUBBER 4 HR	19.40	13.20	7.70	19.00
126	PVC 20 MIN	16.10	7.10	5.90	9.30
127	PVC 1 HR	14.90	11.10	3.80	13.20
128	PVC 4 HR	24.40	4.90	9.90	22.70

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
129	POLYVINYLALCOHOL 20 MIN	11.20	12.40	13.00	12.10
130	POLYVINYLALCOHOL 1 HR	15.30	13.20	13.50	8.80
131	POLYVINYLALCOHOL 4 HR	17.20	13.60	15.40	10.90
132	POLYETHYLENE 20 MIN	16.90	3.30	4.10	8.10
133	POLYETHYLENE 1 HR	17.10	3.10	5.20	8.20
134	POLYETHYLENE 4 HR	24.10	14.90	0.30	24.30
135	VITON 20 MIN	10.90	14.50	3.10	14.10
136	VITON 1 HR	16.50	8.10	8.30	6.60
137	VITON 4 HR	13.60	15.40	8.60	14.40
138	NEOPRENE 20 MIN	17.60	2.50	5.90	6.20
139	NEOPRENE 1 HR	19.00	8.00	0.00	13.20
140	NEOPRENE 4 HR	14.60	13.90	2.30	15.90
141	CH 5100 3 HR	16.60	5.40	4.00	3.80
142	CH 5200 3 HR	16.60	6.00	4.80	3.70
High Temperature Solubility of PVDC					
143	PVDC (110C) SOLUBILITY	17.60	9.10	7.80	3.90
144	PVDC (130C) SOLUBILITY	20.40	10.00	10.20	7.60
Chemical Resistance of High Performance and Other Polymers					
145	PES L C=1	18.70	10.50	7.60	9.10
146	PES L B + C =1	17.70	9.70	6.40	9.30
147	PPS CR 93°C	18.80	4.80	6.80	2.80
148	PPS TS60%12MO	18.70	5.30	3.70	6.70
149	PA6 CR	17.00	3.40	10.60	5.10
150	PA66 SOL	17.40	9.80	14.60	5.10
151	PA11 CR	17.00	4.40	10.60	5.10
152	POMH+POMC CR	17.10	3.10	10.70	5.20
153	PETP CR	18.20	6.40	6.60	5.00
154	PTFE L80 CR	16.20	1.80	3.40	3.90
155	PMMA CR	16.00	5.00	12.00	13.00
156	PE? CR QUESTIONABLE VALUES	16.80	5.40	2.40	4.70
157	PPO CR	17.90	3.10	8.50	8.60
158	PUR CR	18.10	9.30	4.50	9.70
159	ABS CR	16.30	2.70	7.10	7.80
160	PSU CR	16.00	6.00	6.60	9.00
161	VINYL SILANE	16.40	3.70	4.50	10.00
Correlations for Some Barrier-Type Polymers					
162	CELLOPHAN SW	16.10	18.50	14.50	9.30
163	EVOH SOL	20.50	10.50	12.30	7.30
164	SARANEX 4HR	17.70	18.30	0.70	18.40
165	4H 35°C	19.40	13.40	18.00	8.60
166	4H 35°C no plasticizer included	20.50	11.30	10.30	6.70
167	POLYVINYLALCOHOL	15.00	17.20	17.80	10.20
Chemical Resistance Data - Modern Plastics Encyclopedia					
168	ACETAL CELANESE	21.10	9.30	5.90	11.40
169	ACETALHOMO-DUO	19.00	5.00	8.00	5.00
170	CELLULOSE ACETATE	16.90	16.30	3.70	13.70

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
171	CELL. ACET. BUTYRATE	17.20	13.80	2.80	12.60
172	CELL. ACET. PROPIONATE	9.80	13.60	11.40	15.20
173	PCTFE	14.10	2.70	5.50	6.60
174	FEP	19.00	4.00	3.00	4.00
175	FURAN	19.00	6.00	8.00	5.00
176	FURF ALC	19.90	3.90	5.10	3.80
177	PFA(?)	16.70	7.70	-0.50	8.10
178	PHENOLIC	21.60	5.20	18.80	15.40
179	PETG	18.00	3.00	4.00	6.00
180	HDPE 18.00		0.00	2.00	2.00
181	PP	18.00	0.00	1.00	6.00
Poly(Ethylene/Chlorotrifluoroethylene)					
182	PECTFE SOL AT HIGH TEMP.	19.50	7.30	1.70	5.10
Solubility of Polyacrylonitrile					
183	PAN	21.70	14.10	9.10	10.90
PEI - Polyethylene Imide - Environmental Stress Cracking (ESC)					
184	PEI 1200PSI	17.20	6.40	5.20	3.60
185	PEI 2400PSI	17.40	4.60	9.00	7.20
186	PEI 600PSI	17.30	5.30	4.70	3.30
Based on Solvent Range Solubility Data - Not too Reliable					
187	ESTER GUM	16.90	4.50	6.50	9.20
188	ALKYD 45 SOYA	17.50	2.30	7.70	10.00
189	SILICONE DC-1107?	19.60	3.40	10.80	9.80
190	PVETHYLETHER?	15.10	3.10	11.90	12.90
191	PBUTYLACRYLATE	16.20	9.00	3.00	10.10
192	PBMA?	15.90	5.50	5.90	8.50
193	SILICONE DC 23?	16.40	0.00	7.80	5.50
194	PE	16.00	0.80	2.80	3.20
195	GILSONITE	17.10	2.10	3.90	4.90
196	PVINYLBUTYLETHER	17.40	3.40	7.80	8.40
197	NAT RUBBER	16.00	4.00	6.00	1.30
198	HYP 20 CHLOROSULFONATED PE	17.40	3.20	4.00	4.80
199	ETHCEL N22?	22.70	0.50	16.50	20.10
200	CHLORINATED RUBBER	17.90	6.30	5.10	7.60
201	DAMMAR GUM	18.40	4.20	7.80	8.30
202	VERSAMID 100?	18.80	3.00	9.20	7.80
203	PS	18.50	4.50	2.90	5.30
204	PVAC	17.60	2.20	4.00	4.10
205	PVC	17.60	7.80	3.40	8.20
206	PHENOLICS	19.80	7.20	10.80	12.80
207	BUNA N BUTADIENE/ACRYLONITRILE	17.80	3.20	3.40	3.70
208	PMMA	18.10	10.50	5.10	9.50
209	PEO 4000 ? HEATED SAMPLES	21.50	10.90	13.10	15.90
210	POLYETHYLENESULFIDE (GOOD)	17.80	3.80	2.20	4.10
211	PC	18.10	5.90	6.90	5.50
212	PLIOLITE P1230	18.10	4.70	3.70	3.90

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
213	MYLAR PET	18.00	6.20	6.20	5.00
214	VCVA COPOLY	17.30	8.70	6.10	7.80
215	PUR 17.90		6.90	3.70	2.70
216	SAN	16.60	9.80	7.60	4.80
217	VINSOL ROSIN	17.40	10.00	13.00	10.50
218	EPON 1001	17.00	9.60	7.80	7.10
219	SHELLAC	19.70	10.10	15.10	10.70
220	POLYMETHACRYLONITRILE	17.20	14.40	7.60	3.80
221	CELLULOSE ACETATE	18.30	16.50	11.90	8.80
222	CELLULOSE NITRATE	16.90	13.50	10.30	9.90
223	PVOH (NOT GOOD, SEE CHAP. 5)	17.00	9.00	18.00	4.00
224	NYLON 66	16.00	11.00	24.00	3.00
Acrylics - Solvent Range					
225	ACRYLOID B-44	19.40	11.20	4.40	10.50
226	ACRYLOID B-66	18.00	9.00	3.00	9.00
227	ACRYLOID B-72	19.20	11.20	1.80	11.00
228	ACRYLOID B-82	19.10	9.10	3.30	9.00
229	R+H PBA	16.00	8.00	8.00	12.00
230	R+H PiMBA	20.70	4.10	10.70	11.50
231	R+H PNBMA	16.00	6.20	6.60	9.50
232	R+H PEMA	19.00	9.00	8.00	11.00
233	R+H PMAA	25.60	11.20	19.60	20.30
234	R+H PMMA	19.10	11.30	4.10	10.30
235	BMA/AN 80/20	17.50	9.90	4.10	9.50
236	ISOB MALANH/CYCLOL 75/25	16.80	-0.40	7.20	8.50
237	MAA/EA/ST 15/38/47	17.60	5.20	7.00	4.50
238	MAA/MA/VA 15/27.5/57.5	28.50	15.70	18.10	21.50
239	MAA/MA/VA 15/17.5/67.5	25.50	15.70	18.10	21.50
240	MMA/CYCLOL 58/42	18.70	9.90	8.70	8.80
241	MMA/EA 50/50	17.50	9.90	4.10	9.50
242	MMA/EA 25/75	19.00	9.00	15.00	12.00
243	MMA/EA/AGE 40/40/20	17.60	9.80	5.60	9.70
244	MMA/EA/AA	15.90	15.90	11.50	11.10
245	MMA/EA/AN 55/30/15?	16.70	10.90	8.50	8.50
246	MMA/EA/AN 40/40/20	20.40	13.20	11.00	12.30
247	MMA/EA/BAMA 40/40/20	17.90	8.50	11.70	12.90
248	MMA/EA/CYCLOL	17.60	9.80	6.40	9.80
249	MMA/EA/MAM 40/40/20	19.00	9.00	15.00	12.00
250	MMA/EA/MAM 45/45/10?	19.50	11.10	8.70	11.20
251	MMA/EA/BVBE 40/40/20	17.80	10.00	6.60	9.80
Polyesters - Solvent Range					
252	ACID DEGMP	15.30	13.30	14.90	15.60
253	CARB DEG PTH	19.40	13.40	11.60	11.10
254	CRYPLEX 1473-5	19.20	9.40	5.60	8.90
255	DEG ISOPH	19.20	17.20	14.60	11.80
256	DEG PHTH	21.00	15.20	13.20	13.70
257	DPG PHTH	20.10	11.50	6.70	11.60

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
258	DOW ADIP TEREP	17.80	10.40	6.80	9.30
259	DOW X-2635 MALEATE	17.80	5.60	6.80	4.50
260	VITEL PE LINEAR	14.90	10.10	2.90	6.10
261	VITEL PE101-X	21.30	6.30	4.70	7.30
262	HYD BIS A FUM ISPH	17.00	4.40	6.20	5.00
263	HYD BIS A PG FUM ISPH	18.70	8.90	5.50	8.40
264	PENTA BENZ MAL	19.40	12.20	10.20	10.80
265	SOL MYLAR 49001	19.00	5.00	4.00	5.00
266	SOL MYLAR 49002	19.00	5.00	5.00	5.00
267	TEG EG MAL TEREP	18.80	11.40	9.20	10.20
268	TEG MALEATE	18.10	13.90	12.10	9.70
269	VAREZ 123	17.30	10.90	11.90	10.70
Styrene Polymers And Copolymers - Solvent Range					
270	AMOCO 18-290	19.30	3.70	7.90	7.80
271	BUTON 100 BUTAD-STY	17.00	4.00	3.00	7.30
272	BUTON 300	17.30	3.70	7.30	7.00
273	KOPPERS KTPL-A	19.30	3.70	7.90	7.80
274	RUBBER MOD PS	20.00	5.00	1.00	7.00
275	STY MAL ANH	23.40	13.80	15.20	16.50
276	LYTRON 820	21.10	13.10	14.50	14.40
277	MARBON 9200	19.00	4.00	4.00	6.00
278	PARAPOL S-50	17.90	3.90	4.90	3.90
279	PARAPOL S-60	17.90	3.90	4.90	3.90
280	PICCOFLEX 120	17.40	7.80	3.80	7.70
281	SHELL POLYALDEHYDE EX 39	19.60	10.00	3.60	9.40
282	SHELL POLYALDEHYDE EX 40	19.60	10.00	3.60	9.40
283	SHELL X-450	19.30	9.50	11.10	11.10
284	SMA 1430A	18.80	11.40	16.40	14.10
285	SAN 85/15	19.10	9.50	3.10	8.70
286	STY/BUTENOL 85/15	17.40	7.80	3.80	7.70
287	STY/CYCLOL 82/18	18.20	5.60	7.20	5.70
288	STY/2EHA/AA 81/11/8	17.70	4.90	5.90	5.90
289	STY/MAA 90/10	18.70	6.30	7.30	6.70
290	STY/MA 85/15	18.00	9.00	3.00	9.00
291	STY/HALF ESTER MA 60/40	18.90	10.90	10.70	9.70
292	STY/PROP HALF E MA 57/43	18.00	9.80	8.40	10.10
293	STY/VBE 85/15	17.40	7.80	3.80	7.70
294	STYRON 44OM-27 MOD PS	20.00	5.00	1.00	7.00
295	STYRON 475M-27	20.00	5.00	1.00	7.00
296	STYRON 480-27	20.00	6.00	4.00	5.30
Vinyl Resins - Solvent Range					
297	ACRYLOID K120N	17.60	10.00	3.80	9.50
298	DODA 6225	19.00	2.00	1.00	3.00
299	DODA 3457	19.00	2.00	1.00	3.00
300	ELVAX 250	19.00	2.00	1.00	3.00
301	ELVAX 150	18.70	2.30	0.70	6.00
302	ELVAX EOD 3602-1	17.70	3.30	2.70	5.40

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
303	EXON 470 PVC	17.40	7.80	3.80	7.70
304	EXON 471	17.90	8.70	2.50	9.00
305	EXON 473	17.40	7.80	3.80	7.70
306	GEON 121	19.50	6.70	11.10	8.00
307	POLYCYCLOLa	19.00	9.00	15.00	12.00
308	PVBE 16.70		3.70	8.30	8.60
309	PVEE	16.00	4.00	12.00	14.00
310	FORMVAR 7/70E PVFORMAL	22.20	12.60	14.20	14.00
311	FORMVAR 15/95E	22.20	12.60	14.20	14.00
312	PVIBE	16.00	1.00	8.00	10.00
313	SARAN F-120 VCL2/AN?	28.80	16.80	0.80	23.70
314	SARAN F-220 ?	28.80	16.80	0.80	23.70
315	SINCLAIR 3840A	18.40	4.00	9.60	7.30
316	VA/EHA/MA 63/33/4	17.70	6.30	7.70	5.30
317	VA/EHA/CYC/MAA/76/12/8/4	21.20	12.40	13.00	12.60
318	VA/EA/CY 70/20/10	20.00	12.00	11.00	15.00
319	VBE/AN/MAA 46/27/27	18.90	11.70	11.10	9.60
320	VBE/MA/MAC46/27/27	19.40	13.00	13.80	12.30
321	VDC/AA 75/25 ?	20.40	11.00	0.80	11.70
322	VINYLITE AYAA PVAC	22.90	18.30	7.70	20.40
323	VINYLITE VAGH	17.00	7.80	6.80	7.10
324	VINYLITE VMCH	18.30	9.70	7.70	8.50
325	VINYLITE VXCC	18.00	9.40	4.60	8.40
326	VINYLITE VYHH	19.00	11.00	5.00	10.00
327	VINYLITE VYLF	18.00	9.40	4.60	8.40
328	VINYLITE XYHL PVBUTYRAL	19.00	9.00	15.00	12.00
329	VINYLITE XYSG PVBUTYRAL	19.00	9.00	15.00	12.00
330	VYSET 69	17.90	3.50	7.50	5.90
Miscellaneous - Solvent Range					
331	ACRYLAMIDE MONOMER	16.90	18.10	19.90	17.00
332	BAKELITE SULFONE P-47	20.00	3.00	6.00	3.00
333	BECKOLIN 27 MODIF OIL	11.40	0.00	3.00	18.10
334	PEO 4000 ? SAMPLES HEA TED	22.20	11.20	13.20	17.10
335	CHLORINATED RUBBER	18.00	6.00	5.00	7.00
336	CONOCO H-35 HYDROCARBON M	11.40	0.00	3.00	18.10
337	DAMMAR GUM DEWAXED	19.00	2.00	9.00	9.00
338	EPOCRYL E-11 ?	17.30	12.90	12.10	8.50
339	ESTANE X-7 ?? DIO XANE ONLY	19.00	1.80	7.40	1.00
340	HEXADECYL MONOESTER TRIME	19.00	11.60	14.00	11.90
341	HYDR SPERM OIL WX135	20.00	4.00	2.00	5.00
342	HYPALON 20 CHL SULF PE	17.80	3.20	4.40	4.10
343	HYPALON 30	17.80	3.40	3.20	5.10
344	KETONE RESIN S588	18.00	10.80	13.20	12.20
345	SANTOLITE MHP ARYLSULFONA	18.40	12.00	8.40	10.60
346	pTOLSULFONAMIDE-FORMALDEH	24.60	18.60	16.40	20.90

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
Polymer Solubility Data from Various Sources					
347	VYHH-NIF REPT	17.40	9.90	6.70	7.50
348	PVF? (DMF ONLY GOOD SOLVENT)	17.40	13.70	11.30	2.00
349	PES SOL	19.60	10.80	9.20	6.20
Biologically Interesting Systems					
350	LARD 37C	15.90	1.16	5.41	12.03
351	LARD 23C	17.69	2.66	4.36	7.98
352	1%IN WATER -AMINES	15.07	20.44	16.50	18.12
353	1%IN WATER +AMINES	14.96	18.33	15.15	16.22
354	BLOOD SERUM	23.20	22.73	30.60	20.48
355	SUCROSE	21.67	26.26	29.62	20.44
356	UREA	20.90	18.70	26.40	19.42
357	PSORIASIS SCALES	24.64	11.94	12.92	19.04
358	LIGNIN	20.61	13.88	15.25	11.83
359	CHOLESTEROL	20.40	2.80	9.40	12.60
360	CHLOROPHYLL	20.20	15.60	18.20	11.10
361	CELLULOSE-PAPER STRENGTH	25.40	18.60	24.80	21.70
Polysulfone PSU					
362	PSU ULTRASON S	19.70	8.30	8.30	8.00
Barex					
363	BAREX 210 CR	20.10	9.10	12.70	10.90
364	BAREX 210 CR-STYRENE	17.70	8.90	10.90	6.40
Polymers of Interest for Conservation of Paintings					
365	PARALOID B72	17.60	7.40	5.60	9.40
366	ESTIMATE DRIED OIL	16.00	6.00	7.00	5.00
367	DAMMAR DEWAXED	19.00	2.00	9.00	9.00
Permeation of LDPE by Organic Liquids					
368	LDPE PERM>80	16.50	4.50	0.50	6.00
369	LDPE PERM<0.8	15.30	5.30	2.50	10.10
Tolonate Solubility					
370	TOLONATE HDT (RH-POULENC)	19.00	11.00	3.00	12.00
371	TOLONATE HDB (RH-POULENC)	19.00	11.00	2.00	11.30
Chemical Resistance of Elastomers					
372	R ACM	16.80	11.80	11.60	17.00
373	R BUTYL	18.00	0.00	3.00	9.00
374	R ECO	21.30	8.10	6.10	12.00
375	R CSM	28.00	14.00	3.40	28.30
376	R EBONITE (DATA FIT 0.722)	18.70	6.10	2.70	6.60
377	R ETHYLENE/PROPYLENE	16.60	0.00	5.20	9.10
378	R EPDM	18.60	-3.40	4.40	10.70
379	R FQ FL/SI	15.90	20.10	6.90	16.80
380	R FKM (VITON) (0.76 DATA FIT)	11.60	23.00	5.00	21.60
381	R NR NAT RUB	20.80	1.80	3.60	14.00
382	R NBR	19.80	17.80	3.20	19.00

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
383	R CR CHLOROPRENE	24.60	8.60	6.40	20.40
384	R AU ESTER PU	17.90	13.30	10.70	17.10
385	R PEU ETHER PU	17.90	13.30	10.70	17.10
386	R T SULPHIDE	25.30	17.30	6.70	23.60
387	R Q SILICONE (0.748 D ATA FIT)	13.80	5.00	1.20	14.30
388	R SBR	17.20	6.00	4.60	9.80
389	R TFP TETFLPROP (0.744 DATA FIT)	16.60	6.80	0.60	7.90
Chemical Resistance of Plastics					
390	R ABS	17.60	8.60	6.40	10.90
391	R CELLULOSE ACETATE	14.90	7.10	11.10	12.40
392	R CHLORINATED PVC	17.50	6.50	5.50	6.30
393	R DIALLYLPHTHALATE	22.20	12.20	8.60	15.80
394	R POM ACETAL	17.20	9.00	9.80	5.30
395	R PA12	18.50	8.10	9.10	6.30
396	R PA66	18.20	8.80	10.80	5.20
397	R POLYAMIDEIMIDE	18.50	5.70	8.70	4.20
398	R POLYBUTYLENETEREPH	18.00	5.60	8.40	4.50
399	R POLYCARBONATE	19.10	10.90	5.10	12.10
400	R HDPE/LDPE	17.50	4.30	8.30	3.90
401	R PET	19.10	6.30	9.10	4.80
402	R POLYIMIDES	24.30	19.50	22.90	21.60
403	R PMMA	19.30	16.70	4.70	17.40
404	R TPX	18.80	1.40	6.40	7.90
405	R POLYPHENYLENEOXIDE	16.90	8.90	2.70	11.70
406	R POLYSULPHONE	19.80	11.20	6.20	11.30
407	R POLYPROPYLENE	17.20	5.60	-0.40	4.50
408	R EPOXY COLD CURING	16.80	10.80	8.80	8.20
409	R EPOXY HOT CURING	18.30	12.30	9.70	7.30
410	R HET RESIN	17.50	11.30	8.30	8.60
411	R ISOPHTHALIC	19.80	17.40	4.20	18.00
412	R TEREPHTHALIC	19.80	17.40	4.20	18.00
Polymers at Different Test Concentrations - (Conc) Epoxy Polymers					
413	EPIKOTE 828 (60%)	16.60	14.00	2.80	14.90
414	EPIKOTE 828 (30%)	16.30	16.40	1.90	16.70
415	EPIKOTE 1001 (60%)	15.80	16.30	3.30	16.40
416	EPIKOTE 1001 (40%)	16.30	13.10	6.30	10.90
417	EPIKOTE 1001 (20%)	19.80	13.60	8.90	12.00
418	EPIKOTE 1001 (10%)	18.10	11.40	9.00	9.10
419	EPIKOTE 1004 (60%)	17.70	10.10	7.60	9.80
420	EPIKOTE 1004 (30%)	18.50	9.30	8.00	9.60
421	EPIKOTE 1007 (30%)	18.60	10.60	8.10	8.80
422	EPIKOTE 1009 (60%)	17.00	9.60	8.50	7.60
423	EPIKOTE 1009 (30%)	19.80	10.60	10.30	9.70
424	EPIKOTE 1009 (10%)	19.00	9.10	10.70	8.00
Acrylics					
425	PIBMA (10%)	17.00	4.60	7.60	9.50
426	PIBMA (30%)	17.10	5.90	0.70	7.30

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
427	PMMA (10%)	17.80	10.40	2.90	9.60
428	PMMA (30%)	17.20	7.20	3.50	4.80
429	PBMA (10%)	20.60	3.50	7.20	12.80
430	PBMA (30%)	18.10	5.70	0.00	8.50
431	PMMA (10%)	17.60	10.10	5.80	9.40
432	PMMA (30%)	17.50	5.50	3.80	4.50
433	PEMA (10%)	16.50	8.70	5.00	10.40
434	PEMA (30%)	16.90	7.80	0.50	7.30
435	CRODA AC500 THERMOSET (10%)	17.80	6.40	4.70	10.70
436	CRODA AC500 THERMOSET (30%)	21.20	1.40	10.70	12.30
437	CRODA AC550 THERMOSET (10%)	16.30	10.60	7.40	12.90
438	CRODA AC550 THERMOSET (30%)	16.30	10.60	7.40	12.90
Fluorinated Polyethers					
439	LUMFLON LF200 (10%)	18.50	5.40	6.90	9.90
440	LUMFLON LF200 (30%)	20.10	4.40	3.20	8.50
441	LUMFLON LF916 (10%)	17.50	6.80	10.50	12.50
442	LUMFLON LF916 (30%)	18.10	3.90	8.30	8.80
Acrylic Modified Alkyd					
443	PLASTOKYD S27 (30%)	20.10	5.70	5.30	20.00
444	PLASTOKYD SC140 (30%)	25.20	9.20	3.70	20.00
445	PLASTOKYD SC400 (30%)	23.70	0.50	10.30	20.00
446	PLASTOKYD AC4X (30%)	23.90	7.80	8.80	19.90
Chlorinated Rubber					
447	ALLOPRENE R10 (10%)	19.50	9.20	6.90	7.50
448	ALLOPRENE R10 (30%)	17.90	5.60	6.70	5.80
449	ALLOPRENE R10 (60%)	19.60	6.50	5.80	9.10
Chlorosulfonated Polyethylene					
450	HYPALON 20 (30%)	20.30	3.20	0.70	11.30
Polyisoprene Swelling					
451	POLYISOPRENE SW	17.00	4.00	4.00	7.30
Bromobutyl Rubber Swelling					
452	BROMOBUTYL RUBBER S	17.60	1.70	2.00	6.00
453	BROMOBUTYL RUBBER L	17.00	3.40	2.00	6.00
Supplemental Chemical Resistance Correlations					
454	NEOPRENE CR	18.10	4.30	6.70	8.90
455	HYTREL +/- OK	24.20	14.60	13.20	18.80
456	HYTREL +/- NOT OK	26.40	18.80	7.40	26.30
457	HYPALON +/- OK	18.40	3.60	6.40	9.00
458	HYPALON +/- NOT OK	18.40	5.60	6.00	9.40
Ethylene Vinylacetate (EVA) Solubility					
459	EVA 4055 SOL	17.70	3.50	3.70	4.70

TABLE A.2 (CONTINUED)
Hansen Solubility Parameters for Selected Correlations

Number	Polymer	Dispersion	Polar	Hydrogen Bonding	Interaction Radius
COC Solubility					
460	TOPAS 6013 SOL	18.00	3.00	2.00	5.00
461	CZ RESIN SOL	18.00	1.00	3.00	4.00
Miscellaneous					
462	KAURI GUM	18.7	8.1	13.0	8.2
463	POLYVINYLPIRROLIDONE (PVP)	21.4	11.6	21.6	17.3
464	PALM OIL	17.7	3.5	3.7	4.7
465	BETHOXAZIN	22.4	7.6	10.8	13.9
466	CARBON-60	19.7	2.9	2.7	3.9

Appendix A: Table A.3

COMMENTS TO TABLE A.3

The data in Appendix A.3 is for the solubility of the polymers listed in Table 5.2. 0.5 g of polymer was placed in a test tube with 5 ml of the given solvent (except for the Milled Wood Lignin). The evaluation of solubility was made on a scale as follows:

1. Soluble
2. Almost soluble
3. Strongly swollen, slight solubility
4. Swollen
5. Little swelling
6. No visible effect

Note: * indicates a reaction. Data reproduced from Hansen, C.M., "The Three Dimensional Solubility Parameter and Solvent Diffusion Coefficient" doctoral dissertation, Danish Technical Press, Copenhagen, 1967.

Polymer

[illegible]

TABLE A.3 (CONTINUED)
Solubility Data for the Original 33 Polymers and 88 Solvents

Polymer	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	X	Y	Z	A	B	C	D	E	F	G	L	
Methyl Ethyl Ketone	1	1	1	1	1	1	1	1	6	1	1	4	1	1	1	1	1	1	1	4	1	4	1	4	6	1	1	5	1	5	3	1	6	
Methyl Isoamyl Ketone	1	1	1	1	1	1	1	1	6	1	1	6	1	1	1	1	1	6	1	1	1	1	1	1	6	1	1	1	4	2	5	1	6	
Methyl Isobutyl Carbinol	4	5	5	1	1	6	5	1	6	6	6	1	6	1	1	6	4	4	6	5	6	1	6	1	6	1	1	5	4	6	6	6	6	
Methyl Isobutyl Ketone	1	1	1	1	1	1	1	1	6	1	1	6	1	1	1	1	4	1	1	1	1	1	1	4	6	1	1	1	4	4	4	1	6	
Methylal	1	5	1	1	1	1	1	1	6	1	5	5	1	3	1	1	5	1	1	5	1	4	1	3	6	1	1	5	2	3	6	1	6	
Methylene Dichloride	1	1	1	1	1	1	1	1	1	6	1	5	1	6	1	5	3	1	1	1	1	1	1	1	2	1	1	1	1	1	2	1	6	
Morpholine	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	4	1	5	4	5	6	1	1	6	1	6	1	1	
Nitrobenzene	1	1	1	1	1	1	1	1	1	1	1	4	1	5	1	3	2	1	1	1	1	1	1	6	1	6	1	1	5	1	6	2	1	4
Nitroethane	1	1	1	5	5	2	1	6	1	5	5	1	6	3	6	5	1	1	6	1	6	6	6	6	6	3	1	6	3	6	6	2	6	
Nitromethane	1	6	2	5	6	5	5	1	5	1	6	1	2	6	6	6	5	2	1	6	1	6	6	6	6	4	1	6	4	6	6	5	6	
2-Nitropropane	1	1	2	1	6	5	1	1	6	1	3	5	1	6	4	6	4	1	1	6	1	6	6	5	6	1	1	6	3	6	5	1	6	
1-Pentanol	4	6	5	3	1	1	6	2	1	6	6	6	1	1	1	1	6	5	6	6	6	6	1	6	1	1	1	6	4	6	6	6	6	
1-Propanol	5	6	3	5	1	1	6	4	1	6	6	6	1	1	2	1	6	4	2	6	6	6	6	1	5	1	5	1	6	4	6	6	6	
Propylene Carbonate	3	1	1	5	6	6	6	1	6	1	6	3	1	1	6	6	1	1	1	6	4	6	6	6	6	6	1	6	1	6	6	6	6	
Propylene Glycol	6	6	5	6	4	6	6	6	1	6	6	6	6	1	6	4	6	5	1	6	6	6	6	6	6	6	1	6	5	6	6	6	3	
Pyridine	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	5	4	1	5	1	1	5	1	5	2	1	1	
Styrene	1	2	2	1	1	1	1	1	3	6	1	6	1	6	1	4	6	1	1	1	1	1	1	1	1	6	1	1	1	4	1	5	1	5
Tetrahydrofuran	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	3	
Tetrahydronaphthaline	1	3	4	1	1	2	1	5	5	6	1	6	1	6	1	4	6	1	2	1	1	1	1	1	1	5	1	1	5	1	6	1	4	
Toluene	1	2	2	1	1	2	1	1	2	6	1	6	1	6	1	6	6	1	3	1	1	1	1	1	1	6	1	1	5	1	5	1	5	
1,1,1-Trichloroethane	1	5	4	1	1	4	1	1	6	6	1	6	1	6	1	3	6	1	3	1	1	1	1	1	1	6	1	1	4	1	6	1	6	
Trichloroethylene	1	4	2	1	1	1	1	1	1	6	1	6	1	6	1	4	5	1	1	1	1	1	1	1	1	2	1	1	4	1	5	1	5	
Xylene	1	4	3	1	5	3	1	4	5	5	1	6	1	6	1	6	6	1	3	1	1	1	1	1	1	6	1	1	5	1	6	1	6	

Index

A

Absorption
 concentration-dependent, 298
 equations, 294–296
 film formation by solvent evaporation and, 305–306
 plastics, 315–316, 339
 side effects, 304–305
 surface resistance and, 300–301
Acetophenone, 59, 262
Acid-base theories and HSP of pigments, 130–131
Acids
 amines and, 140
 dimerization, 65
 hydrophilic bonding, 133–134
ACLAR®, 251, 252–253
Acrylonitrile/butadiene/styrene (ABS) terpolymer, 120, 300
Active agents, surface, 332–333
Activity coefficient model
 Flory-Huggins models, 82–84
 future challenges, 90–91
 at infinite dilution, 85–8
 mixed solvent-polymer phase equilibria, 88–90
Adenine, 275
Adhesion maximization, physical, 122, 342
Adsorption, controlled, 132–134
Affinity and Hansen solubility parameters, 4–
Alcohol in the bloodstream, 275–276
Alpha-helices, 272
Alternative systems, solvent, 312
Amines, 140
Amino acid side chains and water, 270–271
Ammonia, 57
Amphipathic molecules, 270–271
Anomalous diffusion, 306–308
Aromas and fragrances, 338–339
Asphalt, 152
Asphaltenes, 153, 164–166
 definition, 15
 molecular weight of, 154
 polarity, 155
Availability of HSP data, 321–322
Azeotropes, 212, 213–215, 216–226

B

Barrier polymers
 breakthrough times, 245–247
 concentration-dependent diffusion, 244–245
 human skin as, 250, 316
 laminated, 253–254

RED numbers, 247, 253
solubility parameter correlation of permeation
 coefficients for gases, 251–254
solubility parameter correlation of permeation rates,
 248–250
solubility parameter correlation of polymer swelling,
 250
solubility parameter correlations based on permeation
 phenomena, 245–250
Benzene, 204
Beta sheets, 272
Binders
 coating, 138, 139, 144
 pigment, 128–131
Biological materials
 blood serum and zein, 279, 280–281
 chiral rotation, hydrogen bonding and nanoengineering,
 290–291
 chlorophyll, 279
 cholesterol, 275–277
 DNA, 273–275
 HSP characterization of, 270–271
 human skin, 250, 277–279
 hydrophobic bonding and hydrophilic bonding of,
 271–273
 lard, 277
 lignin, 279, 282–283, 284–288, 316–317
 surface mobility, 290
 urea, 274, 283, 289
 water, 21–22, 289
 wood chemicals and polymers, 279, 281
BISOM test, 170–173
Bitumens
 asphaltenes in, 153, 154–155
 BISOM test, 170–173
 calculation and plotting of Hansen 3D pseudosphere,
 161–164
 components of, 164–166
 Hildebrand solubility parameters, 156, 158
 HSP, 158–164
 hydrocarbons in, 155–156, 164–166
 models of, 152–154
 polymer modified, 15
 polymers and, 166–169
 production process, 151–152
 RED numbers, 160
 solubility parameters of, 155–156
 solubility SPHERE, 159–164
 solvents used for determination of solubility of,
 161–164
 temperature effect on, 168
 testing solubility of, 156, 157
 turbidimetric titrations, 170

uses, 151, 160
 Blanc fi e, 131
 Blistering of coatings, 141, 281
 Blood serum, 279, 280–281
 Boundaries of solubility, 7–8
 Breakthrough times, HSP correlations of, 245–247
 Butadiene block copolymer (SBS), 168
 Butyl rubber, 246

C

Calculations
 versus experimental χ_{12} parameters, 34–39
 polymer HSP, 97–98
 Carbon dioxide, 57, 252
 Hildebrand parameters of, 187–190
 HSP of, 336–337
 ideal solubility of g ases in liquids and, 199–201
 pressure effects on solubility parameters of, 187, 191–196
 solubility data in v arious solvents, 178–186, 187
 solubility in liquid sol vents, 185–186
 solubility parameters, 141–142
 temperature effects on solubility parameters of, 190–191
 three-component solubility parameters, 189–190
 Carbon disulfide, 16–1
 Carbon fiber sur ace characterization, 131–132, 133
 Carbon tetrachloride, 35–36, 204
 Carcinogens, 312
 Castor oil, 214
 Cellulose, 281, 289
 acetate, 303
 Challenge® 5100, 246
 Chemical protective clothing, 315
 Chemical resistance
 acceptable-or-not data on, 232
 effects of solvent molecular size on, 232–233
 HSP characterization of, 231–232, 339
 plastics, 234–237
 special effects with water, 238–239
 tank coatings, 233–234
 tensile strength and, 237–238
 Chiral rotation and HSP, 290–291
 Chlorobenzene, 245, 305
 Chloroform, 35–36
 Chlorophyll, 279
 Cholesterol, 275–277
 Cleaning solvents
 application of HSP methodology to, 212–213
 azeotropes, 212, 213–215, 216–226
 calculating HSP of composites and, 205–206
 HSP of multiple-component soils and, 204–205
 identification of designe , 208
 method for choice of suitable, 206–208
 pathology of soils and, 204
 performance, 210–212
 reference soils, 208, 209–210
 variety of chemicals used in, 204

Coatings
 amines added to, 140
 binders, 138, 139, 144
 blistering in, 141, 281
 chemical resistance and, 231–239, 240
 gases as sol vents in, 141–142
 for hazardous materials, 312
 PET film, 234, 23
 polymer compatibility in, 145–146
 RED numbers, 142–143, 144
 self-stratifying, 120–121
 solvents, 137–142
 and surface phenomena in, 144–145
 SPHERE and, 143
 surface mobility, 333–334
 tank, 233–234
 water-reducible, 139–140
 Cohesion energy parameters, 2, 5, 13–18, 113
 Cohesive energy density, carbon dioxide, 187
 Cohesive energy difference, 27
 polymer segments and sol vents, 30–32
 Composite soils, 204–206
 Compound formation, 9
 Concentration-dependent diffusion, 244–245, 297–298, 304–305
 Constant diffusion coefficients, 296–29
 Controlled adsorption, 132–134
 Controlled release of drugs, 339–340
 Cooling, rapid, 19
 Corresponding states theory (CST), 3, 6
 HSP and, 30–32
 predicting the χ_{12} parameter using, 28
 Critical chi parameter, 32
 Critical surface tensions, 116–117
 Crude oils
 BISOM test, 170–173
 HSP of molecules in, 169–170
 origins and structure, 152–153
 uses, 151–152
 Cyclic olefinic copolymer (COC), 120, 300–30
 Cyclohexanone, 185

D

Danish MAL system, 313–315
 Data, HSP
 availability, 321–322
 quality, 324, 326
 Dendrimers, 90–91
 Designer solvents. *See also* Cleaning solvents
 analysis of capability of, 213–215
 application of HSP methodology to, 212–213
 azeotropes, 212, 213–215, 216–226
 cleaning performance of, 210–212
 defined, 20
 identification of, 20
 variety of chemicals used in, 204
 Desirability function, 20
 Dextran C, 101–102, 322, 323–324, 325–326
 Dibasic esters (DBE), 315

1,2-dibromoethane, 185
Dichloromethane, 185
Diethyl ether, 17, 262
Diffusion
 anomalous, 306–308
 concentration-dependent, 244–245, 297–298, 304–305
 constant coefficients, 296–299
 equations, 294–296
 film formation by solvent evaporation and, 305–306
 side effects, 304–305
 steady state permeation, 296
 surface effects and, 302–304
 surface resistance and, 298–305
 time-dependent, 308
 vapor, 301–302
Dilution, infinite, 85–8
Dimerization, acid, 65
Dimethyl sulfoxide, 289
1,4-dioxane, 262
Dipolar interactions, 50–52, 66
Dipole moments, 8–9
 zero, 16–17
Dispersion solubility parameters, 5, 13–16
 hydrogen bonding, 52–59
Distance, solubility parameter, 7
DNA, 273–275
Dynamic shear rheometer (DSR), 152

E

Elastomers, 245, 333
Electron exchange parameters, 5, 29
Electrostatic repulsion, 140
Energy
 density, cohesive, 13–16
 Gibbs, 92–93
 polar cohesive, 5
 relative difference, 8
 surface free, 113–114
 of vaporization, 13–16
Entropic group contribution model, 78–79, 83
Environmental impact of solvents, 208
Environmental stress cracking (ESC), 107, 120
 incidence of, 259–260
 interpreted using HSP, 260–262
 mechanism for, 259–260, 264–267
 with nonabsorbing stress cracking initiators, 263–264
 RED number and, 260–262
EPDM rubber, 141
Epon® 1001, 233
Epoxies, 138, 141, 233–234
Equation-of-state framework, 46–50, 60–62
 carbon dioxide, 188–189
Ethanol, 4–5, 276
Ethyl acetate, 262
Ethylene glycol monomethyl ether (EGMME), 245
Ethylene vinyl alcohol copolymers (EVOH), 253–254
Evaporation, solvent, 305–306

F

Fickian diffusion, 306–308
Field ionization mass spectrometry (FIMS), 154
Fillers and fibers, 147–14
Film formation by solvent evaporation, 305–306
Films, PET, 234, 235
First-order and second-order groups in solubility parameter
 prediction, 66–73
Floating rates of particles in pigments, 126–127
Flocculation, 153
Flory-Huggins model, 27–28, 77, 341
 versus the GC methods, 84–90
 for multicomponent mixtures, 92–93
 regular solution theory and, 80–82
 using Hildebrand and HSP, 82–84
Fluoropolymers, 16–17
Formamide, 274
Fragrances and aromas, 338–339
Free-volume-based models for polymers, 77–79
 solvent selection for paints, 85–88
Fresh Air Numbers (FAN/MAL) system, 313–315

G

Gases
 characterization problems, 336–337
 HSP of, 336–337
 ideal solubility in liquids, 199–201
 permeation through polymers, 243
 solubility parameter correlation of permeation
 coefficients for, 251–254
 as solvents, 141–142
Gibbs energy, 92–93
Glycol ethers, 138, 140–141
Group-contribution method
 applications, 76–77
 entropic model, 78–79, 83
 Flory-Huggins/Hansen model versus, 84–90
 Flory-Huggins model and regular solution theory,
 80–82
 free-volume-based models for polymers, 77
 reliability of calculation procedures for, 328
 UNIFAC model, 76–78, 83
Guggenheim expression, 47

H

Halar®, 235, 236
Hansen 3D pseudosphere, 161–164
Hansen solubility parameters (HSP)
 acid-base theories and, 130–131
 aromas and fragrances, 338–339
 biological materials, 269–291
 bitumen, 158–164
 blood serum and zein, 279, 280–281
 carbon dioxide, 177–201
 for chemical protective clothing, 315

chemical resistance characterization using, 231–239, 240, 339

chiral rotation and, 290–291

chlorophyll, 279

cholesterol, 275–277

coatings, 145–146

composite soils, 204–206

controlled release drugs, 339–340

correlation of permeation coefficients for gases, 251–254

correlation of permeation rates, 248–250

correlation of polymer swelling, 250

correlation of zeta potential for blank films, 131

correlations for chemical resistance, 234–236

correlations of breakthrough times, 245–247

correlations with surface tension, 113–114

corresponding states theories and, 30–32

data

- availability, 321–322
- quality, 324, 326

defined, 1, 4–

Dextran C, 322, 323–324, 325–326

electron exchange energy and, 29

environmental stress cracking (ESC) and, 260–267

fillers and fibers, 147–1

first-order and second-order groups contribution to, 66–73

Flory-Huggins model and, 81–82

gases, 336–337

GC methods versus the FH/, 84–90

human skin, 277–279, 316

hydrogen bonding and, 290–291

inorganic salts, 337

lard, 277

lignin, 279, 282–283, 284–288

methodology applied to cleaning operations, 212–213

methods and problems in determination of, 6–13

mixtures, 205–206

molecules in crude oils, 169–170

multiple-component soils, 204–206

nanoengineering and, 290–291, 340–341

organic compounds, 177–178

organic salts, 337

organometallic compounds, 338

pigments, 128–131

- fillers and fibers, 125–1

plastics, 315–316, 339

polymers, 29–30, 95–109

solvents, 29–30, 40–41, 137–143

special effects with water, 238–239

surface

- characterization, 113–122, 131–132, 330–332
- liquid adhesion, 342
- mobility and, 263–264, 290

tensile strength and, 237–238

three-component, 189–190

urea, 283, 289

water, 21–22, 289, 326, 334–336

wood chemicals and polymers, 279, 281

χ_{12} parameter and, 32–39

Hazardous materials

- chemical protective clothing for, 315
- Danish MAL system, 313–315
- skin penetration of, 316
- solvent formulation and personal protection for least risk from, 313
- substitutions for, 311–312
- transport phenomena, 316–317
- uptake by plastic containers, 315–316

Heat of vaporization, 50

Hemicelluloses, 281, 289

Henry's law coefficient, 17

Hexane, 116, 120, 204

Hildebrand parameters, 2–4

- of bitumen, 155, 156, 158
- of carbon dioxide, 187–190
- solvent selection for paints, 85–88
- χ_{12} coefficient and, 28, 3

Hoy dispersion parameter, 9, 13

Human skin

- HSP of, 277–279, 316
- penetration of hazardous materials, 316
- as polymeric barrier, 250

Hydrocarbons

- in bitumen, 155–156, 164–166
- carbon dioxide solubility parameters and, 192–194, 336–337
- coatings applications, 138–139
- and corresponding states theories, 29
- in crude oils, 153
- ether solvents partially based on silicon, 208

Hydrochlorides, 339–340

Hydrofluoroethers (HFEs), 20

Hydrogen bonding, 5, 7

- acid dimerization and, 65
- coatings, 138
- cohesive energy differences and, 31–32
- defined, 26
- dipole moments and, 9
- DNA, 273–275
- HSP and, 290–291
- lattice-fluid, 4
- nonrandom, 46, 50–59
- polymer surfaces, 115
- self-stratifying coatings, 120–121
- solubility parameters, 17, 52–59
- techniques for data treatment, 142–143
- temperature increases and, 140–141
- water temperature and, 141

Hydrophilic bonding, 133–134, 271–273, 333–334

Hydrophobic bonding, 271–273, 332

I

Inorganic salts, 337

Interaction radius, 184

Internal pressure, 187

Intrinsic viscosity of polymers, 107–109

J

Joffé effect, 342

K

Keratin, 277, 317
Ketones, 138, 140

L

Laminates, 253–254
Lard, 277
Lattice-fluid hydrogen bonding (LFHB), 46
Lignin, 279, 282–283, 284–288, 316–317
Liquid-liquid phase separation (LLE), 75
Liquids
 absorption by plastic containers, 315–316
 adhesion phenomena, 342
 cohesion energy parameters and surface
 characterization of, 114–116
 environmental stress cracking (ESC) and, 262
 first-order and second-order groups in, 66–7
 ideal solubility of gases in, 199–201
 solubility parameters for, 17–21, 32–33, 52–59
 solvents, CO₂ solubility in, 185–186
 spontaneous spreading, 115–118
 surface tension, 116–117
Lithium, 338
Lorenz-Berthelot mixtures, 41
Low density polyethylene (LDPE), 249
Lower critical solution temperature, 28
Lydersen group constants, 12–13t, 15

M

MAL system, Danish, 313–315
Maltenes, 153, 164–165
MATLAB platform, 163
Melting points of polymers, 106–107
Metallic bonding, 338
Methanol, 19, 141, 238–239
2-methylcyclohexanone, 185
Methylene chloride, 140
Methyl ethyl ketone, 140
Methyl iodide-cellulose acetate, 303
Methyl isobutyl ketone, 262
Methyl methacrylate, 277
Mixtures, solubility parameters of, 205–206, 283, 289
Mobility, surface, 263–264, 290, 333–334
Molar volume and solubility parameters, 7
Molecular size and solubility parameters, 19–20, 32–33, 232–233, 243
Morpholine, 108
Multicomponent mixtures, Flory-Huggins model for, 92–93
Multiple-component soils, 204–206
Multivariable analysis and solubility parameters, 8

N

Nanoengineering, 290–291, 340–341
n-butanol, 138, 140
n-butyl acetate, 262
Neoprene®, 235
New Flory theory, 27–28
n-heptane, 153, 154, 165–166
n-hexane, 243, 262
Nitrates, 337
Nitrile rubber, 246
Nitromethane, 4–5
N-methyl-morpholine-N-oxide, 279
N-methyl-2-pyrrolidone, 204
Nonabsorbing stress cracking initiators, 263–264
Nonbiodegradable substances, 312
Nonpolar interactions in organic materials, 5
Nonrandom hydrogen bonding, 46, 50–59
Normal diffusion, 306–308
Nylon-6, 303

O

Octyl alcohol, 277
Oleic acid, 185
Olive oil, 277
One-component Hildebrand parameter as a function of
 temperature and pressure, 187–189
Organic compounds, HSP and solvents for, 177–178
Organic salts, 337
Organoclays, 340–341
Organometallic compounds, 338
Ostwald coefficient, 17
Ozone depletion potentials (ODP), 208

P

Paints
 hydrophilic bonding in, 272
 older, dried, 142–143
 self-stratifying coatings, 120–121
 solvent selection for, 85–88
Partial solubility parameters determination, 6–13
 dipolar interactions and, 50–52
 for liquids, 17–21, 52–59
Particle suspensions, 126–127
Pathology of soils, 204
P-dioxane, 303
Peat moss, 333
Pentachlorophenol, 316
Permeation
 of liquid or gas through polymers, 243
 rates, 248–250
 size and shape and dependence of diffusion coefficient,
 255–256
 solubility parameter correlation of permeation
 coefficients for gases, 251–254
 solubility parameter correlations based on, 245–250
 steady state, 296